



UNIVERSITEIT VAN AMSTERDAM

Measure Theoretic Probability

NOTES BY

Hengyi Li

LECTURERS

Dr. M. R. Engel

Prof. dr. R. W. J. Meester

Dr. D. E. Terhesiu

INTRODUCTION. The course introduces a rigorous treatment of probability theory based on measure and Lebesgue integration theory. We will develop the necessary measure and integration theory together with the probabilistic interpretation so that the two subjects (measure theory and probability) are developed simultaneously. Measure-theoretic topics include σ -algebras, measures, Lebesgue integrals, convergence theorems, and the *Radon–Nikodym theorem*. Probabilistic topics include conditional probability and expectation, modes of convergence of random variables, characteristic functions, laws of large numbers, and martingales.

Contents

1	Probability	1
1.1	Borel's Normal Number Theorem	1
1.2	Probability Measures	6
1.3	Existence and Extension	9
1.4	Denumerable Probabilities	11
1.5	Simple Random Variables	14
2	Measure	15
2.1	General Measures	15
2.2	Measures in Euclidean Space	17
2.3	Measurable Functions and Mappings	19
2.4	Distribution Functions	21
3	Integration	23
3.1	The Integral	23
3.2	Properties of the Integral	27
3.3	The Integral with Respect to Lebesgue Measure	30
3.4	Product Measure and Fubini's Theorem	31
4	Random Variables and Expected Values	34
4.1	Random Variables and Distributions	34
4.2	Expected Values	37
4.3	Sums of Independent Random Variables	40
5	Convergence of Distributions	42
5.1	Weak Convergence	42
5.2	Characteristic Functions	45
5.3	The Central Limit Theorem	48

6	Derivatives and Conditional Probability	50
6.1	The Radon-Nikodym Theorem	50
6.2	Conditional Probability	51
6.3	Conditional Expectation	55
6.4	Martingales	59
7	Stochastic Processes	64
7.1	Brownian Motion	64
	Appendix	66
A.1	Final Exams	66
A.2	Class Tests	88
A.3	Assignments	94
	References	103

Probability

SECTION 1.1

| Borel's Normal Number Theorem

The theory of discrete probability spaces does not go far enough for the rigorous treatment of problems of two kinds: those involving an infinitely repeated operation, as an infinite sequence of tosses of a coin, and those involving an infinitely fine operation, as the random drawing of a point from a segment.

The Unit Interval

Let Ω denote the unit interval $(0, 1]$. Let ω denote the generic point of Ω . Denote the length of an interval $I = (a, b]$ by

$$|I| = |(a, b]| = b - a.$$

If

$$A = \bigcup_{i=1}^n I_i = \bigcup_{i=1}^n (a_i, b_i],$$

where the intervals $I_i = (a_i, b_i]$ are disjoint and are contained in Ω , assign to A the probability

$$\mathbb{P}(A) = \sum_{i=1}^n |I_i| = \sum_{i=1}^n (b_i - a_i). \quad (1.1)$$

It is important to understand that in this section $\mathbb{P}(A)$ is defined only if A is a finite disjoint union of subintervals of $(0, 1]$.

If A and B are two such finite disjoint unions of intervals, and if A and B are disjoint, then $A \cup B$ is a finite disjoint union of intervals and

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

EXAMPLE 1.1. The definition in (1.1) is the starting point for a mathematical representation of an infinite sequence of tosses of a coin. With each ω associate its non-terminating dyadic expansion

$$\omega = \sum_{n=1}^{\infty} \frac{d_n(\omega)}{2^n} = .d_1(\omega)d_2(\omega)\dots,$$

each $d_n(\omega)$ being 0 or 1. Thus $(d_1(\omega), d_2(\omega), \dots)$ is the sequence of binary digits in the expansion of ω . For definiteness, a point such as $\frac{1}{2} = .1000\dots = .0111\dots$, which has two expansions, takes the non-terminating one; 1 takes the expansion $.111\dots$.

Imagine now a coin with faces labelled 1 and 0 instead of the usual heads and tails. If ω is drawn at random, then $(d_1(\omega), d_2(\omega), \dots)$ behaves as if it resulted from an infinite sequence of tosses of a coin. To see this, consider first the set of ω for which $d_i(\omega) = u_i$ for $i = 1, \dots, n$, where $u_1, \dots, u_n$ is a sequence of 0's and 1's. Such an ω satisfies

$$\sum_{i=1}^n \frac{u_i}{2^i} < \omega \leq \sum_{i=1}^n \frac{u_i}{2^i} + \sum_{i=n+1}^{\infty} \frac{1}{2^i},$$

where the extreme values of ω correspond to $d_i(\omega) = 0$ and $d_i(\omega) = 1$ for $i > n$. The second case can be achieved, but since the binary expansions represented by the $d_i(\omega)$ are non-terminating, the first cannot, and ω must actually exceed $\sum_{i=1}^n \frac{u_i}{2^i}$. Thus

$$\{\omega : d_i(\omega) = u_i, i = 1, \dots, n\} = \left(\sum_{i=1}^n \frac{u_i}{2^i}, \sum_{i=1}^n \frac{u_i}{2^i} + \frac{1}{2^n} \right]. \quad (1.2)$$

In the model for coin tossing, the set in (1.2) represents the event that the first n tosses give the outcomes $u_1, \dots, u_n$ in sequence. By (1.1) and (1.2),

$$\mathbb{P}(\omega : d_i(\omega) = u_i, i = 1, \dots, n) = \frac{1}{2^n},$$

which is what probabilistic intuition requires.

The probabilities of various familiar events can be written down immediately. The sum $\sum_{i=1}^n d_i(\omega)$ is the number of 1's among $d_1(\omega), \dots, d_n(\omega)$, to be thought of as the number of heads in n tosses of a fair coin. The usual binomial formula is

$$\mathbb{P}\left(\omega : \sum_{i=1}^n d_i(\omega) = k\right) = \binom{n}{k} \frac{1}{2^n}, \quad 0 \leq k \leq n.$$

The Weak Law of Large Numbers

THEOREM 1.2 (Weak Law of Large Numbers).

For each $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\omega : \left| \frac{1}{n} \sum_{i=1}^n d_i(\omega) - \frac{1}{2} \right| \geq \epsilon\right) = 0.$$

Proof.

□

The Strong Law of Large Numbers

Define a subset A of Ω to be *negligible*¹ if for each positive ϵ there exists a finite or countable² collection $I_1, I_2, \dots$ of intervals (they may overlap) satisfying

$$A \subset \bigcup_k I_k, \quad (1.3)$$

$$\sum_k |I_k| < \epsilon. \quad (1.4)$$

THEOREM 1.3 (Borel's Normal Number Theorem).

The set of normal numbers

$$N = \left\{ \omega : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n d_i(\omega) = \frac{1}{2} \right\} \quad (1.5)$$

has negligible complement.

Proof.

□

EXAMPLE 1.4. Consider the set N of normal numbers in the form (1.5). Then N can be put in the form

$$N = \bigcap_{k=1}^{\infty} \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} \left\{ \omega : \left| \frac{1}{n} \sum_{i=1}^n d_i(\omega) - \frac{1}{2} \right| < \frac{1}{k} \right\}, \quad k \in \mathbb{N}.$$

Exercises

EXERCISE 1.1. Define a set A to be *trifling*³ if for each ϵ there exists a finite sequence of intervals I_k satisfying (1.3) and (1.4). This definition and the definition of negligibility apply as they stand to all sets on the real line, not just to subsets of $(0, 1]$.

- (i) Show that a trifling set is negligible.
- (ii) Show that the closure of a trifling set is also trifling.
- (iii) Find a bounded negligible set that is not trifling.
- (iv) Show that the closure of a negligible set may not be negligible.
- (v) Show that finite unions of trifling sets are trifling but that this can fail for countable unions.

SOLUTION. (ii) Suppose A is trifling and let $\bar{A}$ be its closure. Given ϵ choose intervals $(a_k, b_k]$, $k = 1, \dots, n$, such that

$$A \subset \bigcup_{k=1}^n (a_k, b_k], \quad \sum_{k=1}^n (b_k - a_k) < \frac{\epsilon}{2}.$$

If $x_k = a_k - \frac{\epsilon}{2n}$, then $\bar{A} \subset \bigcup_{k=1}^n (x_k, b_k]$ and $\sum_{k=1}^n (b_k - x_k) < \epsilon$.

¹ The term *negligible* is introduced for the purposes of this section only. The negligible sets will reappear later as the sets of Lebesgue measure 0.

² *Countably infinite* is unambiguous. *Countable* will mean finite or countably infinite, although it will sometimes for emphasis be expanded as here to *finite* or *countable*.

³ Like negligible, *trifling* is a nonce word used only here. The trifling sets are exactly the sets of content 0.

- (iii) The Cantor set C has Lebesgue measure 0, which means it is negligible. It is bounded since it is a subset of $(0, 1]$. But it cannot be covered by a finite collection of intervals, because we have 2^n disjoint intervals with each length 3^{-n} .
- (iv) Consider the set of rationals in $(0, 1)$, denoted $A = \mathbb{Q} \cap (0, 1)$. Since the rational numbers in $(0, 1)$ are countable, for any $\epsilon > 0$, we can cover each point q_i with an interval $(q_i - \delta_i, q_i + \delta_i)$, where $\delta_i > 0$ is chosen such that

$$\sum_{i=1}^{\infty} 2\delta_i < \epsilon.$$

This satisfies the conditions for A to be negligible, as $A \subset \bigcup_{i=1}^{\infty} (q_i - \delta_i, q_i + \delta_i)$. The rationals are dense in $[0, 1]$; therefore, the closure $\bar{A} = [0, 1]$ is not negligible.

- (v) Let $A_1, \dots, A_m$ be a finite collection of trifling sets. By definition, for any $\epsilon > 0$, there exists a sequence of intervals $\{I_k^i\}$ for each A_i such that

$$A_i \subset \bigcup_k I_k^i, \quad \sum_k |I_k^i| < \frac{\epsilon}{m}.$$

Consider the union $A = \bigcup_{i=1}^m A_i$. The collection of intervals $\bigcup_{i=1}^m \{I_k^i\}$ covers A , and the total length of these intervals is less than ϵ .

For countable unions, we consider $\mathbb{Q} \cap (0, 1)$. Each singleton $\{q\}$, where $q \in \mathbb{Q}$ is trifling, since it can be covered by intervals of arbitrarily small length. But the union cannot be covered by a finite collection of intervals.

□

EXERCISE 1.2. For $i = 0, \dots, r-1$, let $A_r(i)$ be the set of numbers in $(0, 1]$ whose non-terminating expansions in the base r do not contain the digit i . Show that $A_r(i)$ is trifling.

SOLUTION. Cover $A_r(i)$ by $(r-1)^n$ intervals of length r^{-n} .

□

EXERCISE 1.3. The Cantor set C can be defined as the closure of $A_3(1)$.

- (i) Show that C is uncountable but trifling.
- (ii) From $[0, 1]$, remove the open middle third $(\frac{1}{3}, \frac{2}{3})$; from the remainder, a union of two closed intervals, remove the two open middle thirds $(\frac{1}{9}, \frac{2}{9})$ and $(\frac{7}{9}, \frac{8}{9})$. Show that C is what remains when this process is continued ad infinitum.
- (iii) Show that C is perfect⁴.

SOLUTION. (i) The set $A_3(1)$ is itself uncountable, since a point in it is specified by a sequence of 0's and 2's (excluding the countably many that end in 0's).

- (ii) For sequences $u_1, \dots, u_n$ of 0's, 1's, and 2's, let $M_{u_1, \dots, u_n}$ consist of the points in $(0, 1]$ whose non-terminating base-3 expansions start out with those digits. Then

$$A_3(1) = (0, 1] - \bigcup M_{u_1, \dots, u_n},$$

where the union extends over sequences $u_1, \dots, u_n$ containing at least one 1. The set described in part (ii) is $(0, 1] - \bigcup M_{u_1, \dots, u_n}^0$, where the union is as before, and this is the closure of $A_3(1)$.

⁴ A set A is defined to be *perfect* if it is closed and for each x in A and positive ϵ , there is a y in A such that $0 < |x - y| < \epsilon$. An equivalent requirement is that A be closed and for each x in A there exists a sequence $\{x_n\}$ in A such that $x_n \neq x$ and $x_n \rightarrow x$. The Cantor set is uncountable, nowhere dense, and perfect.

From this representation of C , it is not hard to deduce that it can be defined as the set of points in $[0, 1]$ that can be written in base 3 without any 1's if terminating expansions are also allowed. For example, C contains

$$\frac{2}{3} = 0.1222\ldots = 0.2000\ldots,$$

because it is possible to avoid 1 in the expansion.

- (iii) Given ϵ and an ω in C , choose ω' in $A_3(1)$ within $\frac{\epsilon}{2}$ of ω ; now define ω'' by changing from 2 to 0 some digit of ω' far enough out that ω'' differs from ω' by at most $\frac{\epsilon}{2}$.

□

SECTION 1.2

| Probability Measures

Classes of Sets

DEFINITION 1.5 (Fields).

Consider now a completely arbitrary non-empty space Ω . A class $\mathcal{F}$ of subsets of Ω is called a *field* (or *algebra*) if it contains Ω itself and is closed under the formation of complements and finite unions:

- (i) $\Omega \in \mathcal{F}$.
- (ii) $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$.
- (iii) $A, B \in \mathcal{F}$ implies $A \cup B \in \mathcal{F}$.

Remark. Since Ω and the empty set $\emptyset$ are complementary, (i) is the same in the presence of (ii) as the assumption $\emptyset \in \mathcal{F}$. In fact, (i) simply ensures that $\mathcal{F}$ is non-empty: If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ by (ii) and $\Omega = A \cup A^c \in \mathcal{F}$ by (iii).

By DeMorgan's law,

$$A \cap B = (A^c \cup B^c)^c, \quad A \cup B = (A^c \cap B^c)^c.$$

If $\mathcal{F}$ is closed under complementation, therefore, it is closed under the formation of finite unions if and only if it is closed under the formation of finite intersections. Thus (iii) can be replaced by the requirement

$$A, B \in \mathcal{F} \implies A \cap B \in \mathcal{F}.$$

DEFINITION 1.6 (σ -Fields).

A class $\mathcal{F}$ of subsets of Ω is a σ -field if it is a field and if it is also closed under the formation of countable unions:

- (iv) $A_1, A_2, \dots \in \mathcal{F}$ implies $A_1 \cup A_2 \cup \dots \in \mathcal{F}$.

Remark. Note that (iv) implies (iii) because one can take $A_1 = A$ and $A_n = B$ for $n \geq 2$. By the infinite form of DeMorgan's law, assuming (iv) is the same thing as assuming

$$A_1, A_2, \dots \in \mathcal{F} \implies A_1 \cap A_2 \cap \dots \in \mathcal{F}.$$

For a class $\mathcal{A}$, we write $\sigma(\mathcal{A})$ for the smallest σ field that contains $\mathcal{A}$.

EXAMPLE 1.7. (i) Consider the finite disjoint unions of subintervals of $\Omega = (0, 1]$. A field $\mathcal{B}_0$: finite disjoint unions where each intersection here is an interval or else the empty set, is a field but not a σ -field.

- (ii) Let $S^n = S \times \dots \times S$ be the Cartesian product of n copies of S ; it consists of the n -long sequences $(u_1, \dots, u_n)$ of elements of S . For such a sequence, the set

$$\{\omega : (z_1(\omega), \dots, z_n(\omega)) = (u_1, \dots, u_n)\}$$

represents the event that the first n repetitions of the experiment give the outcomes $u_1, \dots, u_n$ in sequence. A *cylinder of rank n* is a set of the form

$$A = \{\omega : (z_1(\omega), \dots, z_n(\omega)) \in H\},$$

where $H \subset S^n$. Let C_0 be the class of cylinders of all ranks. Then C_0 is a field.

Probability Measures

DEFINITION 1.8.

A *set function* is a real-valued function defined on some class of subsets of Ω . A set function $\mathbb{P}$ on a field $\mathcal{F}$ is a *probability measure* if it satisfies these conditions:

- (i) $0 \leq \mathbb{P}(A) \leq 1$ for $A \in \mathcal{F}$.
- (ii) $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$.
- (iii) If $A_1, A_2, \dots$ is a disjoint sequence of $\mathcal{F}$ -sets and if $\bigcup_{k=1}^{\infty} A_k \in \mathcal{F}$, then

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(A_k).$$

PROPOSITION 1.9. (i) *Monotone:* $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subset B$.

- (ii) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- (iii) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- (iv) *Countable subadditivity:* If $A_1, A_2, \dots$ and $\bigcup_{k=1}^{\infty} A_k$ lie in $\mathcal{F}$ (the A_k need not be disjoint), then

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) \leq \sum_{k=1}^{\infty} \mathbb{P}(A_k).$$

- (v) *Continuity:* If A_n and A lie in $\mathcal{F}$ and $A_n \uparrow A$, then $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$; if A_n and A lie in $\mathcal{F}$ and $A_n \downarrow A$, then $\mathbb{P}(A_n) \downarrow \mathbb{P}(A)$.

Proof. (ii) As a special case of $\mathbb{P}(B - A) = \mathbb{P}(B) - \mathbb{P}(A)$.

- (iii) $\mathbb{P}(A) + \mathbb{P}(B) = \mathbb{P}(A \cup B) + \mathbb{P}(A \cap B)$. The common value of the two sides being $\mathbb{P}(A \cup B^c) + 2\mathbb{P}(A \cap B) + \mathbb{P}(A^c \cap B)$.

- (iv) If $B_1 = A_1$ and $B_k = A_k \cap A_1^c \cap \dots \cap A_{k-1}^c$, then the B_k are disjoint and $\bigcup_{k=1}^n A_k = \bigcup_{k=1}^n B_k$, so that

$$\mathbb{P}\left(\bigcup_{k=1}^n A_k\right) = \sum_{k=1}^n \mathbb{P}(B_k),$$

where $\mathbb{P}(B_k) \leq \mathbb{P}(A_k)$ by monotonicity.

- (v) Put $B_1 = A_1$ and $B_k = A_k - A_{k-1}$. Then the B_k are disjoint,

$$A = \bigcup_{k=1}^{\infty} B_k, \quad A_n = \bigcup_{k=1}^n B_k,$$

so that by countable and finite additivity,

$$\mathbb{P}(A) = \sum_{k=1}^{\infty} \mathbb{P}(B_k) = \lim_n \sum_{k=1}^n \mathbb{P}(B_k) = \lim_n \mathbb{P}(A_n).$$

Observe that $A_n \downarrow A$ implies $A_n^c \uparrow A^c$, so that $1 - \mathbb{P}(A_n) \uparrow 1 - \mathbb{P}(A)$.

□

EXAMPLE 1.10. If $\mathbb{P}$ is a finitely additive probability measure on the field $\mathcal{F}$, and if $A_n \downarrow \emptyset$ for sets A_n in $\mathcal{F}$ implies $\mathbb{P}(A_n) \downarrow 0$, then $\mathbb{P}$ is countably additive.

Lebesgue Measure on the Unit Interval

DEFINITION 1.11 (Lebesgue Measures).

If $A = \bigcup_{i=1}^n I_i$, the I_i being disjoint I -sets,

$$\lambda(A) = \sum_{i=1}^n \lambda(I_i) = \sum_{i=1}^n |I_i|$$

defines a set function λ on $\mathcal{B}_0$ called *Lebesgue measure*.

| Exercises

EXERCISE 1.4. (i) Suppose that $\Omega \in \mathcal{F}$ and that $A, B \in \mathcal{F}$ implies $A - B = A \cap B^c \in \mathcal{F}$. Show that $\mathcal{F}$ is a field.

(ii) Suppose that $\Omega \in \mathcal{F}$ and that $\mathcal{F}$ is closed under the formation of complements and finite *disjoint* unions. Show that $\mathcal{F}$ need not be a field.

SOLUTION. (ii) Let Ω consist of four points, and let $\mathcal{F}$ consist of the empty set, Ω itself, and all six of the two-point sets.

□

EXERCISE 1.5. Let $\mathcal{F}_1, \mathcal{F}_2, \dots$ be classes of sets in a common space Ω .

(i) Suppose that $\mathcal{F}_n$ are fields satisfying $\mathcal{F}_n \subset \mathcal{F}_{n+1}$. Show that $\bigcup_{n=1}^{\infty} \mathcal{F}_n$ is a field.

(ii) Suppose that $\mathcal{F}_n$ are σ -fields satisfying $\mathcal{F}_n \subset \mathcal{F}_{n+1}$. Show by example that $\bigcup_{n=1}^{\infty} \mathcal{F}_n$ need not be a σ -field.

SOLUTION. (ii) For example, take Ω to consist of the integers, and let $\mathcal{F}_n$ be the σ -field generated by the singletons $\{k\}$ with $k \leq n$. As a matter of fact, any example in which $\mathcal{F}_n$ is a proper subclass of $\mathcal{F}_{n+1}$ for all n will do, because it can be shown that in this case $\bigcup_n \mathcal{F}_n$ necessarily fails to be a σ -field.

□

EXERCISE 1.6. (i) Let $\mathcal{F}$ be the field consisting of the finite and the cofinite sets in an infinite Ω , and define $\mathbb{P}$ on $\mathcal{F}$ by taking $\mathbb{P}(A)$ to be 0 or 1 as A is finite or cofinite. (Note that $\mathbb{P}$ is not well defined if Ω is finite.) Show that $\mathbb{P}$ is finitely additive.

(ii) Show that this $\mathbb{P}$ is not countably additive if Ω is countably infinite.

(iii) Show that this $\mathbb{P}$ is countably additive if Ω is uncountable.

EXERCISE 1.7. Suppose that $\mathbb{P}$ is a probability measure on a field $\mathcal{F}$, that $A_1, A_2, \dots$, and $A = \bigcup_n A_n$ lie in $\mathcal{F}$, and that the A_n are nearly disjoint in the sense that $\mathbb{P}(A_m \cap A_n) = 0$ for $m \neq n$. Show that $\mathbb{P}(A) = \sum_n \mathbb{P}(A_n)$.

SECTION 1.3

| Existence and Extension

THEOREM 1.12.

A probability measure on a field has a unique extension to the generated σ -field.

Construction of the Extension

DEFINITION 1.13 (Outer Measures).

For each subset A of Ω , define its *outer measure* by

$$\mathbb{P}^*(A) = \inf \sum_n \mathbb{P}(A_n),$$

where the infimum extends over all finite and infinite sequences $A_1, A_2, \dots$ of $\mathcal{F}_0$ -sets satisfying $A \subset \bigcup_n A_n$.

Uniqueness and the π - λ TheoremDEFINITION 1.14 (π -Systems).

A class $\mathcal{P}$ of subsets of Ω is a π -system if it is closed under the formation of *finite intersections*:

$$A, B \in \mathcal{P} \implies A \cap B \in \mathcal{P}.$$

DEFINITION 1.15 (λ -Systems).

A class $\mathcal{L}$ is a λ -system if it contains Ω and is closed under the formation of *complements* and of finite and countable *disjoint* unions:

- (i) $\Omega \in \mathcal{L}$.
- (ii) $A \in \mathcal{L} \implies A^c \in \mathcal{L}$.
- (iii) $A_1, A_2, \dots \in \mathcal{L}$ and $A_n \cap A_m = \emptyset$ for $m \neq n$ imply $\bigcup_n A_n \in \mathcal{L}$.

Remark. Because of the disjointness condition in (iii), the definition of λ -system is weaker (more inclusive) than that of σ -field. In the presence of (i) and (ii), which imply $\emptyset \in \mathcal{L}$, the countably infinite case of (iii) implies the finite one.

PROPOSITION 1.16.

In the presence of (i) and (iii), (ii) is equivalent to the condition that $\mathcal{L}$ is closed under the formation of proper differences:

$$(ii') \quad A, B \in \mathcal{L}, A \subset B \implies B - A \in \mathcal{L}.$$

Proof. Suppose, in fact, that $\mathcal{L}$ satisfies (ii) and (iii). If $A, B \in \mathcal{L}$ and $A \subset B$, then $\mathcal{L}$ contains B^c , the disjoint union $A \cup B^c$, and its complement $(A \cup B^c)^c = B - A$. Hence (ii'). On the other hand, if $\mathcal{L}$ satisfies (i) and (ii'), then $A \in \mathcal{L}$ implies $A^c = \Omega - A \in \mathcal{L}$. Hence (ii). $\square$

Although a σ -field is a λ -system, the reverse is not true⁵. But the connection is close.

⁵ In a four-point space take $\mathcal{L}$ to consist of $\emptyset, \Omega$, and the six two-point sets.

LEMMA 1.17.

A class that is both a π -system and a λ -system is a σ -field.

THEOREM 1.18 (Dynkin's π - λ Theorem).

If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a λ -system, then $\mathcal{P} \subset \mathcal{L}$ implies $\sigma(\mathcal{P}) \subset \mathcal{L}$.

THEOREM 1.19.

Suppose that $\mathbb{P}_1$ and $\mathbb{P}_2$ are probability measures on $\sigma(\mathcal{P})$, where $\mathcal{P}$ is a π -system. If $\mathbb{P}_1$ and $\mathbb{P}_2$ agree on $\mathcal{P}$, then they agree on $\sigma(\mathcal{P})$.

| Exercises

EXERCISE 1.8. Show by example that a finitely additive probability measure on a field may not be countably subadditive. Show in fact that if a finitely additive probability measure is countably subadditive, then it is necessarily countably additive as well.

SOLUTION. For a specific example consider $\bigcup_n (2^{-1} + n^{-1}, 1]$. But an example is provided by every $\mathbb{P}$ that is finitely but not countably additive. If $\mathbb{P}$ is finitely additive on a field $\mathcal{F}_0$ and A_n are disjoint $\mathcal{F}_0$ -sets whose union A also lies in $\mathcal{F}_0$, then monotonicity (which requires finite additivity only) gives

$$\sum_{k \leq n} \mathbb{P}(A_k) = \mathbb{P}\left(\bigcup_{k \leq n} A_k\right) \leq \mathbb{P}(A),$$

and hence $\sum_k \mathbb{P}(A_k) \leq \mathbb{P}(A)$. Countable subadditivity will ensure that there is equality here. $\square$

EXERCISE 1.9. Deduce directly by countable additivity that the Cantor set has Lebesgue measure 0.

SECTION 1.4

| Denumerable Probabilities

General Formulas

DEFINITION 1.20 (Conditional Probability).

If $\mathbb{P}(A) > 0$, the *conditional probability* of B given A is defined in the usual way as

$$\mathbb{P}(B | A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}.$$

PROPOSITION 1.21.

There are the chain-rule formulas

$$\begin{aligned}\mathbb{P}(A \cap B) &= \mathbb{P}(A)\mathbb{P}(B | A), \\ \mathbb{P}(A \cap B \cap C) &= \mathbb{P}(A)\mathbb{P}(B | A)\mathbb{P}(C | A \cap B).\end{aligned}$$

If $A_1, A_2, \dots$ partition Ω , then

$$\mathbb{P}(B) = \sum_n \mathbb{P}(A_n)\mathbb{P}(B | A_n).$$

Limit Sets

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For a sequence of sets $A_1, A_2, \dots \in \mathcal{F}$, we define the *limits superior* and *inferior*:

$$\limsup_{n \rightarrow \infty} A_n := \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \in \mathcal{F}, \quad (1.6)$$

$$\liminf_{n \rightarrow \infty} A_n := \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \in \mathcal{F}, \quad (1.7)$$

where ω lies in (1.6) if and only if for each n there is some $k \geq n$ for which $\omega \in A_k$, i.e., *infinitely many* of the A_n ; ω lies in (1.7) if and only if there is some n such that $\omega \in A_k$ for all $k \geq n$, i.e., *all but finitely many* of the A_n .

Note that

$$\bigcap_{k=n}^{\infty} A_k \uparrow \liminf_n A_n, \quad \bigcup_{k=n}^{\infty} A_k \downarrow \limsup_n A_n.$$

For every m and n , $\bigcap_{k=m}^{\infty} A_k \subset \bigcup_{k=n}^{\infty} A_k$, because for $i \geq \max\{m, n\}$, A_i contains the first of these sets and is contained in the second. If (1.6) and (1.7) are equal, write

$$\lim_n A_n = \liminf_n A_n = \limsup_n A_n.$$

THEOREM 1.22. (i) For each sequence $\{A_n\}$,

$$\mathbb{P}\left(\liminf_n A_n\right) \leq \liminf_n \mathbb{P}(A_n) \leq \limsup_n \mathbb{P}(A_n) \leq \mathbb{P}\left(\limsup_n A_n\right).$$

(ii) If $A_n \rightarrow A$, then $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$.

Proof.

□

Independent Events

DEFINITION 1.23.

Events A and B are *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. More generally, a finite collection $A_1, \dots, A_n$ of events is *independent* if

$$\mathbb{P}(A_{k_1} \cap \dots \cap A_{k_j}) = \mathbb{P}(A_{k_1}) \dots \mathbb{P}(A_{k_j})$$

for $2 \leq j \leq n$ and $1 \leq k_1 < \dots < k_j \leq n$. An infinite (perhaps uncountable) collection of events is defined to be *independent* if each of its finite sub-collections is.

DEFINITION 1.24.

Define classes $\mathcal{A}_1, \dots, \mathcal{A}_n$ in the basic σ -field $\mathcal{F}$ to be *independent* if for each choice of A_i from $\mathcal{A}_i$, $i = 1, \dots, n$, the events $A_1, \dots, A_n$ are independent.

THEOREM 1.25.

If $\mathcal{A}_1, \dots, \mathcal{A}_n$ are independent and each $\mathcal{A}_i$ is a π -system, then $\sigma(\mathcal{A}_1), \dots, \sigma(\mathcal{A}_n)$ are independent.

The Borel-Cantelli Lemmas

THEOREM 1.26 (Borel-Cantelli Lemmas). (i) If $\sum_n \mathbb{P}(A_n)$ converges, then $\mathbb{P}(\limsup_n A_n) = 0$.

(ii) If $\{A_n\}$ is an independent sequence of events and $\sum_n \mathbb{P}(A_n)$ diverges, then $\mathbb{P}(\limsup_n A_n) = 1$.

Remark. If the probabilities of (A_n) sum up to a finite number, almost surely only finitely many A_n will occur. If the sum diverges, almost surely infinitely many A_n will occur.

Proof. (i)

(ii)

□

EXAMPLE 1.27. A coin is tossed independently again and again such that the probability of a head at the n -th toss is $\mathbb{P}(A_n)$.

(i) If $\mathbb{P}(A_n) = \frac{1}{n}$, $n \geq 1$, infinitely many heads will occur by the *second Borel-Cantelli Lemma*.

(ii) If $\mathbb{P}(A_n) = \frac{1}{n^2}$, $n \geq 1$, only finitely many heads will occur by the *first Borel-Cantelli Lemma*.

| Exercises

EXERCISE 1.10.

EXERCISE 1.11.

EXERCISE 1.12.

EXERCISE 1.13.

EXERCISE 1.14.

SECTION 1.5

| Simple Random Variables

Convergence of Random Variables

DEFINITION 1.28.

For a sequence of random variables $X_1, X_2, \dots$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

(i)

(ii)

*Remark.*THEOREM 1.29. (i) *There is convergence $\lim_n X_n = X$ with probability 1 if and only if (5.6) holds for each ϵ .*(ii) *Convergence with probability 1 implies convergence in probability.**Proof.*

□

EXAMPLE 1.30.

Measure

SECTION 2.1

| General Measures

Measures

DEFINITION 2.1 (Measures).

A set function μ on a field $\mathcal{F}$ in Ω is a *measure* if it satisfies these conditions:

- (i) $\mu(A) \in [0, \infty]$ for $A \in \mathcal{F}$.
- (ii) $\mu(\emptyset) = 0$.
- (iii) If $A_1, A_2, \dots$ is a disjoint sequence of $\mathcal{F}$ -sets and if $\bigcup_{k=1}^{\infty} A_k \in \mathcal{F}$, then

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k).$$

Remark. The measure μ is finite or infinite as $\mu(\Omega) < \infty$ or $\mu(\Omega) = \infty$; it is a probability measure if $\mu(\Omega) = 1$.

THEOREM 2.2.

Let μ be a measure on a field $\mathcal{F}$.

- (i) *Continuity from below:* If A_n and A lie in $\mathcal{F}$ and $A_n \uparrow A$, then $\mu(A_n) \uparrow \mu(A)$.
- (ii) *Continuity from above:* If A_n and A lie in $\mathcal{F}$ and $A_n \downarrow A$, and if $\mu(A_1) < \infty$, then $\mu(A_n) \downarrow \mu(A)$.

Proof. (ii) If $\mu(A_1) < \infty$, subtraction is possible and $A_1 - A_n \uparrow A_1 - A$ implies that

$$\mu(A_1) - \mu(A_n) = \mu(A_1 - A_n) \uparrow \mu(A_1 - A) = \mu(A_1) - \mu(A).$$

□

| Exercises

EXERCISE 2.1. Show that if conditions (i) and (iii) in *Definition 2.1* hold, and if $\mu(A) < \infty$ for some $A \in \mathcal{F}$, then condition (ii) holds.

EXERCISE 2.2. On the σ -field of all subsets of $\Omega = \{1, 2, \dots\}$ put $\mu(A) = \sum_{k \in A} 2^{-k}$ if A is finite and $\mu(A) = \infty$ otherwise. Is μ finitely additive? Countably additive?

EXERCISE 2.3.

EXERCISE 2.4. The natural generalisation of (4.9) is

$$\mu\left(\liminf_n A_n\right) \leq \liminf_n \mu(A_n) \leq \limsup_n \mu(A_n) \leq \mu\left(\limsup_n A_n\right).$$

Show that the left-hand inequality always holds. Show that the right-hand inequality holds if $\mu(\bigcup_{k \geq n} A_k) < \infty$ for some n but can fail otherwise.

EXERCISE 2.5. *Example 10.5* shows that *Theorem 10.3* fails without the σ -finiteness condition. Construct other examples of this kind.

SOLUTION.

□

SECTION 2.2

Measures in Euclidean Space

Lebesgue Measure

The k -dimensional Lebesgue measure λ_k on the class $\mathbb{R}^k$ of k -dimensional Borel sets is specified by the requirement that bounded rectangles have measure

$$\lambda_k\{x : a_i < x_i \leq b_i, i = 1, \dots, k\} = \prod_{i=1}^k (b_i - a_i).$$

A basic property of λ_k is *translation invariance*.

THEOREM 2.3.

If $A \in \mathbb{R}^k$, then $A + x = \{a + x : a \in A\} \in \mathbb{R}^k$ and $\lambda_k(A) = \lambda_k(A + x)$ for all x .

THEOREM 2.4.

If $T : \mathbb{R}^k \rightarrow \mathbb{R}^k$ is linear and nonsingular, then $A \in \mathbb{R}^k$ implies that $TA \in \mathbb{R}^k$ and

$$\lambda_k(TA) = |\det T| \cdot \lambda_k(A).$$

Specifying Measures on the Line

Define a real function F by

$$F(x) = \begin{cases} \mu(0, x], & x \geq 0 \\ -\mu(x, 0], & x \leq 0 \end{cases}.$$

It is because $\mu(A) < \infty$ for bounded A that F is a finite function. For finite μ , it is customary to standardize F by defining it by

$$F(x) = \mu(-\infty, x].$$

Clearly, F is non-decreasing. Suppose that $x_n \downarrow x$. If $x \geq 0$, apply part (ii) of [Theorem 2.2](#),

$$F(x_n) = \mu(0, x_n) \downarrow \mu(0, x] = F(x).$$

If $x < 0$, apply part (i),

$$F(x_n) = -\mu(x_n, 0] \uparrow -\mu(x, 0] = F(x).$$

Thus F is continuous from the right. Finally,

$$\mu(a, b] = F(b) - F(a). \tag{2.1}$$

THEOREM 2.5.

If F is a non-decreasing, right-continuous real function on the line, there exists on $\mathbb{R}$ a unique measure μ satisfying (2.1) for all a and b .

| Exercises

EXERCISE 2.6. Suppose that μ is a measure on $\mathbb{R}$ that is finite for bounded sets and is translation-invariant: $\mu(A + x) = \mu(A)$. Show that $\mu(A) = a\lambda(A)$ for some $a \geq 0$. Extend to $\mathbb{R}^k$.

EXERCISE 2.7. Consider the set $\mathbb{N} = \{1, 2, 3, \dots\}$. Let, for each $k = 1, 2, 3, \dots$, the set A_k be defined as

$$A_k := \{k, 2k, 3k, \dots\}.$$

Finally we define

$$\mathcal{H} := \emptyset \cup \{A_k; k = 1, 2, 3, \dots\}.$$

- (i) Show that $\mathcal{H}$ is closed under finite intersections.
- (ii) Prove that $\sigma(\mathcal{H})$ is the collection $\mathcal{P}(\mathbb{N})$ of all subsets of $\mathbb{N}$.
- (iii) Show that two finite measures on $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$ that agree on $\mathcal{H}$ are the same.

EXERCISE 2.8. Consider the Borel sigma-algebra $\mathcal{B}$ on $\mathbb{R}$. Consider the collection

$$\mathcal{H} = \{B \in \mathcal{B} : -B \in \mathcal{B}\},$$

so $\mathcal{H}$ is the collection of sets in $\mathcal{B}$ for which the reflection in the origin is also in $\mathcal{B}$.

- (i) Show that $\mathcal{H}$ is a sigma-algebra.
- (ii) Use (i) to show that $\mathcal{H} = \mathcal{B}$.
- (iii) Now let λ be Lebesgue measure on $\mathcal{B}$ and define a second measure ν on the same sigma-algebra by

$$\nu(B) := \lambda(-B).$$

Show that $\lambda = \nu$.

SECTION 2.3

| Measurable Functions and Mappings

Measurable Mappings

Let $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$ be two measurable spaces. For a mapping $T : \Omega \rightarrow \Omega'$, consider the inverse images $T^{-1}A' = \{\omega \in \Omega : T\omega \in A'\}$ for $A' \subset \Omega'$. The mapping T is $\mathcal{F}/\mathcal{F}'$ -measurable if $T^{-1}A' \in \mathcal{F}$ for each $A' \in \mathcal{F}'$.

THEOREM 2.6. (i) If $T^{-1}A' \in \mathcal{F}$ for each $A' \in \mathcal{A}'$ and $\mathcal{A}'$ generates $\mathcal{F}'$, then T is $\mathcal{F}/\mathcal{F}'$ -measurable.

(ii) If T is $\mathcal{F}/\mathcal{F}'$ -measurable and T' is $\mathcal{F}'/\mathcal{F}''$ -measurable, then $T'T$ is $\mathcal{F}/\mathcal{F}''$ -measurable.

Proof. (i) Since

$$T^{-1}(\Omega' - A') = \Omega - T^{-1}A', \quad T^{-1}\left(\bigcup_n A'_n\right) = \bigcup_n T^{-1}A'_n.$$

And since $\mathcal{F}$ is a σ -field in Ω , the class $\{A' : T^{-1}A' \in \mathcal{F}\}$ is a σ -field in Ω' . If this σ -field contains $\mathcal{A}'$, it must also contain $\sigma(\mathcal{A}')$, and (i) follows. □

Mappings into $\mathbb{R}^k$

THEOREM 2.7.

If $f : \mathbb{R}^i \rightarrow \mathbb{R}^k$ is continuous, then it is measurable.

Limits and Measurability

THEOREM 2.8.

Suppose that $f_1, f_2, \dots$ are $\mathcal{F}$ -measurable real functions. The functions $\sup_n f_n, \inf_n f_n, \limsup_n f_n$, and $\liminf_n f_n$ are $\mathcal{F}$ -measurable.

Proof. Clearly,

$$\left\{ \sup_n f_n \leq x \right\} = \bigcap_n \{f_n \leq x\}$$

lies in $\mathcal{F}$ even for $x = \infty$ and $x = -\infty$, and so $\sup_n f_n$ is measurable. The measurability of $\inf_n f_n$ follows the same way, and hence

$$\limsup_n f_n = \inf_n \sup_{k \geq n} f_k, \quad \liminf_n f_n = \sup_n \inf_{k \geq n} f_k$$

are measurable. □

DEFINITION 2.9 (Simple Functions).

A *simple* real function is one with finite range, which can be put in the form

$$f(\omega) = \sum x_i \mathbb{1}_{A_i}(\omega),$$

where the A_i form a finite decomposition of Ω .

THEOREM 2.10.

If f is real and $\mathcal{F}$ -measurable, there exists a sequence $\{f_n\}$ of simple functions, each $\mathcal{F}$ -measurable such that

$$\begin{aligned} 0 &\leq f_n(\omega) \uparrow f(\omega), & f(\omega) &\geq 0, \\ 0 &\geq f_n(\omega) \downarrow f(\omega), & f(\omega) &\leq 0. \end{aligned}$$

Transformations of Measures

Let $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$ be measurable spaces, and suppose that the mapping $T : \Omega \rightarrow \Omega'$ is $\mathcal{F}/\mathcal{F}'$ -measurable. Given a measure μ on $\mathcal{F}$, define a set function μT^{-1} on $\mathcal{F}'$ by

$$\mu T^{-1}(A') = \mu(T^{-1}A'), \quad A' \in \mathcal{F}'.$$

Thus μT^{-1} is a measure.

Exercises

EXERCISE 2.9. Let $\mathcal{F}$ be a σ -field in $\mathbb{R}$. Show that $\mathcal{B}^1 \subseteq \mathcal{F}$ if and only if every continuous function is measurable $\mathcal{F}$. Thus $\mathcal{B}^1$ is the smallest σ -field with respect to which all the continuous functions are measurable.

EXERCISE 2.10. Consider Lebesgue measure λ restricted to the class $\mathcal{B}$ of Borel sets in $(0, 1]$. For a fixed permutation $n_1, n_2, \dots$ of the positive integers, if x has dyadic expansion $x = .x_1x_2\dots$, take $Tx = .x_{n_1}x_{n_2}\dots$. Show that T is measurable $\mathcal{B}/\mathcal{B}$ and that $\lambda T^{-1} = \lambda$.

SECTION 2.4

| Distribution Functions

Distribution Functions

A random variable is a measurable real function X on a probability measure space $(\Omega, \mathcal{F}, \mathbb{P})$.

DEFINITION 2.11 (Distribution Functions).

The *distribution function* of X is defined by

$$F(x) = \mu(-\infty, x] = \mathbb{P}(X \leq x)$$

for real x .

THEOREM 2.12.

If F is a non-decreasing, right-continuous function satisfying

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow \infty} F(x) = 1,$$

then there exists on some probability space a random variable X for which $F(x) = \mathbb{P}(X \leq x)$.

| Exercises

EXERCISE 2.11. The general non-decreasing function F has at most countably many discontinuities. Prove this by considering the open intervals

$$\left(\sup_{u < x} F(u), \inf_{v > x} F(v) \right),$$

each non-empty one contains a rational.

EXERCISE 2.12. (i) Suppose that X has a continuous, strictly increasing distribution function F . Show that the random variable $F(X)$ is uniformly distributed over the unit interval in the sense that $P[F(X) \leq u] = u$ for $0 \leq u \leq 1$. Passing from X to $F(X)$ is called the *probability transformation*.

(ii) Show that the function $\varphi(u)$ defined by (14.5) satisfies $F(\varphi(u)-) \leq u \leq F(\varphi(u))$ and that, if F is continuous (but not necessarily strictly increasing), then $F(\varphi(u)) = u$ for $0 < u < 1$.

(iii) Show that $P[F(X) < u] = F(\varphi(u)-)$ and hence that the result in part (i) holds as long as F is continuous.

SOLUTION.

□

EXERCISE 2.13. The *Lévy distance* $d(F, G)$ between two distribution functions is the infimum of those ϵ such that

$$G(x - \epsilon) - \epsilon \leq F(x) \leq G(x + \epsilon) + \epsilon$$

for all x . Verify that this is a metric on the set of distribution functions. Show that a necessary and sufficient condition for $F_n \Rightarrow F$ is that $d(F_n, F) \rightarrow 0$.

SOLUTION.

□

Integration

SECTION 3.1

| The Integral

The conventions about infinity are

$$\begin{aligned} 0 \cdot \infty &= \infty \cdot 0 = 0, \\ x \cdot \infty &= \infty \cdot x = \infty, \quad 0 < x < \infty, \\ \infty \cdot \infty &= \infty. \end{aligned}$$

On a measurable space $(\Omega, \mathcal{F}, \mu)$, for general $f : \Omega \rightarrow \mathbb{R} \cup \{-\infty, \infty\}$, consider its *positive part*,

$$f^+(\omega) = \begin{cases} f(\omega), & 0 \leq f(\omega) \leq \infty \\ 0, & -\infty \leq f(\omega) \leq 0 \end{cases}$$

and its *negative part*,

$$f^-(\omega) = \begin{cases} -f(\omega), & -\infty \leq f(\omega) \leq 0 \\ 0, & 0 \leq f(\omega) \leq \infty \end{cases}.$$

These functions are non-negative and measurable, and $f = f^+ - f^-$.

DEFINITION 3.1. (i) The *integral* of $f \geq 0$ is defined as the supremum of the sums:

$$\int f \, d\mu = \sup_{\mathcal{P}} \sum_i \left(\inf_{\omega \in A_i} f(\omega) \right) \mu(A_i), \quad (3.1)$$

where $\mathcal{P}$ denotes all finite decompositions $\{A_i\}$ of Ω into $\mathcal{F}$ -sets.

(ii) The general integral is defined by

$$\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu. \quad (3.2)$$

Remark. If $\int f^+ \, d\mu$ and $\int f^- \, d\mu$ are both finite, then f is integrable, and has (3.2) as its definite integral. If $\int f^+ \, d\mu = \infty$ and $\int f^- \, d\mu < \infty$, then f is not integrable but in accordance with (3.2) is assigned ∞ as its definite integral. Similarly, if $\int f^+ \, d\mu < \infty$ and $\int f^- \, d\mu = \infty$, then f is not integrable but has definite integral $-\infty$.

Non-Negative Functions

THEOREM 3.2. (i) If $f = \sum_i x_i \mathbb{1}_{A_i}$ is a non-negative simple function, $\{A_i\}$ being a finite decomposition of Ω into $\mathcal{F}$ -sets, then

$$\int f d\mu = \sum_i x_i \mu(A_i).$$

(ii) If $0 \leq f(\omega) \leq g(\omega)$ for all ω , then

$$\int f d\mu \leq \int g d\mu.$$

(iii) If $0 \leq f_n(\omega) \uparrow f(\omega)$ for all ω , then

$$0 \leq \int f_n d\mu \uparrow \int f d\mu.$$

(iv) For non-negative functions f and g and non-negative constants α and β ,

$$\int (\alpha f + \beta g) d\mu = \alpha \int f d\mu + \beta \int g d\mu.$$

Proof. (i)

(ii)

(iii)

(iv)

□

EXAMPLE 3.3. Consider the line $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ with Lebesgue measure. Suppose that $-\infty < a_0 \leq a_1 \leq \dots \leq a_m < \infty$, and let

$$f = \begin{cases} x_i \geq 0, & (a_{i-1}, a_i], i = 1, \dots, m \\ 0, & (-\infty, a_0] \cup (a_m, \infty) \end{cases}.$$

By [Theorem 3.2\(i\)](#),

$$\int f d\lambda = \sum_{i=1}^m x_i (a_i - a_{i-1}).$$

THEOREM 3.4.

Suppose that f and g are non-negative. Recall that almost everywhere (a.e.) means outside a set of measure 0.

(i) If $f = 0$ a.e., then $\int f d\mu = 0$.

(ii) If $\mu(\omega : f(\omega) > 0) > 0$, then $\int f d\mu > 0$.

(iii) If $\int f d\mu < \infty$, then $f < \infty$ a.e.

(iv) If $f \leq g$ a.e., then $\int f d\mu \leq \int g d\mu$.

(v) If $f = g$ a.e., then $\int f d\mu = \int g d\mu$.

Proof. (i)

(ii)

(iii)

(iv)

(v)

□

Exercises

These problems outline alternative definitions of the integral and clarify the role measurability plays. Call (3.1) the *lower integral*, and write it as

$$\int_* f d\mu = \sup_i \left(\sum_i \inf_{\omega \in A_i} f(\omega) \right) \mu(A_i) \quad (3.3)$$

to distinguish it from the *upper integral*

$$\int^* f d\mu = \inf_i \left(\sum_i \sup_{\omega \in A_i} f(\omega) \right) \mu(A_i). \quad (3.4)$$

The infimum in (3.4), like the supremum in (3.3), extends over all finite partitions $\{A_i\}$ of Ω into $\mathcal{F}$ -sets.

EXERCISE 3.1. Suppose that f is measurable and non-negative. Show that $\int^* f d\mu = \infty$ if $\mu\{\omega : f(\omega) > 0\} = \infty$ or if $\mu\{\omega : f(\omega) > a\} > 0$ for all a .

SOLUTION.

□

There are many functions familiar from calculus that ought to be integrable but are of the types in the preceding problem and hence have infinite upper integral. Examples are $x^{-2}\mathbb{1}_{(1,\infty)}(x)$ and $x^{-\frac{1}{2}}\mathbb{1}_{(0,1)}(x)$. Therefore, (3.4) is inappropriate as a definition of $\int f d\mu$ for non-negative f . The only problem with (3.4), however, is that it treats infinity the wrong way. To see this, and to focus on essentials, assume that $\mu(\Omega) < \infty$ and that f is bounded, although not necessarily non-negative or measurable $\mathcal{F}$.

EXERCISE 3.2. (i) Show that

$$\sum_i \left[\inf_{\omega \in A_i} f(\omega) \right] \mu(A_i) \leq \sum_j \left[\inf_{\omega \in B_j} f(\omega) \right] \mu(B_j)$$

if $\{B_j\}$ refines $\{A_i\}$. Prove a dual relation for the sums in (3.4) and conclude that

$$\int_* f d\mu \leq \int^* f d\mu.$$

(ii) Now assume that f is measurable $\mathcal{F}$ and let M be a bound for $|f|$. Consider the partition $A_i = \{\omega : i\epsilon < f(\omega) \leq (i+1)\epsilon\}$, where i ranges from $-N$ to N and N is large enough that $N\epsilon > M$. Show that

$$\sum_i \left[\sup_{\omega \in A_i} f(\omega) \right] \mu(A_i) - \sum_i \left[\inf_{\omega \in A_i} f(\omega) \right] \mu(A_i) \leq \epsilon \mu(\Omega).$$

Conclude that

$$\int_* f d\mu = \int^* f d\mu. \quad (3.5)$$

To define the integral as the common value in (3.5) is the Darboux-Young approach. The advantage of (15.3) as a definition is that (in the non-negative case) it applies at once to unbounded f and infinite μ .

EXERCISE 3.3. For $A \subseteq \Omega$, define $\mu^*(A)$ and $\mu_*(A)$ by (3.9) and (3.10) with μ in place of P . Show that $\int^* I_A d\mu = \mu^*(A)$ and $\int_* I_A d\mu = \mu_*(A)$ for every A . Therefore, (3.5) can fail if f is not measurable $\mathcal{F}$. (Where was measurability used in the proof of (3.5)?)

SECTION 3.2

| Properties of the Integral

Equalities and Inequalities

The requirement for integrability of f is that $\int f^+ d\mu$ and $\int f^- d\mu$ both be finite, which is the same as the requirement that

$$\int f^+ d\mu + \int f^- d\mu < \infty,$$

and hence is the same as the requirement that

$$\int (f^+ + f^-) d\mu < \infty$$

by [Theorem 3.2\(iv\)](#). Since $f^+ + f^- = |f|$, f is integrable if and only if

$$\int |f| d\mu < \infty.$$

If $\mu(\Omega) < \infty$, a bounded f is integrable.

THEOREM 3.5. (i) *Monotonicity: If f and g are integrable and $f \leq g$ a.e., then*

$$\int f d\mu \leq \int g d\mu.$$

(ii) *Linearity: If f and g are integrable and α, β are finite real numbers, then $\alpha f + \beta g$ is integrable and*

$$\int (\alpha f + \beta g) d\mu = \alpha \int f d\mu + \beta \int g d\mu.$$

Proof.

□

Integration to the Limit

THEOREM 3.6 (Monotone Convergence Theorem).

If $0 \leq f_n \uparrow f$ a.e., then

$$\int f_n d\mu \uparrow \int f d\mu.$$

Proof. If there is a set $A \in \mathcal{F}$ with $\mu(A^c) = 0$ and $0 \leq f_n(\omega) \uparrow f(\omega)$ for all $\omega \in A$, then $0 \leq f_n \mathbb{1}_A \uparrow f \mathbb{1}_A$ holds everywhere on Ω , and it follows by [Theorem 3.2\(iii\)](#) and the remark following [Theorem 3.4\(v\)](#) that

$$\int f_n d\mu = \int f_n \mathbb{1}_A d\mu \uparrow \int f \mathbb{1}_A d\mu = \int f d\mu.$$

□

EXAMPLE 3.7. If μ is a measure on $\mathcal{F}$, and $\mathcal{F}_0$ is a σ -field contained in $\mathcal{F}$, then the restriction μ_0 of μ to $\mathcal{F}_0$ is another measure. If $f = \mathbb{1}_A$ and $A \in \mathcal{F}_0$, then

$$\int f d\mu = \int f d\mu_0,$$

the common value being $\mu(A) = \mu_0(A)$. The same is true by linearity for non-negative simple $\mathcal{F}_0$ -measurable functions. It holds by [Theorem 3.6](#) for all non-negative f that are $\mathcal{F}_0$ -measurable because ([Theorem 2.10](#)) $0 \leq f_n \uparrow f$ for simple functions f_n that are $\mathcal{F}_0$ -measurable. For $\mathcal{F}_0$ -measurable functions, integration with respect to μ is thus the same thing as integration with respect to μ_0 .

THEOREM 3.8 (Fatou's Lemma).

For a non-negative sequence f_n ,

$$\int \liminf_n f_n d\mu \leq \liminf_n \int f_n d\mu. \quad (3.6)$$

Proof. If $g_n = \inf_{k \geq n} f_k$, then

$$0 \leq g_n \uparrow g := \liminf_n f_n$$

on Ω . [Theorem 3.5\(i\)](#) and [Theorem 3.6](#) give

$$\int f_n d\mu \geq \int g_n d\mu \rightarrow \int g d\mu.$$

□

EXAMPLE 3.9. On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, the functions $f_n = n^2 \mathbb{1}_{(0, n^{-1})}$ and $f = 0$ satisfy $f_n(x) \rightarrow f(x)$ for each x , but

$$\int f d\lambda = 0, \quad \int f_n d\lambda = n \rightarrow \infty.$$

This shows that the inequality in (3.6) can be strict and that it is not always possible to integrate to the limit.

THEOREM 3.10 (Lebesgue's Dominated Convergence Theorem).

If $|f_n| \leq g$ a.e., where g is integrable, and if $f_n \rightarrow f$ a.e., then f and the f_n are integrable and

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Proof.

□

THEOREM 3.11 (Bounded Convergence Theorem).

If $\mu(\Omega) < \infty$ and the f_n are uniformly bounded, then $f_n \rightarrow f$ a.e. implies

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Proof.

□

Integration over Sets

The integral of f over a set A in $\mathcal{F}$ is defined by

$$\int_A f d\mu = \int \mathbb{1}_A f d\mu.$$

THEOREM 3.12. (i) If f and g are non-negative and $\int_A f d\mu = \int_A g d\mu$ for all A in $\mathcal{F}$, and if μ is σ -finite, then $f = g$ a.e.

- (ii) If f and g are integrable and $\int_A f d\mu = \int_A g d\mu$ for all A in $\mathcal{F}$, then $f = g$ a.e.
- (iii) If f and g are integrable and $\int_A f d\mu = \int_A g d\mu$ for all A in $\mathcal{P}$, where $\mathcal{P}$ is a π -system generating $\mathcal{F}$ and Ω is a finite or countable union of $\mathcal{P}$ -sets, then $f = g$ a.e.

Proof. □

Exercises

EXERCISE 3.4. Suppose that $\mu(\Omega) < \infty$ and f_n are uniformly bounded.

- (i) Assume $f_n \rightarrow f$ uniformly and deduce $\int f_n d\mu \rightarrow \int f d\mu$ from (16.5).
- (ii) Use part (i) and *Egoroff's theorem* to give another proof of [Theorem 3.11](#).

EXERCISE 3.5. Prove that if $0 \leq f_n \rightarrow f$ almost everywhere and $\int f_n d\mu \leq A < \infty$, then f is integrable and $\int f d\mu \leq A$.¹

EXERCISE 3.6 (Beppo Levi's Theorem). Suppose that f_n are integrable and $\sup_n \int f_n d\mu < \infty$. Show that, if $f_n \uparrow f$, then f is integrable and $\int f_n d\mu \rightarrow \int f d\mu$.

SOLUTION. □

EXERCISE 3.7. (i) Suppose that functions a_n, b_n, f_n converge almost everywhere to functions a, b, f , respectively. Suppose that the first two sequences may be integrated to the limit—that is, the functions are all integrable and $\int a_n d\mu \rightarrow \int a d\mu$, $\int b_n d\mu \rightarrow \int b d\mu$. Suppose, finally, that the first two sequences enclose the third: $a_n \leq f_n \leq b_n$ a.e. Show that the third may be integrated to the limit.

- (ii) Deduce *Lebesgue's dominated convergence theorem* from part (i).

SOLUTION. □

EXERCISE 3.8. Show that if f is integrable, then for each ϵ there is a δ such that $\mu(A) < \delta$ implies $\int_A |f| d\mu < \epsilon$.

SOLUTION. □

¹ This is essentially the same as *Fatou's lemma* and is sometimes called by that name.

SECTION 3.3

| The Integral with Respect to Lebesgue Measure

The Lebesgue Integral in $\mathbb{R}^k$

Consider the k -dimensional Lebesgue integral in $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k), \lambda_k)$. Suppose that $T : U \rightarrow \mathbb{R}^k$, where U is an open set in $\mathbb{R}^k$. The map has the form

$$Tx = (t_1(x), \dots, t_k(x)).$$

It is by definition continuously differentiable if the partial derivatives $t_{ij}(x) = \frac{\partial t_i}{\partial x_j}$ exist and are continuous in U . Let $D_x = \{t_{ij}(x)\}$ be the *Jacobian matrix*, let $J(x) = \det D_x$ be the *Jacobian determinant*, and let $V = TU$.

THEOREM 3.13 (Change of Variable Formula).

Let T be a continuously differentiable map of the open set U onto V . Suppose that T is one-to-one and that $J(x) \neq 0$ for all x . If f is non-negative, then

$$\int_U f(Tx)|J(x)|dx = \int_V f(y)dy. \quad (3.7)$$

EXAMPLE 3.14. In $\mathbb{R}^2$ take $U = \{(\rho, \theta) : \rho > 0, 0 < \theta < 2\pi\}$ and $T(\rho, \theta) = (\rho \cos \theta, \rho \sin \theta)$. The Jacobian is $J(\rho, \theta) = \rho$, and (3.7) gives the formula for integrating in polar coordinates:

$$\iint_{\substack{\rho > 0 \\ 0 < \theta < 2\pi}} f(\rho \cos \theta, \rho \sin \theta) \rho d\rho d\theta = \iint_{\mathbb{R}^2} f(x, y) dx dy. \quad (3.8)$$

SECTION 3.4

| Product Measure and Fubini's Theorem

Product Spaces

Let $(X, \mathcal{X})$ and $(Y, \mathcal{Y})$ be measurable spaces. In the product space $X \times Y$ a *measurable rectangle* is a product $A \times B$ for which $A \in \mathcal{X}$ and $B \in \mathcal{Y}$. The natural class of sets in $X \times Y$ to consider is the *product σ -field* $\mathcal{X} \times \mathcal{Y}$ generated by the measurable rectangles.

EXAMPLE 3.15. Suppose that $X = Y = \mathbb{R}$ and $\mathcal{X} = \mathcal{Y} = \mathcal{B}(\mathbb{R})$. As the measurable rectangles do include the ordinary ones and the latter generate $\mathcal{R}^2$, it follows that $\mathcal{B}(\mathbb{R}^2) \subset \mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})$. On the other hand, if A is an interval, $\{B : A \times B \in \mathcal{R}^2\}$ is a σ -field containing the intervals and hence the Borel sets. Therefore, $\{A : A \times B \in \mathcal{R}^2\}$ contains the intervals and hence, being a σ -field, contains the Borel sets. Thus all the measurable rectangles are in $\mathcal{R}^2$, and so $\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^2)$ consists exactly of the two-dimensional Borel sets.

THEOREM 3.16. (i) If $E \in \mathcal{X} \times \mathcal{Y}$, then for each x the set $\{y : (x, y) \in E\}$ lies in $\mathcal{Y}$ and for each y the set $\{x : (x, y) \in E\}$ lies in $\mathcal{X}$.

(ii) If f is $\mathcal{X} \times \mathcal{Y}$ -measurable, then for each fixed x the function $f(x, \cdot)$ is $\mathcal{Y}$ -measurable, and for each fixed y the function $f(\cdot, y)$ is $\mathcal{X}$ -measurable.

Remark. The set $\{y : (x, y) \in E\}$ is the section of E determined by x , and $f(x, \cdot)$ is the section of f determined by x .

Proof. □

Product Measure

Suppose that $(X, \mathcal{X}, \mu)$ and $(Y, \mathcal{Y}, \nu)$ are measure spaces, and suppose for the moment that μ and ν are finite. By Theorem 3.16, $\nu\{y : (x, y) \in E\}$ is a well-defined function of x , and is $\mathcal{X}/\mathcal{X} \times \mathcal{Y}$ -measurable by the π - λ theorem. It follows that

$$\pi'(E) = \int_X \nu\{y : (x, y) \in E\} \mu(dx), \quad E \in \mathcal{X} \times \mathcal{Y}, \quad (3.9)$$

$$\pi''(E) = \int_Y \mu\{x : (x, y) \in E\} \nu(dy), \quad E \in \mathcal{X} \times \mathcal{Y}. \quad (3.10)$$

are finite measures on $\mathcal{X} \times \mathcal{Y}$. The class of E in $\mathcal{X} \times \mathcal{Y}$ for which $\pi'(E) = \pi''(E)$ thus contains the measurable rectangles.

THEOREM 3.17.

If $(X, \mathcal{X}, \mu)$ and $(Y, \mathcal{Y}, \nu)$ are σ -finite measure spaces, $\pi(E) = \pi'(E) = \pi''(E)$ defines a σ -finite measure on $\mathcal{X} \times \mathcal{Y}$; it is the only measure such that $\pi(A \times B) = \mu(A) \cdot \nu(B)$ for measurable rectangles.

Proof. □

Remark. The π thus defined is called *product measure*; it is usually denoted $\mu \times \nu$.

Fubini's Theorem

Integrals with respect to π are usually computed via the formulas

$$\int_{X \times Y} f(x, y) \pi(d(x, y)) = \int_X \left(\int_Y f(x, y) \nu(dy) \right) \mu(dx), \quad (3.11)$$

$$\int_{X \times Y} f(x, y) \pi(d(x, y)) = \int_Y \left(\int_X f(x, y) \mu(dx) \right) \nu(dy). \quad (3.12)$$

The left side here is a *double integral*, and the right sides are *iterated integrals*.

THEOREM 3.18.

Let $(X, \mathcal{X}, \mu)$ and $(Y, \mathcal{Y}, \nu)$ be σ -finite measure spaces.

(i) *Fubini's theorem: For non-negative f the functions*

$$\int_Y f(x, y) \nu(dy), \quad \int_X f(x, y) \mu(dx) \quad (3.13)$$

are $\mathcal{X}$ - and $\mathcal{Y}$ -measurable, respectively, and (3.11)-(3.12) hold.

(ii) *Tonelli's theorem: If f (not necessarily non-negative) is integrable with respect to π , then the two functions (3.13) are finite and measurable on sets*

$$A_0 = \left\{ x \in X \mid \int_Y |f(x, y)| \nu(dy) < \infty \right\},$$

$$B_0 = \left\{ y \in Y \mid \int_X |f(x, y)| \mu(dx) < \infty \right\},$$

respectively, where $\mu(X - A_0) = \nu(Y - B_0) = 0$, and again (3.11)-(3.12) hold.

Proof.

□

EXAMPLE 3.19. Let D_r be the closed disk in the plane with centre at the origin and radius r . By (3.8),

$$\lambda_2(D_r) = \iint_{D_r} dx dy = \iint_{\substack{0 < \rho \leq r \\ 0 < \theta < 2\pi}} \rho d\rho d\theta.$$

The last integral can be evaluated by *Fubini's theorem*:

$$\lambda_2(D_r) = 2\pi \int_0^r \rho d\rho = \pi r^2.$$

Integration by Parts

Let F and G be two non-decreasing, right-continuous functions on an interval $[a, b]$, and let μ and ν be the corresponding measures:

$$\mu(x, y] = F(y) - F(x), \quad \nu(x, y] = G(y) - G(x), \quad a \leq x \leq y \leq b.$$

We write $dF(x)$ and $dG(x)$ in place of $\mu(dx)$ and $\nu(dx)$.

THEOREM 3.20.

If F and G have no common points of discontinuity in $(a, b]$, then

$$\int_{(a, b]} G(x) dF(x) = F(b)G(b) - F(a)G(a) - \int_{(a, b]} F(x) dG(x).$$

Proof.

□

| Exercises

EXERCISE 3.9. Show by *Theorem 3.16* that if $A \times B$ is non-empty and lies in $\mathcal{X} \times \mathcal{Y}$, then $A \in \mathcal{X}$ and $B \in \mathcal{Y}$.

EXERCISE 3.10. The assumption of σ -finiteness in *Theorem 3.17* is essential: Let μ be Lebesgue measure on the line, let ν be counting measure on the line, and take

$$E = \{(x, y) : x = y\}.$$

Then (3.9) and (3.10) do not agree.

EXERCISE 3.11. Suppose that f is non-negative on a σ -finite measure space $(\Omega, \mathcal{F}, \mu)$. Show that

$$\int_{\Omega} f d\mu = (\mu \times \lambda)\{(\omega, y) \in \Omega \times \mathbb{R} : 0 \leq y \leq f(\omega)\}.$$

Prove that the set on the right is measurable. This gives the *area under the curve*. Given the existence of $\mu \times \lambda$ on $\Omega \times \mathbb{R}$, one can use the right side of this equation as an alternative definition of the integral.

EXERCISE 3.12. Suppose that $\nu[y : (x, y) \in E] = \nu[y : (x, y) \in F]$ for all x , and show that

$$(\mu \times \nu)(E) = (\mu \times \nu)(F).²$$

EXERCISE 3.13. Exhibit a case in which (18.12) fails because F and G have a common point of discontinuity.

EXERCISE 3.14. Prove for distribution functions F that

$$\int_{-\infty}^{\infty} (F(x+c) - F(x)) dx = c.$$

² This is a general version of *Cavalieri's principle*.

Random Variables and Expected Values

SECTION 4.1

| Random Variables and Distributions

Random Variables and Vectors

DEFINITION 4.1.

A *random variable* on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a $\mathcal{F}$ -measurable real-valued function $X = X(\omega)$. A *random vector* is a $\mathcal{F}$ -measurable mapping from Ω to $\mathbb{R}^k$:

$$\omega \rightarrow X(\omega) = (X_1(\omega), \dots, X_k(\omega)),$$

where $X_1, \dots, X_k$ are random variables.

Subfields

THEOREM 4.2.

Let $X = (X_1, \dots, X_k)$ be a random vector.

- (i) The σ -field $\sigma(X) = \sigma(X_1, \dots, X_k)$ consists exactly of the sets $\{\omega : X(\omega) \in H\}$ for $H \in \mathcal{B}(\mathbb{R}^k)$.
- (ii) In order that a random variable Y be measurable $\sigma(X) = \sigma(X_1, \dots, X_k)$ it is necessary and sufficient that there exists a measurable map $f : \mathbb{R}^k \rightarrow \mathbb{R}$ such that $Y(\omega) = f(X_1(\omega), \dots, X_k(\omega))$ for all ω .

Proof.

□

Distributions

The *distribution* or *law* of a random variable X was defined as the probability measure on the line given by $\mu = \mathbb{P}X^{-1}$, or

$$\mu(A) = \mathbb{P}\{X \in A\}, \quad A \in \mathcal{B}(\mathbb{R}).$$

The distribution function of X was defined by

$$F(x) = \mu(-\infty, x] = \mathbb{P}\{X \leq x\}$$

for real x . A *support* for μ is a Borel set S for which $\mu(S) = 1$.

DEFINITION 4.3. (i) A random variable and its distribution function are *discrete* if μ has a countable support $S = \{x_1, x_2, \dots\}$.

(ii) A random variable and its distribution have *density* f with respect to Lebesgue measure if f is a non-negative Borel function on $\mathbb{R}$ and

$$\mathbb{P}(X \in A) = \mu(A) = \int_A f(x) dx, \quad A \in \mathcal{B}(\mathbb{R}).$$

EXAMPLE 4.4. (i) A familiar discrete distribution is the *binomial*:

$$\mathbb{P}(X = r) = \mu(r) = \binom{n}{r} p^r (1-p)^{n-r}, \quad r = 0, 1, \dots, n.$$

Another common discrete distribution is the *Poisson* distribution with parameter $\lambda > 0$:

$$\mathbb{P}(X = r) = \mu(r) = e^{-\lambda} \frac{\lambda^r}{r!}, \quad r = 0, 1, \dots$$

(ii) For the *exponential* distribution with parameter $\alpha > 0$, the density is

$$f(x) = \begin{cases} 0, & x < 0 \\ \alpha e^{-\alpha x}, & x \geq 0 \end{cases}.$$

The corresponding distribution function

$$F(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-\alpha x}, & x \geq 0 \end{cases}.$$

For the *normal* distribution with parameters $m \in \mathbb{R}$ and $\sigma > 0$,

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}. \quad (4.1)$$

For the *uniform* distribution over an interval $(a, b]$,

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x \leq b \\ 0, & \text{otherwise} \end{cases}.$$

Multidimensional Distributions

DEFINITION 4.5.

For a k -dimensional random vector $X = (X_1, \dots, X_k)$, the *joint distribution* μ (a probability measure on $\mathcal{B}(\mathbb{R}^k)$) and the *joint distribution function* F (a real function on $\mathbb{R}^k$) are defined by

$$\mu(A) = \mathbb{P}((X_1, \dots, X_k) \in A), \quad A \in \mathcal{B}(\mathbb{R}^k),$$

$$F(x_1, \dots, x_k) = \mathbb{P}(X_1 \leq x_1, \dots, X_k \leq x_k) = \mu(S_x).$$

The *marginal distributions* of μ are

$$\mu_j(A) = \mu((x_1, \dots, x_k) : x_j \in A) = \mathbb{P}(X_j \in A).$$

Remark. If μ has a density f in $\mathbb{R}^k$, then μ_j has over the line the density

$$f_j(x) = \int_{\mathbb{R}^{k-1}} f(x_1, \dots, x_{j-1}, x, x_{j+1}, \dots, x_k) dx_1 \dots dx_{j-1} dx_{j+1} \dots dx_k,$$

by Fubini's theorem.

Independence

DEFINITION 4.6 (Independence).

Random variables $X_1, \dots, X_k$ are defined to be *independent* if the σ -fields $\sigma(X_1), \dots, \sigma(X_k)$ they generate are independent.

PROPOSITION 4.7.

Independence is also equivalent to:

(i) For all linear Borel sets $H_1, \dots, H_k$:

$$\mathbb{P}(X_1 \in H_1, \dots, X_k \in H_k) = \mathbb{P}(X_1 \in H_1) \dots \mathbb{P}(X_k \in H_k).$$

(ii) For all real $x_1, \dots, x_k$:

$$\mathbb{P}(X_1 \leq x_1, \dots, X_k \leq x_k) = \mathbb{P}(X_1 \leq x_1) \dots \mathbb{P}(X_k \leq x_k).$$

(iii) For $(X_1, \dots, X_k)$ with distribution μ and distribution function F :

$$\mu = \mu_1 \times \dots \times \mu_k, \quad F(x_1, \dots, x_k) = F_1(x_1) \dots F_k(x_k).$$

Suppose that each μ_i has density f_i , by Fubini's theorem:

$$f(\curvearrowright) = f_1(x_1) \dots f_k(x_k).$$

| Exercises

EXERCISE 4.1. If X is a positive random variable with density f , then X^{-1} has density

$$\frac{f(1/x)}{x^2}.$$

Prove this by (20.16) and by a direct argument.

EXERCISE 4.2. Show that $X_1, X_2, \dots$ are independent if $\sigma(X_1, \dots, X_{n-1})$ and $\sigma(X_n)$ are independent for each n .

SECTION 4.2

| Expected Values

Expected Value as Integral

DEFINITION 4.8 (Expected Values).

The *expected value* of a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$ is the integral of X with respect to the measure $\mathbb{P}$:

$$\mathbb{E}[X] = \int X d\mathbb{P} = \int_{\Omega} X(\omega) \mathbb{P}(d\omega).$$

Remark. X is integrable if and only if $\mathbb{E}[|X|] < \infty$. The integral $\int_A X d\mathbb{P}$ over a set A is defined as $\mathbb{E}[\mathbf{1}_A X]$.

Expected Values and Distributions

Suppose that X has distribution μ . If g is a real function of a real variable, then by the change-of-variable formula,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) \mu(dx).$$

Moments

DEFINITION 4.9.

μ and F determine all the *absolute moments* of X :

$$\mathbb{E}[|X|^k] = \int_{-\infty}^{\infty} |x|^k \mu(dx) = \int_{-\infty}^{\infty} |x|^k dF(x), \quad k = 1, 2, \dots$$

X has k -th moment

$$\mathbb{E}[X^k] = \int_{-\infty}^{\infty} x^k \mu(dx) = \int_{-\infty}^{\infty} x^k dF(x).$$

Remark. If the first two moments of X are finite and $\mathbb{E}[X] = m$, then the variance is

$$\text{Var}[X] = \mathbb{E}[(X - m)^2] = \int_{-\infty}^{\infty} (x - m)^2 \mu(dx) = \mathbb{E}[X^2] - m^2.$$

EXAMPLE 4.10. Consider the normal density (4.1) with $m = 0$ and $\sigma = 1$. Integration by parts shows that

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^k e^{-\frac{x^2}{2}} dx = \frac{k-1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^{k-2} e^{-\frac{x^2}{2}} dx, \quad k = 2, 3, \dots$$

Of course, $\mathbb{E}[X] = 0$ by symmetry and $\mathbb{E}[X^0] = 1$. It follows by induction that

$$\mathbb{E}[X^{2k}] = 1 \times 3 \times 5 \times \dots \times (2k-1), \quad k = 1, 2, \dots,$$

and that the odd moments all vanish. From a change of variable, it follows that a random variable with the normal density has mean m and variance σ^2 .

Inequalities

THEOREM 4.11 (Markov's Inequality).

Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.

(i) If X is non-negative, then for any $\alpha > 0$,

$$\mathbb{P}(X \geq \alpha) \leq \frac{1}{\alpha} \mathbb{E}[X].$$

(ii) For $\alpha > 0$ and $k \geq 0$,

$$\mathbb{P}(|X| \geq \alpha) \leq \frac{1}{\alpha^k} \mathbb{E}[|X|^k].$$

Proof.

□

COROLLARY 4.12 (Chebyshev's Inequality).

As a special case of Markov's Inequality:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \alpha) \leq \frac{1}{\alpha^2} \text{Var}(X)$$

Proof.

□

THEOREM 4.13 (Jensen's Inequality).

If φ is convex on an interval containing the range of X and if X and $\varphi(X)$ both have expected values,

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

THEOREM 4.14 (Hölder's Inequality).

$$\mathbb{E}[|XY|] \leq \mathbb{E}^{\frac{1}{p}}[|X|^p] \mathbb{E}^{\frac{1}{q}}[|Y|^q], \quad \frac{1}{p} + \frac{1}{q} = 1.$$

COROLLARY 4.15 (Schwarz's Inequality).

The case $p = q = 2$:

$$\mathbb{E}[|XY|] \leq \mathbb{E}^{\frac{1}{2}}[X^2] \mathbb{E}^{\frac{1}{2}}[Y^2].$$

If X and Y have second moments, then XY must have a first moment.

Independence and Expected Value

THEOREM 4.16.

If $X_1, \dots, X_k$ are independent and integrable, then the product $X_1 \dots X_k$ is also integrable and

$$\mathbb{E}[X_1 \dots X_k] = \mathbb{E}[X_1] \dots \mathbb{E}[X_k].$$

Proof.

□

| Exercises

EXERCISE 4.3. Show that, if X has the standard normal distribution, then

$$E[|X|^{2n+1}] = \frac{2^n n! \sqrt{2/\pi}}{(2n+1)!!}.$$

EXERCISE 4.4. Prove (21.9) by *Fubini's theorem*.

EXERCISE 4.5. Suppose that X and Y are random variables with distribution functions F and G .

- (i) Show that if F and G have no common jumps, then $E[F(Y)] + E[G(X)] = 1$.
- (ii) If F is continuous, then $E[F(X)] = \frac{1}{2}$.
- (iii) Even if F and G have common jumps, if X and Y are taken to be independent, then $E[F(Y)] + E[G(X)] = 1 - P[X = Y]$.
- (iv) Even if F has jumps, $E[F(X)] = \frac{1}{2} + \frac{1}{2} \sum_x P^2[X = x]$.

EXERCISE 4.6. Suppose that X and Y are independent and that $f(x, y)$ is nonnegative. Put $g(x) = E[f(x, Y)]$ and show that $E[g(X)] = E[f(X, Y)]$. Show more generally that $\int_{X \in A} g(X) dP = \int_{X \in A} f(X, Y) dP$. Extend to f that may be negative.

SOLUTION. □

EXERCISE 4.7. The integrability of $X + Y$ does not imply that of X and Y separately. Show that it does if X and Y are independent.

SOLUTION. □

SECTION 4.3

| Sums of Independent Random Variables

Let S_n denote the partial sum $\sum_{k=1}^n X_k$ ($S_0 = 0$) of the random variables $(X_n)_{n=1}^\infty$.

The Strong Law of Large Numbers

THEOREM 4.17 (Strong Law of Large Numbers).

If $X_1, X_2, \dots$ are i.i.d. and have finite mean, then $\frac{S_n}{n} \rightarrow \mathbb{E}[X_1]$ with probability 1.

Proof.

□

COROLLARY 4.18.

Suppose that $X_1, X_2, \dots$ are i.i.d. and

$$\mathbb{E}[X_1^-] < \infty, \quad \mathbb{E}[X_1^+] = \infty,$$

where X_1^- and X_1^+ denote the negative and positive part respectively (so that $\mathbb{E}[X_1] = \infty$). Then $n^{-1} \sum_{k=1}^n X_k \rightarrow \infty$ with probability 1.

Proof.

□

Remark.

Kolmogorov's Zero-One Law

THEOREM 4.19.

Suppose that $\{X_n\}$ is independent and that

$$A \in \mathcal{T} = \bigcap_{n=1}^{\infty} \sigma(X_n, X_{n+1}, \dots).$$

Then either $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$.

Proof.

□

| Exercises

EXERCISE 4.8.

EXERCISE 4.9. (i) Generalise the *Borel-Cantelli lemmas*: Suppose X_n are non-negative. If $\sum \mathbb{E}[X_n] < \infty$, then $\sum X_n$ converges with probability 1. If the X_n are independent and uniformly bounded, and if $\sum \mathbb{E}[X_n] = \infty$, then $\sum X_n$ diverges with probability 1.

(ii) Construct independent, non-negative X_n such that $\sum X_n$ converges with probability 1 but $\sum \mathbb{E}[X_n]$ diverges. For an extreme example, arrange that $P[X_n > 0 \text{ i.o.}] = 0$ but $\mathbb{E}[X_n] = \infty$.

EXERCISE 4.10. If $X_1, X_2, \dots$ are *i.i.d.*, and if $\mathbb{P}(X_1 \geq 0) = 1$ and $\mathbb{P}(X_1 > 0) > 0$, then $\sum_n X_n = \infty$ with probability 1. Deduce this from [Theorem 4.17](#) and its corollary and also directly: find a positive ϵ such that $X_r > \epsilon$ infinitely often with probability 1.

SOLUTION. □

EXERCISE 4.11.

Convergence of Distributions

SECTION 5.1

| Weak Convergence

DEFINITION 5.1 (Weak Convergence). (i) Distribution functions F_n *converges weakly* to the distribution function F if

$$\lim_n F_n(x) = F(x) \quad (5.1)$$

for every continuity point x of F , writing $F_n \Rightarrow F$.

(ii) If μ_n and μ are the probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ corresponding to F_n and F , then $F_n \Rightarrow F$ if and only if

$$\lim_n \mu_n(A) = \mu(A) \quad (5.2)$$

for every A of the form $A = (-\infty, x]$ for which $\mu(x) = 0$. In this case the distributions themselves *converge weakly* written by $\mu_n \Rightarrow \mu$.

EXAMPLE 5.2. (i)

(ii)

(iii)

DEFINITION 5.3 (Convergence in Distribution).

Let X_n and X be random variables with respective distribution functions F_n and F . If $F_n \Rightarrow F$, then X_n is said to *converge in distribution* or *in law* to X , written $X_n \Rightarrow X$.

Remark. (i) $X_n \Rightarrow X$ if and only if

$$\lim_n \mathbb{P}(X_n \leq x) = \mathbb{P}(X \leq x)$$

for every x such that $\mathbb{P}(X = x) = 0$.

(ii) By [Theorem 2.12](#) there does exist a random variable for any prescribed distribution.

EXAMPLE 5.4.

DEFINITION 5.5 (Convergence in Probability).

Suppose that $X, X_1, X_2, \dots$ are random variables all defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. If $X_n \rightarrow X$ with probability 1, then $\mathbb{P}(|X_n - X| \geq \epsilon \text{ i.o.}) = 0$ for $\epsilon > 0$, and hence

$$\lim_n \mathbb{P}(|X_n - X| > \epsilon) = 0$$

by *Theorem 1.22*. Thus there is convergence in probability $X_n \xrightarrow{\mathbb{P}} X$.

THEOREM 5.6.

Suppose that X_n and X are random variables on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- (i) If $X_n \rightarrow X$ with probability 1, then $X_n \xrightarrow{\mathbb{P}} X$.
- (ii) If $X_n \xrightarrow{\mathbb{P}} X$, then $X_n \Rightarrow X$.

Proof.

□

Remark.

Fundamental Theorems

THEOREM 5.7 (Skorokhod Representation Theorem).

Suppose that μ_n and μ are probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $\mu_n \Rightarrow \mu$. There exist random variables Y_n and Y on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that Y_n has distribution μ_n , Y has distribution μ , and $Y_n(\omega) \rightarrow Y(\omega)$ for all $\omega \in \Omega$.

Proof.

□

THEOREM 5.8 (Portmanteau Theorem).

The following three conditions are equivalent.

- (i) $\mu_n \Rightarrow \mu$.
- (ii) $\int f d\mu_n \rightarrow \int f d\mu$ for every bounded, continuous real function f .
- (iii) $\mu_n(A) \rightarrow \mu(A)$ for every μ -continuity set A .

Proof.

□

Helly's Theorem

THEOREM 5.9 (Helly Selection Theorem).

For every sequence $\{F_n\}$ of distribution functions there exists a subsequence $\{F_{n_k}\}$ and a non-decreasing, right-continuous function F such that $F_{n_k}(x) \rightarrow F(x)$ at continuity points x of F .

Proof.

□

Remark.

DEFINITION 5.10 (Tightness).

A sequence of probability measures μ_n on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is *tight* if for all $\epsilon > 0$, there exists a finite interval $(a, b]$ such that $\mu_n(a, b] > 1 - \epsilon$ for all n .

THEOREM 5.11 (Prokhorov's Theorem).

Tightness is a necessary and sufficient condition that for every subsequence $\{\mu_{n_k}\}$ there exists a further subsequence $\{\mu_{n_{k(j)}}\}$ and a probability measure μ such that $\mu_{n_{k(j)}} \Rightarrow \mu$ as $j \rightarrow \infty$.

Proof.

□

COROLLARY 5.12.

If $\{\mu_n\}$ is a tight sequence of probability measures, and if every convergent subsequence converges weakly to the same probability measure μ , then $\mu_n \Rightarrow \mu$ as $n \rightarrow \infty$.

| Exercises

EXERCISE 5.1. (i) Show by example that distribution functions having densities can converge weakly even if the densities do not converge.

Hint: Consider $f_n(x) = 1 + \cos 2\pi nx$ on $[0, 1]$.

- (ii) Show that distributions with densities can converge weakly to a limit that has no density (even to a unit mass).
- (iii) Show that discrete distributions can converge weakly to a distribution that has a density.
- (iv) Construct an example, like that of [Example 5.2](#), in which $\mu_n(A) \rightarrow \mu(A)$ fails but in which all the measures come from continuous densities on $[0, 1]$.

SOLUTION.

□

EXERCISE 5.2. Show that for each probability measure μ on the line there exist probability measures μ_n with finite support such that $\mu_n \Rightarrow \mu$. Show further that $\mu_n(x)$ can be taken rational and that each point in the support can be taken rational. Thus there exists a countable set of probability measures such that every μ is the weak limit of some sequence from the set. The space of distribution functions is thus separable in the Lévy metric (see Problem 14.5).

EXERCISE 5.3. Show that (25.5) implies that

$$P([X \leq x] \Delta [X_n \leq x]) \rightarrow 0$$

if $P[X = x] = 0$.

EXERCISE 5.4. Prove weak convergence in [Example 5.2](#) by using [Theorem 5.8](#) and the theory of the Riemann integral.

EXERCISE 5.5. Suppose that $\sup_n \int f d\mu_n < \infty$ for a nonnegative f such that $f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$. Show that (μ_n) is tight.

SECTION 5.2

| Characteristic Functions

DEFINITION 5.13 (Characteristic Functions).

The *characteristic function* of a probability measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is defined for real t by

$$\varphi(t) = \int_{-\infty}^{\infty} e^{itx} \mu(dx) = \int_{-\infty}^{\infty} \cos(tx) \mu(dx) + i \int_{-\infty}^{\infty} \sin(tx) \mu(dx).$$

A random variable X with distribution μ has *characteristic function*

$$\varphi(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} \mu(dx).$$

Remark. The characteristic function always exists because e^{itx} is bounded. In non-probabilistic contexts, it is called the *Fourier transform*.

Moments and Derivatives

Of course, $\varphi(0) = 1$, and $|\varphi(t)| \leq 1$ for all t . By [Theorem 3.10](#), $\varphi(t)$ is continuous in t . In fact,

$$|\varphi(t+h) - \varphi(t)| \leq \int |e^{ihx} - 1| \mu(dx),$$

and so it follows by the *bounded convergence theorem* that $\varphi(t)$ is *uniformly continuous*.

PROPOSITION 5.14. (i) For any t satisfying

$$\lim_n \frac{|t|^n \mathbb{E}[|X|^n]}{n!} = 0,$$

$\varphi(t)$ must therefore have the expansion

$$\varphi(t) = \sum_{k=0}^{\infty} \frac{(it)^k}{k!} \mathbb{E}[X^k]. \quad (5.3)$$

(ii) If

$$\sum_{k=0}^{\infty} \frac{|t|^k}{k!} \mathbb{E}[|X|^k] = \mathbb{E}[e^{|tX|}] < \infty,$$

then (5.3) must hold.

Proof.

□

EXAMPLE 5.15.

THEOREM 5.16 (Riemann-Lebesgue Theorem).

If μ has a density, then $\varphi(t) \rightarrow 0$ as $|t| \rightarrow \infty$.

Proof.

□

Independence

PROPOSITION 5.17.

If $X_1, \dots, X_n$ are independent, then

$$\mathbb{E}\left[e^{it \sum_{k=1}^n X_k}\right] = \prod_{k=1}^n \mathbb{E}\left[e^{it X_k}\right].$$

Proof.

□

Remark. If X has characteristic function $\varphi(t)$, then $aX + b$ has characteristic function

$$\mathbb{E}\left[e^{it(aX+b)}\right] = e^{itb} \varphi(at).$$

In particular, $-X$ has characteristic function $\varphi(-t)$, which is the complex conjugate of $\varphi(t)$.

Inversion and the Uniqueness Theorem

THEOREM 5.18 (Inversion and the Uniqueness Theorem).

If the probability measure μ has characteristic function φ , and if $\mu\{a\} = \mu\{b\} = 0$, then

$$\mu(a, b] = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T \frac{e^{-ita} - e^{-itb}}{it} \varphi(t) dt.$$

Distinct measures cannot have the same characteristic function.

Proof.

□

EXAMPLE 5.19.

The Continuity Theorem

THEOREM 5.20 (Lévy's Continuity Theorem).

Let μ_n, μ be probability measures with characteristic functions φ_n, φ . A necessary and sufficient condition for $\mu_n \Rightarrow \mu$ is that $\varphi_n(t) \rightarrow \varphi(t)$ for each t .

Proof.

□

Exercises

EXERCISE 5.6. (i) Show that a necessary condition for X to have a lattice distribution is that $|\varphi(t)| = 1$ for some $t \neq 0$.

(ii) Show that the condition is sufficient as well.

(iii) Suppose that $|\varphi(t)| = |\varphi(t')| = 1$ for incommensurable t and t' ($t \neq 0, t' \neq 0, t/t'$ irrational). Show that $P[X = c] = 1$ for some constant c .

SOLUTION. (i) For a lattice distribution, the characteristic function can be written as

$$\varphi(t) = \mathbb{E}\left[e^{it(a+nb)}\right] = e^{ita} \mathbb{E}\left[e^{itnb}\right],$$

since a is a constant and n takes values in $\{0, \pm 1, \pm 2, \dots\}$. Notice that the exponential function $e^{it(a+nb)}$ has period $\frac{2\pi}{b}$. Taking $t = \frac{2\pi}{b}$, we have

$$\phi\left(\frac{2\pi}{b}\right) = e^{i(2\pi/b)a} \mathbb{E}[e^{2\pi i n}],$$

where $\mathbb{E}[e^{2\pi i n}] = \mathbb{E}[1] = 1$. Therefore, $|\phi(t)| = 1$ for $t = \frac{2\pi}{b} \neq 0$. □

EXERCISE 5.7. Show for independent random variables uniformly distributed over $(-1, +1)$ that $X_1 + \dots + X_n$ has density

$$\pi^{-1} \int_0^\infty \left(\frac{\sin t}{t}\right)^n \cos tx dt$$

for $n \geq 2$.

EXERCISE 5.8. Suppose that $F_n \Rightarrow F$ and that the characteristic functions are dominated by an integrable function. Show that F has a density that is the limit of the densities of the F_n .

EXERCISE 5.9.

SOLUTION. To show $a_n X_n + b_n \Rightarrow aX + b$, we need to prove that

$$\phi_{a_n X_n + b_n}(t) \rightarrow \phi_{aX + b}(t)$$

as $n \rightarrow \infty$. By definition,

$$\phi_{a_n X_n + b_n}(t) = \mathbb{E}[e^{it(a_n X_n + b_n)}] = \mathbb{E}[e^{it a_n X_n}] e^{it b_n},$$

and

$$\phi_{aX + b}(t) = \mathbb{E}[e^{it(aX + b)}] = \mathbb{E}[e^{it a X}] e^{it b}.$$

Since $b_n \rightarrow b$, we have $e^{it b_n} \rightarrow e^{it b}$. It remains to show that $\mathbb{E}[e^{it a_n X_n}] \rightarrow \mathbb{E}[e^{it a X}]$, or equivalently, $\phi_{X_n}(a_n t) \rightarrow \phi_X(at)$. We can write

$$|\phi_{X_n}(a_n t) - \phi_X(at)| \leq |\phi_{X_n}(a_n t) - \phi_X(a_n t)| + |\phi_X(a_n t) - \phi_X(at)|.$$

The second term tends to zero because ϕ_X is uniformly continuous and $a_n \rightarrow a$. Together with $X_n \Rightarrow X$, this proves $a_n X_n + b_n \Rightarrow aX + b$. □

SECTION 5.3

| The Central Limit Theorem

Identically Distributed Summands

THEOREM 5.21 (Central Limit Theorem).

Suppose that $\{x_n\}$ is an i.i.d sequence of random variables with mean μ and finite positive variance σ^2 . If $S_n = X_1 + \cdots + X_n$, then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \Rightarrow N.$$

Proof.

□

EXAMPLE 5.22.

LEMMA 5.23.

Let $z_1, \dots, z_m$ and $w_1, \dots, w_m$ be complex numbers of modulus at most 1. Then

$$|z_1 \cdots z_m - w_1 \cdots w_m| \leq \sum_{k=1}^m |z_k - w_k|.$$

EXAMPLE 5.24.

EXAMPLE 5.25 (Delta Method).

The Lindeberg and Lyapounov Theorems

DEFINITION 5.26 (Triangular Arrays).

Suppose that for each n , $X_{n1}, \dots, X_{nr_n}$ are independent, the probability space for the sequence may change with n . Such a collection is called a *triangular array* of random variables.

Remark.

EXAMPLE 5.27.

Put $S_n = X_{n1} + \cdots + X_{nr_n}$. Suppose that the means are 0 and the variances are finite¹, write

$$\mathbb{E}[X_{nk}] = 0, \quad \sigma_{nk}^2 = \mathbb{E}[X_{nk}^2] < \infty, \quad s_n^2 = \sum_{k=1}^{r_n} \sigma_{nk}^2. \quad (5.4)$$

Assume $s_n^2 > 0$ for large n . A successful remainder estimate is possible under the assumption of the *Lindeberg condition*:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^{r_n} \frac{1}{s_n^2} \int_{|X_{nk}| \geq \epsilon s_n} X_{nk}^2 d\mathbb{P} = 0, \quad \epsilon > 0. \quad (5.5)$$

¹ The assumption that X_{nk} has mean 0 entails no loss of generality.

THEOREM 5.28 (Lindeberg Theorem).

Suppose that for each $n \in \mathbb{N}$ the sequence $X_{n1}, \dots, X_{nr_n}$ is independent and satisfies (5.4). If (5.5) holds for all positive ϵ , then $\frac{S_n}{s_n} \Rightarrow N$.

Proof. □

Suppose that the $|X_{nk}|^{2+\delta}$ are integrable for some positive δ and that Lyapunov's condition

$$\lim_n \sum_{k=1}^{r_n} \frac{1}{s_n^{2+\delta}} \mathbb{E}[|X_{nk}|^{2+\delta}] = 0 \quad (5.6)$$

holds. Then Lindeberg's condition follows because the sum in (5.5) is bounded by

$$\sum_{k=1}^{r_n} \frac{1}{s_n^2} \int_{|X_{nk}| \geq \epsilon s_n} \frac{|X_{nk}|^{2+\delta}}{\epsilon^\delta s_n^\delta} d\mathbb{P} \leq \frac{1}{\epsilon^\delta} \sum_{k=1}^{r_n} \frac{1}{s_n^{2+\delta}} \mathbb{E}[|X_{nk}|^{2+\delta}].$$

COROLLARY 5.29 (Lyapunov's Theorem).

Suppose that for each n the sequence $X_{n1}, \dots, X_{nr_n}$ is independent and satisfies (5.4). If (5.6) holds for some positive δ , then $\frac{S_n}{s_n} \Rightarrow N$.

EXAMPLE 5.30.

Exercises

EXERCISE 5.10. If $\{X_n\}$ is independent and the X_n all have the same distribution with finite first moment, then $n^{-1}S_n \rightarrow E[X_1]$ with probability 1 (Theorem 22.1), so that $n^{-1}S_n \Rightarrow E[X_1]$. Prove the latter fact by characteristic functions. *Hint: Use (27.5).*

EXERCISE 5.11. For a Poisson variable Y_λ with mean λ , show that $(Y_\lambda - \lambda)/\sqrt{\lambda} \Rightarrow N$ as $\lambda \rightarrow \infty$. Show that (22.3) fails for $t = 1$.

EXERCISE 5.12. Suppose that $|X_{nk}| \leq M_n$ with probability 1 and $M_n/s_n \rightarrow 0$. Verify Lyapunov's condition and then Lindeberg's condition.

EXERCISE 5.13. Suppose that the random variables in any single row of the triangular array are identically distributed. To what do Lindeberg's and Lyapunov's conditions reduce?

EXERCISE 5.14. Construct an example where Lindeberg's condition holds but Lyapunov's does not.

Derivatives and Conditional Probability

SECTION 6.1

| The Radon-Nikodym Theorem

Absolute Continuity and Singularity

DEFINITION 6.1 (Absolute Continuity).

The measure ν is *absolutely continuous* with respect to μ if for each A in $\mathcal{F}$, $\mu(A) = 0$ implies $\nu(A) = 0$.

Remark. In this case ν is also said to be dominated by μ , and the relation is indicated by $\nu \ll \mu$. If $\nu \ll \mu$ and $\mu \ll \nu$, the measures are equivalent, indicated by $\nu \equiv \mu$.

The Radon-Nikodym Theorem

If $\nu(A) = \int_A f d\mu$, then certainly $\nu \ll \mu$. The Radon-Nikodym theorem goes in the opposite direction.

THEOREM 6.2 (Radon-Nikodym Theorem).

If μ and ν are σ -finite measures such that $\nu \ll \mu$, then there exists a non-negative f , a density, such that $\nu(A) = \int_A f d\mu$ for all $A \in \mathcal{F}$. For two such densities f and g , $\mu(f \neq g) = 0$.

Proof. The uniqueness of the density up to sets of μ -measure 0 is settled by [Theorem 3.12](#). It is only the existence that must be proved. $\square$

SECTION 6.2

| Conditional Probability

The Discrete Case

Consider first the conditional probability of a set A with respect to another set B . It is defined of course by

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

unless $\mathbb{P}(B)$ vanishes, in which case it is not defined at all. Let

$$f(\omega) = \begin{cases} \mathbb{P}(A | B), & \omega \in B \\ \mathbb{P}(A | B^c), & \omega \in B^c \end{cases}.$$

Let $B_1, B_2, \dots$ be a finite or countable partition of Ω into $\mathcal{F}$ -sets, and let $\mathcal{G}$ consist of all the unions of the B_i . Then $\mathcal{G}$ is the σ -field generated by the B_i . For A in $\mathcal{F}$, consider the function with values

$$f(\omega) = \mathbb{P}(A | B_i) = \frac{\mathbb{P}(A \cap B_i)}{\mathbb{P}(B_i)}, \quad \omega \in B_i, \quad i = 1, 2, \dots,$$

which is called the *conditional probability of A given $\mathcal{G}$* and is denoted $\mathbb{P}(A | \mathcal{G})$. This is written $\mathbb{P}(A | \mathcal{G})_\omega$ whenever the argument ω needs to be explicitly shown.

The General Case

Fix an A in $\mathcal{F}$ and define a finite measure ν on $\mathcal{G}$ by

$$\nu(G) = \mathbb{P}(A \cap G), \quad G \in \mathcal{G}.$$

Then $\mathbb{P}(G) = 0$ implies that $\nu(G) = 0$. The *Radon-Nikodym theorem* can be applied to the measures ν and $\mathbb{P}$ on the measurable space $(\Omega, \mathcal{G})$ because the first one is absolutely continuous with respect to the second. It follows that there exists a function or random variable f , measurable $\mathcal{G}$ and integrable with respect to $\mathbb{P}$, such that

$$\mathbb{P}(A \cap G) = \nu(G) = \int_G f d\mathbb{P}, \quad \forall G \in \mathcal{G}.$$

Denote this function f by $\mathbb{P}(A | \mathcal{G})$.

PROPOSITION 6.3. (i) $\mathbb{P}(A | \mathcal{G})$ is $\mathcal{G}$ -measurable and integrable.

(ii) $\mathbb{P}(A | \mathcal{G})$ satisfies the functional equation

$$\int_G \mathbb{P}(A | \mathcal{G}) d\mathbb{P} = \mathbb{P}(A \cap G), \quad G \in \mathcal{G}.$$

Remark. Condition (ii) has a gambling interpretation. Suppose that the observer, after he has learned the outcome of $\mathcal{G}$, is offered the opportunity to bet on the event A (unless A lies in $\mathcal{G}$, he does not yet know whether or not it occurred). He is required to pay an entry fee of $\mathbb{P}(A | \mathcal{G})$ units and will win 1 unit if A occurs and nothing otherwise. If the observer decides to bet and pays his fee, he gains $\mathbb{1}_A - \mathbb{P}(A | \mathcal{G})$.

If he declines to bet, his gain is of course 0. Suppose that he adopts the strategy of betting if G occurs but not otherwise, where G is some set in $\mathcal{G}$. He can actually carry out this strategy, since after learning the outcome of the experiment $\mathcal{G}$ he knows whether or not G occurred. His expected gain with this strategy is his gain integrated over G :

$$\int_G (\mathbb{1}_A - \mathbb{P}(A \mid \mathcal{G})) d\mathbb{P} = \mathbb{P}(A \cap G) - \int_G \mathbb{P}(A \mid \mathcal{G}) d\mathbb{P} = 0.$$

Condition (ii) requires then that each strategy be fair in the sense that the observer stands neither to win nor to lose on the average.

EXAMPLE 6.4. (i) Suppose that $A \in \mathcal{G}$, which will always hold if $\mathcal{G}$ coincides with the whole σ -field $\mathcal{F}$. Then $\mathbb{P}(A \mid \mathcal{G}) = \mathbb{1}_A$ with probability 1. To know the outcome of $\mathcal{G}$ viewed as an experiment is in particular to know whether or not A has occurred.

- (ii) If $\mathcal{G}$ is $\{\emptyset, \Omega\}$, the smallest possible σ -field, every $\mathcal{G}$ -measurable function must be constant. Therefore, $\mathbb{P}(A \mid \mathcal{G})_\omega = \mathbb{P}(A)$ for all ω in this case.
- (iii) Let Ω be the plane $\mathbb{R}^2$ and let $\mathcal{F}$ be the class $\mathbb{R}^2$ of planar Borel sets. A point of Ω is a pair (x, y) of reals. Let $\mathcal{G}$ be the σ -field consisting of the vertical strips, the product sets

$$E \times \mathbb{R}^1 = ((x, y) : x \in E),$$

where E is a linear Borel set. If the observer knows for each strip $E \times \mathbb{R}^1$ whether or not it contains (x, y) , then, as he knows this for each one-point set E , he knows the value of x . Thus the experiment $\mathcal{G}$ consists in the determination of the first coordinate of the sample point. Suppose now that $\mathbb{P}$ is a probability measure on $\mathbb{R}^2$ having a density $f(x, y)$ with respect to planar Lebesgue measure:

$$\mathbb{P}(A) = \iint_A f(x, y) dx dy.$$

Let A be a horizontal strip

$$\mathbb{R}^1 \times F = ((x, y) : y \in F),$$

F being a linear Borel set. Put

$$\varphi(x, y) = \frac{\int_F f(x, t) dt}{\int_{\mathbb{R}^1} f(x, t) dt}. \quad (6.1)$$

Set $\varphi(x, y) = 0$, say, at points where the denominator here vanishes; these points form a set of $\mathbb{P}$ -measure 0. Since $\varphi(x, y)$ is a function of x alone, it is measurable $\mathcal{G}$. The general element of $\mathcal{G}$ being $E \times \mathbb{R}^1$, it will follow that φ is a version of $\mathbb{P}(A \mid \mathcal{G})$ if it is shown that

$$\int_{E \times \mathbb{R}^1} \varphi(x, y) d\mathbb{P}(x, y) = \mathbb{P}(A \cap (E \times \mathbb{R}^1)).$$

Since $A = \mathbb{R}^1 \times F$, the right side here is $\mathbb{P}(E \times F)$. Since $\mathbb{P}$ has density f , the change-of-measure formula for densities and Fubini's theorem reduce the left side to

$$\begin{aligned} \int_E \int_{\mathbb{R}^1} \varphi(x, y) f(x, y) dy dx &= \int_E \int_F f(x, t) dt dx \\ &= \iint_{E \times F} f(x, y) dx dy = \mathbb{P}(E \times F). \end{aligned}$$

Thus (6.1) does give a version of $\mathbb{P}(\mathbb{R}^1 \times F \mid \mathcal{G})$.

(iv) The set A is by definition independent of the σ -field $\mathcal{G}$ if it is independent of each G in $\mathcal{G}$:

$$\mathbb{P}(A \cap G) = \mathbb{P}(A)\mathbb{P}(G), \quad \forall G \in \mathcal{G}.$$

This being the same thing as

$$\mathbb{P}(A \cap G) = \int_G \mathbb{P}(A) d\mathbb{P},$$

A is independent of $\mathcal{G}$ if and only if $\mathbb{P}(A \mid \mathcal{G}) = \mathbb{P}(A)$ with probability 1.

Remark. The σ -field $\sigma(X)$ generated by a random variable X consists of the sets $\{\omega : X(\omega) \in H\}$ for $H \in \mathbb{R}$. The conditional probability of A given X is defined as $\mathbb{P}(A \mid \sigma(X))$ and is denoted $\mathbb{P}(A \mid X)$. Thus $\mathbb{P}(A \mid X) = \mathbb{P}(A \mid \sigma(X))$ by definition.

THEOREM 6.5.

Let $\mathcal{P}$ be a π -system generating the σ -field $\mathcal{G}$, and suppose that Ω is a finite or countable union of sets in $\mathcal{P}$. An integrable function f is a version of $\mathbb{P}(A \mid \mathcal{G})$ if it is measurable $\mathcal{G}$ and if

$$\int_G f d\mathbb{P} = \mathbb{P}(A \cap G) \quad (6.2)$$

holds for all G in $\mathcal{P}$.

EXAMPLE 6.6. Suppose that X and Y are independent random variables with a common distribution function F that is positive and continuous. What is the conditional probability of $X \leq x$ given the random variable $M = \max\{X, Y\}$?

Proof. As it should clearly be 1 if $M \leq x$, suppose that $M > x$. Since $X \leq x$ requires $M = Y$, the chance of which is $\frac{1}{2}$ by symmetry, the conditional probability of $X \leq x$ should by independence be

$$\frac{1}{2} \cdot \frac{F(x)}{F(m)} = \frac{1}{2} \mathbb{P}(X \leq x \mid X \leq m)$$

with the random variable M substituted for m . Intuition thus gives

$$\mathbb{P}(X \leq x \mid M)_\omega = \mathbf{1}_{\{M \leq x\}}(\omega) + \frac{1}{2} \mathbf{1}_{\{M > x\}}(\omega) \frac{F(x)}{F(M(\omega))}.$$

It suffices to check (6.2) for sets $G = \{M \leq m\}$, because these form a π -system generating $\sigma(M)$. The functional equation reduces to

$$\mathbb{P}(M \leq \min\{x, m\}) + \frac{1}{2} \int_{x < M \leq m} \frac{F(x)}{F(M)} d\mathbb{P} = \mathbb{P}(M \leq m, X \leq x). \quad (6.3)$$

Since the other case is easy, suppose that $x < m$. Since the distribution of (X, Y) is product measure, it follows by *Fubini's theorem* and the assumed continuity of F that

$$\int_{x < M \leq m} \frac{1}{F(M)} d\mathbb{P} = \iint_{\substack{u \leq v \\ x < v \leq m}} \frac{1}{F(v)} dF(u) dF(v) + \iint_{\substack{v < u \\ x < u \leq m}} \frac{1}{F(u)} dF(u) dF(v) = 2(F(m) - F(x)),$$

which gives (6.3). $\square$

Properties of Conditional Probability

PROPOSITION 6.7.

With probability 1, $\mathbb{P}(\emptyset \mid \mathcal{G}) = 0$, $\mathbb{P}(\Omega \mid \mathcal{G}) = 1$; and $0 \leq \mathbb{P}(A \mid \mathcal{G}) \leq 1$ for each A . If $A_1, A_2, \dots$ is a

finite or countable sequence of disjoint sets, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n \mid \mathcal{G}\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n \mid \mathcal{G})$$

with probability 1.

Proof. If the A_n are disjoint and if G lies in $\mathcal{G}$, it follows from *Theorem 3.6* that

$$\begin{aligned} \int_G \sum_n \mathbb{P}(A_n \mid \mathcal{G}) d\mathbb{P} &= \sum_n \int_G \mathbb{P}(A_n \mid \mathcal{G}) d\mathbb{P} \\ &= \sum_n \mathbb{P}(A_n \cap G) = \mathbb{P}\left(\left(\bigcup_n A_n\right) \cap G\right). \end{aligned}$$

Thus $\sum_n \mathbb{P}(A_n \mid \mathcal{G})$, which is certainly measurable $\mathcal{G}$, satisfies the functional equation for $\mathbb{P}(\bigcup_n A_n \mid \mathcal{G})$, and so must coincide with it except perhaps on a set of $\mathbb{P}$ -measure 0. $\square$

Remark. Additional useful facts can be established by similar arguments. If $A \subset B$, then

$$\mathbb{P}(B - A \mid \mathcal{G}) = \mathbb{P}(B \mid \mathcal{G}) - \mathbb{P}(A \mid \mathcal{G}), \quad \mathbb{P}(A \mid \mathcal{G}) \leq \mathbb{P}(B \mid \mathcal{G}).$$

If $A_n \uparrow A$, then $\mathbb{P}(A_n \mid \mathcal{G}) \uparrow \mathbb{P}(A \mid \mathcal{G})$, and if $A_n \downarrow A$, then $\mathbb{P}(A_n \mid \mathcal{G}) \downarrow \mathbb{P}(A \mid \mathcal{G})$. Further, $\mathbb{P}(A) = 1$ implies that $\mathbb{P}(A \mid \mathcal{G}) = 1$, and $\mathbb{P}(A) = 0$ implies that $\mathbb{P}(A \mid \mathcal{G}) = 0$.

| Exercises

EXERCISE 6.1.

EXERCISE 6.2.

EXERCISE 6.3.

EXERCISE 6.4.

SECTION 6.3

| Conditional Expectation

DEFINITION 6.8 (Conditional Expectation).

Suppose that X is an *integrable* random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ and that $\mathcal{G}$ is a σ -algebra in $\mathcal{F}$. There exists a random variable $\mathbb{E}[X | \mathcal{G}]$, called the *conditional expected value* of X given $\mathcal{G}$, having these two properties:

- (i) $\mathbb{E}[X | \mathcal{G}]$ is $\mathcal{G}$ -measurable and integrable.
- (ii) For every $G \in \mathcal{G}$,

$$\int_G \mathbb{E}[X | \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P}. \quad (6.4)$$

Remark. (i) The only requirement on X is integrability, it does not have to be $\mathcal{G}$ -measurable in general. If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X | \mathcal{G}] = X$ a.s.

(ii) $\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X]$ by taking $G = \Omega$ in (ii).

(iii) This definition can be related to the definition of conditional probability by taking $X = \mathbb{1}_A$ for $A \in \mathcal{F}$:

$$\int_G \mathbb{E}[\mathbb{1}_A | \mathcal{G}] d\mathbb{P} = \int_G \mathbb{1}_A d\mathbb{P} = \mathbb{P}(A \cap G),$$

i.e., $\mathbb{P}(A | \mathcal{G}) := \mathbb{E}[\mathbb{1}_A | \mathcal{G}]$.

EXAMPLE 6.9. Suppose that $B_1, B_2, \dots$ is a finite or countable partition of Ω generating the σ -field $\mathcal{G}$. Then $\mathbb{E}[X | \mathcal{G}]$ must, since it is measurable $\mathcal{G}$, have some constant value over B_i , say α_i . Then (6.4) for $G = B_i$ gives

$$\alpha_i \mathbb{P}(B_i) = \int_{B_i} X d\mathbb{P}.$$

Thus

$$\mathbb{E}[X | \mathcal{G}]_\omega = \frac{1}{\mathbb{P}(B_i)} \int_{B_i} X d\mathbb{P}, \quad \omega \in B_i,$$

where $\mathbb{P}(B_i) > 0$. If $\mathbb{P}(B_i) = 0$, the value of $\mathbb{E}[X | \mathcal{G}]$ over B_i is constant but arbitrary.

Properties of Conditional Expectation

THEOREM 6.10.

Let $\mathcal{P}$ be a π -system generating the σ -field $\mathcal{G}$, and suppose that Ω is a finite or countable union of sets in $\mathcal{P}$. An integrable function f is a version of $\mathbb{E}[X | \mathcal{G}]$ if it is $\mathcal{G}$ -measurable and if

$$\int_G f d\mathbb{P} = \int_G X d\mathbb{P}$$

holds for all G in $\mathcal{P}$.

PROPOSITION 6.11.

Suppose that X, Y, X_n are integrable.

- (i) If $X = a$ with probability 1, then $\mathbb{E}[X | \mathcal{G}] = a$.

- (ii) For constants a and b , $\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}]$.
- (iii) If $X \leq Y$ with probability 1, then $\mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}]$.
- (iv) $|\mathbb{E}[X \mid \mathcal{G}]| \leq \mathbb{E}[|X| \mid \mathcal{G}]$.
- (v) If $\lim_n X_n = X$ with probability 1, $|X_n| \leq Y$, and Y is integrable, then $\lim_n \mathbb{E}[X_n \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{G}]$ with probability 1.

THEOREM 6.12.

If X is $\mathcal{G}$ -measurable, and if Y and XY are integrable, then

$$\mathbb{E}[XY \mid \mathcal{G}] = X\mathbb{E}[Y \mid \mathcal{G}] \quad (6.5)$$

with probability 1.

Proof. It will be shown first that the right side of (6.5) is a version of the left side if $X = \mathbb{1}_{G_0}$ and $G_0 \in \mathcal{G}$. Since $\mathbb{1}_{G_0}\mathbb{E}[Y \mid \mathcal{G}]$ is certainly $\mathcal{G}$ -measurable, it suffices to show that it satisfies the functional equation

$$\int_G \mathbb{1}_{G_0}\mathbb{E}[Y \mid \mathcal{G}]d\mathbb{P} = \int_G \mathbb{1}_{G_0}Yd\mathbb{P}, \quad G \in \mathcal{G}.$$

But this reduces to

$$\int_{G \cap G_0} \mathbb{E}[Y \mid \mathcal{G}]d\mathbb{P} = \int_{G \cap G_0} Yd\mathbb{P},$$

which holds by the definition of $\mathbb{E}[Y \mid \mathcal{G}]$. Thus (6.5) holds if X is the indicator of an element of $\mathcal{G}$.

It follows by Proposition 6.11(ii) that (6.5) holds if X is a $\mathcal{G}$ -measurable simple function. For the general X that is $\mathcal{G}$ -measurable, there exist $\mathcal{G}$ -measurable simple functions X_n such that $|X_n| \leq |X|$ and $\lim_n X_n = X$ (Theorem 2.10). Since $|X_n Y| \leq |XY|$ and $|XY|$ is integrable, Proposition 6.11(v) implies that $\lim_n \mathbb{E}[X_n Y \mid \mathcal{G}] = \mathbb{E}[XY \mid \mathcal{G}]$ with probability 1. But $\mathbb{E}[X_n Y \mid \mathcal{G}] = X_n \mathbb{E}[Y \mid \mathcal{G}]$ by the case already treated, and of course $\lim_n X_n \mathbb{E}[Y \mid \mathcal{G}] = X \mathbb{E}[Y \mid \mathcal{G}]$.¹ Thus (6.5) holds in general. Notice that X has not been assumed integrable. $\square$

THEOREM 6.13 (Law of Total Expectation).

If X is integrable and the σ -fields $\mathcal{G}_1$ and $\mathcal{G}_2$ satisfy $\mathcal{G}_1 \subset \mathcal{G}_2$, then

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}_2] \mid \mathcal{G}_1] = \mathbb{E}[X \mid \mathcal{G}_1] \quad (6.6)$$

with probability 1.

Proof. The left side of (6.6) is $\mathcal{G}_1$ -measurable, and so to prove that it is a version of $\mathbb{E}[X \mid \mathcal{G}_1]$. If $G \in \mathcal{G}_1$, then $G \in \mathcal{G}_2$, then

$$\int_G \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}_2] \mid \mathcal{G}_1]d\mathbb{P} = \int_G \mathbb{E}[X \mid \mathcal{G}_2]d\mathbb{P} = \int_G Xd\mathbb{P}.$$

$\square$

Remark.

LEMMA 6.14 (Fatou's Lemma for Conditional Expectations).

Let $X_1, X_2, \dots$ be non-negative integrable random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ -algebra. Then

$$\mathbb{E}\left[\liminf_{n \rightarrow \infty} X_n \mid \mathcal{G}\right] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n \mid \mathcal{G}].$$

¹ Note that $|X_n \mathbb{E}[Y \mid \mathcal{G}]| = |\mathbb{E}[X_n Y \mid \mathcal{G}]| \leq \mathbb{E}[|X_n Y| \mid \mathcal{G}] \leq \mathbb{E}[|XY| \mid \mathcal{G}]$, so that the limit $X \mathbb{E}[Y \mid \mathcal{G}]$ is integrable.

Proof. For each k , let $Y_k = \inf_{n \geq k} X_n$. Then $0 \leq Y_k \uparrow \liminf_{n \rightarrow \infty} X_n$. By monotonicity of conditional expectation,

$$\mathbb{E}[Y_k | \mathcal{G}] \leq \inf_{n \geq k} \mathbb{E}[X_n | \mathcal{G}].$$

Taking $k \rightarrow \infty$ and applying the monotone convergence theorem for conditional expectations gives

$$\mathbb{E}\left[\liminf_{n \rightarrow \infty} X_n | \mathcal{G}\right] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n | \mathcal{G}].$$

□

| Exercises

EXERCISE 6.5. In the context of [Example 6.4](#), show that the conditional expected value of Y (if it is integrable) given X is $g(X)$, where

$$g(x) = \frac{\int_{-\infty}^{\infty} f(x, y) y dy}{\int_{-\infty}^{\infty} f(x, y) dy}.$$

EXERCISE 6.6. Show that the independence of X and Y implies that $\mathbb{E}[Y | X] = \mathbb{E}[Y]$, which in turn implies that $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. Show by examples in an Ω of three points that the reverse implications are both false.

SOLUTION. (i) Since $\mathbb{E}[Y | X]$ is constant, it is measurable with respect to any σ -field, including $\sigma(X)$.

Let $G \in \sigma(X)$, we compute

$$\int_G Y d\mathbb{P} = \mathbb{E}[Y \mathbf{1}_G] = \mathbb{E}[Y] \mathbb{E}[\mathbf{1}_G] = \mathbb{E}[Y] \mathbb{P}(G).$$

Because X and Y are independent, we have

$$\int_G \mathbb{E}[Y | X] d\mathbb{P} = \int_G \mathbb{E}[Y] d\mathbb{P} = \mathbb{E}[Y] \mathbb{P}(G).$$

Therefore, $\mathbb{E}[Y | X] = \mathbb{E}[Y]$ *a.s.*

(ii)

(iii)

□

EXERCISE 6.7.

EXERCISE 6.8. The equation (34.5) was proved by showing that the left side is a version of the right side. Prove it by showing that the right side is a version of the left side.

EXERCISE 6.9. Prove for bounded X and Y that

$$E[YE[X | \mathcal{G}] | \mathcal{G}] = E[XE[Y | \mathcal{G}] | \mathcal{G}].$$

EXERCISE 6.10. (i) Show for non-negative X that $E[X | \mathcal{G}] = \int_0^\infty P[X > t | \mathcal{G}]dt$ with probability 1.

(ii) Generalise Markov's inequality: $P(|X| \geq \alpha | \mathcal{G}) \leq \alpha^{-k} E[|X|^k | \mathcal{G}]$ with probability 1.

EXERCISE 6.11. Define $\text{Var}[X | \mathcal{G}] = E[(X - E[X | \mathcal{G}])^2 | \mathcal{G}]$. Prove that

$$\text{Var}[X] = E[\text{Var}[X | \mathcal{G}]] + \text{Var}[E[X | \mathcal{G}]].$$

SECTION 6.4

| Martingales

DEFINITION 6.15 (Martingales).

Let $X_1, X_2, \dots$ be a sequence of random variables on a probability space $(\Omega, \mathcal{F}, P)$, and let $\mathcal{F}_1, \mathcal{F}_2, \dots$ be a sequence of σ -algebras in $\mathcal{F}$. The sequence $(X_n, \mathcal{F}_n)$ is a *martingale* if these four conditions hold:

- (i) $\mathcal{F}_n \subset \mathcal{F}_{n+1}$.
- (ii) X_n is $\mathcal{F}_n$ -measurable.
- (iii) $\mathbb{E}[|X_n|] < \infty$.
- (iv) $\mathbb{E}[X_{n+1} | \mathcal{F}_n] = X_n$ with probability 1.

Remark. If σ -fields $\mathcal{G}_n = \sigma(X_1, \dots, X_n)$, then

$$\mathbb{E}[X_{n+1} | \mathcal{G}_n] = \mathbb{E}[\mathbb{E}[X_{n+1} | \mathcal{F}_n] | \mathcal{G}_n] = \mathbb{E}[X_n | \mathcal{G}_n] = X_n$$

by (6.6). It reduces to

$$\mathbb{E}[X_{n+1} | X_1, \dots, X_n] = X_n.$$

EXAMPLE 6.16. (i) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let ν be a finite measure on $\mathcal{F}$, and let $\mathcal{F}_1, \mathcal{F}_2, \dots$ be a non-decreasing sequence of σ -fields in $\mathcal{F}$. Suppose that $A \in \mathcal{F}_n$ and $\mathbb{P}(A) = 0$ together imply that $\nu(A) = 0$. There is then a density or Radon-Nikodym derivative X_n of ν with respect to $\mathbb{P}$ satisfies

$$\int_A X_n d\mathbb{P} = \nu(A), \quad A \in \mathcal{F}_n.$$

If $A \in \mathcal{F}_n$, then $A \in \mathcal{F}_{n+1}$ as well, so that $\int_A X_{n+1} d\mathbb{P} = \nu(A)$. Thus the X_n are a martingale with respect to the $\mathcal{F}_n$.

- (ii) Suppose that Z is an integrable random variable on $(\Omega, \mathcal{F}, P)$ and that $\mathcal{F}_n$ are non-decreasing σ -fields in $\mathcal{F}$. If $X_n = \mathbb{E}[Z | \mathcal{F}_n]$, then the first three conditions in the martingale definition are satisfied, and by (6.6),

$$\mathbb{E}[X_{n+1} | \mathcal{F}_n] = \mathbb{E}[\mathbb{E}[Z | \mathcal{F}_{n+1}] | \mathcal{F}_n] = \mathbb{E}[Z | \mathcal{F}_n] = X_n.$$

Thus X_n is a martingale relative to $\mathcal{F}_n$.

- (iii) Let N_{nk} , $n, k = 1, 2, \dots$, be a *i.i.d.* random variables assuming the values $0, 1, 2, \dots$. Define $Z_0, Z_1, Z_2, \dots$ inductively by $Z_0(\omega) = 1$ and

$$Z_n(\omega) = N_{n,1}(\omega) + \dots + N_{n,Z_{n-1}(\omega)}(\omega),$$

where $Z_n(\omega) = 0$ if $Z_{n-1}(\omega) = 0$. If N_{nk} is thought of as the number of progeny of an organism, and if Z_{n-1} represents the size at time $n-1$ of a population of these organisms, then Z_n represents the size at time n . If the expected number of progeny is $\mathbb{E}[N_{nk}] = m$, then

$$\mathbb{E}[Z_n | Z_{n-1}] = Z_{n-1}m,$$

so that $X_n = \frac{Z_n}{m^n}$, $n = 0, 1, 2, \dots$, is a martingale. The sequence $Z_0, Z_1, \dots$ is a *branching process*.

(iv) Let $X_0 = 0$ and

$$X_{n+1} = \begin{cases} X_n + 1, & \frac{1}{2} \\ X_n - 1, & \frac{1}{2} \end{cases}.$$

We claim this is a martingale with respect to the natural filtration $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Let $x = \mathbb{P}(\text{reach 1 at some time})$, then consider the first step starting from $X_0 = 0$,

$$x = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot x^2$$

solving for x gives $x = 1$. So the random walk will reach 1 at some time *a.s.* However, consider the expected time for the walk to reach 1. Let e be the expected number of steps to reach 1 starting from 0. Then, by similar reasoning,

$$e = \frac{1}{2} \cdot 1 + \frac{1}{2}(1 + 2e) \implies e = 1 + e,$$

which is a contradiction unless $e = \infty$.

Submartingales

DEFINITION 6.17 (Submartingales).

Random variables X_n are a *submartingale* relative to σ -fields $\mathcal{F}_n$ if (i), (ii), and (iii) of the definition above hold and if

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n$$

holds in place of (iv).

Remark. The requirement is the same thing as

$$\int_A X_{n+1} d\mathbb{P} \geq \int_A X_n d\mathbb{P}, \quad A \in \mathcal{F}_n.$$

This extends inductively, and so

$$\mathbb{E}[X_{n+k} \mid \mathcal{F}_n] \geq X_n$$

for $k \geq 1$.

Functions of Martingales

THEOREM 6.18. (i) If $X_1, X_2, \dots$ is a martingale relative to $\mathcal{F}_1, \mathcal{F}_2, \dots$, if φ convex, and if $\varphi(X_n)$ are integrable, then $\varphi(X_1), \varphi(X_2), \dots$ is a martingale relative to $\mathcal{F}_1, \mathcal{F}_2, \dots$.

(ii) If $X_1, X_2, \dots$ is a submartingale relative to $\mathcal{F}_1, \mathcal{F}_2, \dots$, the same holds providing φ is non-decreasing.

Proof.

□

Stopping Times

DEFINITION 6.19.

Let τ be a random variable taking as values positive integers or the special value ∞ . It is a *stopping time* with respect to $\{\mathcal{F}_n\}$ if $\{\tau = k\} \in \mathcal{F}_k$ for each finite k . Then we define the

stopping sequence as

$$X_n^* = \begin{cases} X_n, & n \leq \tau \\ X_\tau, & n \geq \tau \end{cases},$$

where $X_\tau = X_{\tau(\omega)}(\omega)$.

PROPOSITION 6.20.

X_n^* is a martingale.

Proof. Need to show that for all n ,

$$\mathbb{E}[X_{n+1}^* | \mathcal{F}_n] = X_n^*$$

i.e., for any $G \in \mathcal{F}_n$,

$$\int_G X_{n+1}^* d\mathbb{P} = \int_G X_n^* d\mathbb{P}$$

Note that

$$X_{n+1}^* = X_{\min(n+1, \tau)} = X_{n+1} \mathbb{1}_{\{\tau > n\}} + X_\tau \mathbb{1}_{\{\tau \leq n\}}.$$

Since this a martingale, we know $\mathbb{E}[X_n^*] = \mathbb{E}[X_0^*]$. □

Define

$$\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \leq k\} \in \mathcal{F}_k, 1 \leq k < \infty\}.$$

This is a σ -field. Suppose now that τ_1 and τ_2 are two stopping times and $\tau_1 \leq \tau_2$. If $A \in \mathcal{F}_{\tau_1}$, then $A \cap \{\tau_1 \leq k\} \in \mathcal{F}_k$ and hence

$$A \cap \{\tau_2 \leq k\} = A \cap \{\tau_1 \leq k\} \cap \{\tau_2 \leq k\} \in \mathcal{F}_k \implies \mathcal{F}_{\tau_1} \subset \mathcal{F}_{\tau_2}.$$

THEOREM 6.21 (Optional Sampling Theorem).

If $X_1, \dots, X_n$ is a submartingale with respect to $\mathcal{F}_1, \dots, \mathcal{F}_n$ and τ_1, τ_2 are stopping times satisfying $1 \leq \tau_1 \leq \tau_2 \leq n$, then X_{τ_1}, X_{τ_2} is a submartingale with respect to $\mathcal{F}_{\tau_1}, \mathcal{F}_{\tau_2}$, and $\mathbb{E}[X_{\tau_2} | \mathcal{F}_{\tau_1}] \geq X_{\tau_1}$.

Proof. Since the X_{τ_i} are dominated by $\sum_{k=1}^n |X_k|$, they are integrable. It is required to show that $\mathbb{E}[X_{\tau_2} | \mathcal{F}_{\tau_1}] \geq X_{\tau_1}$, or

$$\int_A (X_{\tau_2} - X_{\tau_1}) d\mathbb{P} \geq 0, \quad A \in \mathcal{F}_{\tau_1}.$$

But $A \in \mathcal{F}_{\tau_1}$ implies that

$$A \cap \{\tau_1 < k \leq \tau_2\} = (A \cap \{\tau_1 \leq k-1\}) \cap \{\tau_2 \leq k-1\}^c$$

lies in $\mathcal{F}_{k-1}$. If $\Delta_k = X_k - X_{k-1}$, then

$$\int_A (X_{\tau_2} - X_{\tau_1}) d\mathbb{P} = \int_A \sum_{k=1}^n \mathbb{1}_{\{\tau_1 < k \leq \tau_2\}} \Delta_k d\mathbb{P} = \sum_{k=1}^n \int_{A \cap \{\tau_1 < k \leq \tau_2\}} \Delta_k d\mathbb{P} \geq 0$$

by the submartingale property. □

Inequalities

THEOREM 6.22 (Kolmogorov's Inequality).

If $X_1, \dots, X_n$ is a submartingale, then for $\alpha > 0$,

$$\mathbb{P}\left(\max_{1 \leq i \leq n} X_i \geq \alpha\right) \leq \frac{1}{\alpha} \mathbb{E}[|X_n|].$$

Proof.

□

THEOREM 6.23 (Upcrossing Lemma).

For a submartingale $X_1, \dots, X_n$, the number U of upcrossings of $[\alpha, \beta]$ satisfies

$$\mathbb{E}[U] \leq \frac{\mathbb{E}[|X_n|] + |\alpha|}{\beta - \alpha}.$$

Convergence Theorems

THEOREM 6.24 (Martingale Convergence Theorem).

Let $X_1, X_2, \dots$ be a submartingale. If $K = \sup_n \mathbb{E}[|X_n|] < \infty$, then $X_n \rightarrow X$ with probability 1, where X is a random variable satisfying $\mathbb{E}[|X|] \leq K$.

Proof.

□

COROLLARY 6.25.

If the X_n form a non-negative martingale, then $\mathbb{E}[|X_n|] = \mathbb{E}[X_n] = \mathbb{E}[X_1]$, and K is necessarily finite.

EXAMPLE 6.26. The X_n in [Example 6.16\(iii\)](#) are non-negative, and so $X_n = \frac{Z_n}{m^n} \rightarrow X$, where X is non-negative and integrable. If $m < 1$, then, since Z_n is an integer, $Z_n = 0$ for large n , and the population dies out. In this case, $X = 0$ with probability 1. Since $\mathbb{E}[X_n] = \mathbb{E}[X_0] = 1$, this shows that $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$ may fail in [Theorem 6.24](#).

Exercises

EXERCISE 6.12. Suppose that $\Delta_1, \Delta_2, \dots$ are independent random variables with mean 0. Let

$$X_1 = \Delta_1, \quad X_{n+1} = X_n + \Delta_{n+1} f_n(X_1, \dots, X_n),$$

and suppose that the X_n are integrable. Show that (X_n) is a martingale. The martingales of gambling have this form.

EXERCISE 6.13. Let $Y_1, Y_2, \dots$ be independent random variables with mean 0 and variance σ^2 . Let

$$X_n = \left(\sum_{k=1}^n Y_k \right)^2 - n\sigma^2.$$

Show that (X_n) is a martingale.

EXERCISE 6.14. Suppose that (Y_n) is a finite-state Markov chain with transition matrix p_{ij} . Suppose that $\sum_j p_{ij}x(j) = \lambda x(i)$ for all i .² Put $X_n = \lambda^{-n}x(Y_n)$ and show that (X_n) is a martingale.

² The $x(i)$ are the components of a right eigenvector of the transition matrix.

EXERCISE 6.15. Suppose that $Y_1, Y_2, \dots$ are independent, positive random variables and that $\mathbb{E}Y_n = 1$. Put $X_n = Y_1 \cdots Y_n$.

- (i) Show that (X_n) is a martingale and converges with probability 1 to an integrable X .
- (ii) Suppose specifically that Y_n assumes the values $\frac{1}{2}$ and $\frac{3}{2}$ with probability $\frac{1}{2}$ each. Show that $X = 0$ with probability 1. This gives an example where $\mathbb{E}\left[\prod_{n=1}^{\infty} Y_n\right] \neq \prod_{n=1}^{\infty} \mathbb{E}Y_n$ for independent, integrable, positive random variables. Show, however, that $\mathbb{E}\left[\prod_{n=1}^{\infty} Y_n\right] \leq \prod_{n=1}^{\infty} \mathbb{E}Y_n$ always holds.

EXERCISE 6.16. Show that a submartingale (X_n) can be represented as $X_n = Y_n + Z_n$, where (Y_n) is a martingale and $0 \leq Z_1 \leq Z_2 \leq \dots$.

Hint: Take $X_0 = 0$ and $\Delta_n = X_n - X_{n-1}$, and define $Z_n = \sum_{k=1}^n \mathbb{E}[\Delta_k \mid \mathcal{F}_{k-1}]$, where $\mathcal{F}_0 = \{\emptyset, \Omega\}$.

EXERCISE 6.17. Let $X_1, X_2, \dots$ be a martingale and assume that $|X_1(\omega)|$ and $|X_n(\omega) - X_{n-1}(\omega)|$ are bounded by a positive constant K , independent of ω and n . Let τ be a stopping time with $\mathbb{E}[\tau] < \infty$. Write

$$\int_{\tau \leq k} X_{\tau} d\mathbb{P} = \sum_{i=1}^k \left(\int_{\tau \geq i} X_i d\mathbb{P} - \int_{\tau \geq i+1} X_i d\mathbb{P} \right).$$

- (i) Use the fact that $X_1, X_2, \dots$ is a martingale to rewrite the last integral and to show that

$$\int_{\tau \leq k} X_{\tau} d\mathbb{P} = \mathbb{E}[X_1] - \int_{\tau > k} X_{k+1} d\mathbb{P}.$$

- (ii) Explain why $\int_{\tau > k} \tau d\mathbb{P} \rightarrow 0$ as $k \rightarrow \infty$.
- (iii) Show that $\int_{\tau > k} X_{k+1} d\mathbb{P} \rightarrow 0$ as $k \rightarrow \infty$.
- (iv) Show that X_{τ} is integrable and that $\mathbb{E}[X_{\tau}] = \mathbb{E}[X_1]$.

SOLUTION. (i) For every $n \geq 1$:

$$|X_n| \leq |X_1| + \sum_{j=2}^n |X_j - X_{j-1}| \leq K + (n-1)K = nK.$$

(ii)

(iii)

(iv)

□

Stochastic Processes

SECTION 7.1

| Brownian Motion

DEFINITION 7.1 (Brownian Motion).

A *Brownian motion* or *Wiener process* is a stochastic process $\{W_t : t \geq 0\}$, on some $(\Omega, \mathcal{F}, \mathbb{P})$, with these three properties:

- (i) The process starts at 0: $\mathbb{P}(W_0 = 0) = 1$.
- (ii) The increments are independent: If $0 \leq t_0 \leq t_1 \leq \dots \leq t_k$, then

$$\mathbb{P}(W_{t_i} - W_{t_{i-1}} \in H_i, i \leq k) = \prod_{i \leq k} \mathbb{P}(W_{t_i} - W_{t_{i-1}} \in H_i).$$

- (iii) For $0 \leq s < t$ the increment $W_t - W_s$ is normally distributed with mean 0 and variance $t - s$:

$$\mathbb{P}(W_t - W_s \in H) = \frac{1}{\sqrt{2\pi(t-s)}} \int_H e^{-\frac{x^2}{2(t-s)}} dx. \quad (7.1)$$

If $0 \leq s \leq t$, then by the independence of the increments,

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s(W_t - W_s)] + \mathbb{E}[W_s^2] = \mathbb{E}[W_s] \mathbb{E}[W_t - W_s] + \mathbb{E}[W_s^2] = s.$$

If $0 < t_1 < \dots < t_k$, the joint density of $(W_{t_1}, W_{t_2} - W_{t_1}, \dots, W_{t_k} - W_{t_{k-1}})$ is the product of the corresponding normal densities. By the Jacobian formula, $(W_{t_1}, \dots, W_{t_k})$ has density

$$f_{t_1, \dots, t_k}(x_1, \dots, x_k) = \prod_{i=1}^k \frac{1}{\sqrt{2\pi(t_i - t_{i-1})}} \exp \left[-\frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})} \right],$$

where $t_0 = x_0 = 0$.

Continuity of Paths

Measurable Processes

THEOREM 7.2.

Brownian motion is measurable.

Irregularity of Brownian Motion Paths

THEOREM 7.3.

For ω outside a set of probability 0, $W(\cdot; \omega)$ is nowhere differentiable.

Proof.

□

Appendix

SECTION A.1

| Final Exams

Semester 2025–2026

EXERCISE A.1. Suppose that for $n = 1, 2, \dots$, $0 \leq p_n \leq 1$, and let $\alpha_n = \min\{p_n, 1 - p_n\}$. Suppose that on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ we have independent events $A_1, A_2, \dots$ such that $\mathbb{P}(A_n) = p_n$ for all n . We assume throughout that

$$\sum_{n=1}^{\infty} \alpha_n = \infty.$$

- (i) Let B_i denote either A_i or A_i^c . Show that for all n and for all choices of the B_i we have $\mathbb{P}(B_1 \cap B_2 \cap \dots \cap B_n) \leq e^{-\sum_{i=1}^n \alpha_i}$.
- (ii) Show that Ω cannot be countable.
- (iii) Let

$$A_x := \left\{ \omega \in \Omega : \sum_n \mathbb{1}_{A_n}(\omega) 2^{-n} \leq x \right\}.$$

Show that the function $f(x) = \mathbb{P}(A_x)$ ($x \in [0, 1]$) is continuous.

- (iv) Show that for all $x \in [0, 1]$ there is a set D_x such that $\mathbb{P}(D_x) = x$.

SOLUTION. (i) Since $B_i \in \{A_i, A_i^c\}$,

$$\mathbb{P}(B_i) \in \{p_i, 1 - p_i\} \leq \max\{p_i, 1 - p_i\} = 1 - \min\{p_i, 1 - p_i\} = 1 - \alpha_i.$$

By independence,

$$\mathbb{P}\left(\bigcap_{i=1}^n B_i\right) = \prod_{i=1}^n \mathbb{P}(B_i) \leq \prod_{i=1}^n (1 - \alpha_i) \leq \prod_{i=1}^n e^{-\alpha_i} = e^{-\sum_{i=1}^n \alpha_i},$$

where $1 - x \leq e^{-x}$ for $x \in [0, 1]$.

- (ii) Assume Ω is countable by $\Omega := \{\omega_1, \omega_2, \dots\}$. Fix k . Define

$$B_i^{(k)} := \begin{cases} A_i, & \omega_k \in A_i \\ A_i^c, & \omega_k \notin A_i \end{cases}.$$

Then for every n , $\omega_k \in \bigcap_{i=1}^n B_i^{(k)}$, so by (i),

$$\mathbb{P}(\{\omega_k\}) \leq \mathbb{P}\left(\bigcap_{i=1}^{\infty} B_i^{(k)}\right) = \mathbb{P}\left(\lim_{n \rightarrow \infty} \bigcap_{i=1}^n B_i^{(k)}\right) \leq \lim_{n \rightarrow \infty} e^{-\sum_{i=1}^n \alpha_i} = 0.$$

Therefore,

$$1 = \mathbb{P}(\Omega) = \mathbb{P}\left(\bigcup_{k=1}^{\infty} \{\omega_k\}\right) = \sum_{k=1}^{\infty} \mathbb{P}(\{\omega_k\}) = 0,$$

which is a contradiction.

(iii) Let

$$X_n := \mathbb{1}_{A_n}, \quad S(\omega) := \sum_{n=1}^{\infty} X_n(\omega) 2^{-n}.$$

Then $A_x = \{\omega \in \Omega : S(\omega) \leq x\}$, and $f(x) = \mathbb{P}(S \leq x)$. The jump at x equals

$$f(x) - f(x^-) = f(x) - \lim_{y \uparrow x} f(y) = \mathbb{P}(S \leq x) - \mathbb{P}(S < x) = \mathbb{P}(S = x).$$

Since f is right-continuous, it suffices to show $\mathbb{P}(S = x) = 0$ for every $x \in [0, 1]$. Choose a binary expansion

$$x = \sum_{n=1}^{\infty} d_n(x) 2^{-n}.$$

For dyadic rationals, pick the non-terminating expansion. Define

$$B_n(x) := \begin{cases} A_n, & d_n(x) = 1 \\ A_n^c, & d_n(x) = 0 \end{cases}.$$

If $S(\omega) = x$, then

$$\sum_{n=1}^{\infty} X_n(\omega) 2^{-n} = \sum_{n=1}^{\infty} d_n(x) 2^{-n} \implies X_n(\omega) = d_n(x)$$

for all n , hence

$$\{S = x\} \subseteq \bigcap_{n \geq 1} B_n.$$

By (i),

$$\mathbb{P}\left(\bigcap_{n \geq 1} B_n\right) = \lim_{m \rightarrow \infty} \mathbb{P}\left(\bigcap_{n=1}^m B_n\right) \leq \lim_{m \rightarrow \infty} e^{-\sum_{n=1}^m \alpha_n} = 0.$$

(iv) From (iii), f is continuous and non-decreasing. Also,

$$f(1) = \mathbb{P}(S \leq 1) = 1, \quad f(0) = \mathbb{P}(S \leq 0) = \mathbb{P}(S = 0) = 0.$$

By the *intermediate value property*, for each $x \in [0, 1]$ there exists $t \in [0, 1]$ such that $f(t) = x$. Define

$$D_x := A_t = \{\omega : S(\omega) \leq t\}.$$

Then $\mathbb{P}(D_x) = \mathbb{P}(A_t) = f(t) = x$.

□

EXERCISE A.2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A_1, A_2, \dots, A \in \mathcal{F}$.

(i) Show that

$$\limsup_{n \rightarrow \infty} (A_n \cap A) = \left(\limsup_{n \rightarrow \infty} A_n \right) \cap A.$$

(ii) Show that if

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n \cap A) < \infty,$$

then

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right) \leq 1 - \mathbb{P}(A).$$

SOLUTION. (i) By definition,

$$\begin{aligned} \limsup_{n \rightarrow \infty} (A_n \cap A) &= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} (A_k \cap A) \\ &= \bigcap_{n=1}^{\infty} \left(\left(\bigcup_{k=n}^{\infty} A_k \right) \cap A \right) \\ &= \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right) \cap A = \left(\limsup_{n \rightarrow \infty} A_n \right) \cap A. \end{aligned}$$

(ii) By the *first Borel-Cantelli lemma* and (i),

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} (A_n \cap A)\right) = \mathbb{P}\left(\left(\limsup_{n \rightarrow \infty} A_n\right) \cap A\right) = 0.$$

Therefore,

$$\begin{aligned} \mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right) &= \mathbb{P}\left(\left(\limsup_{n \rightarrow \infty} A_n\right) \cap A\right) + \mathbb{P}\left(\left(\limsup_{n \rightarrow \infty} A_n\right) \cap A^c\right) \\ &\leq 0 + \mathbb{P}(A^c) = 1 - \mathbb{P}(A). \end{aligned}$$

□

EXERCISE A.3. Let $X_1, X_2, \dots$ be a sequence of *i.i.d.* standard normal random variables. Set

$$Y_n = \frac{1}{n} \sum_{k=1}^n \sqrt{k} X_k.$$

(i) Derive the characteristic function of a random variable Y that is normally distributed with mean 0 and variance $\frac{1}{2}$.

Hint:

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{itx} e^{-\frac{x^2}{2}} dx = e^{-\frac{t^2}{2}}.$$

(ii) Show that as $n \rightarrow \infty$, Y_n converges in distribution to a random variable Y as in (i).

SOLUTION. (i) The characteristic function is

$$\begin{aligned} \varphi_Y(t) &= \mathbb{E}[e^{itY}] \\ &= \int_{-\infty}^{\infty} e^{itx} f_Y(x) dx \\ &= \int_{-\infty}^{\infty} e^{itx} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} dx \\ &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{itx} e^{-x^2} dx \\ &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{itx} e^{-\frac{(\sqrt{2}x)^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{itu}{\sqrt{2}}} e^{-\frac{u^2}{2}} du = e^{-\frac{t^2}{4}}. \end{aligned} \quad (u = \sqrt{2}x)$$

(ii) For $X \sim \mathcal{N}(0, 1)$, $\mathbb{E}[e^{itX}] = e^{-\frac{t^2}{2}}$. By independence,

$$\begin{aligned}
 \varphi_{Y_n}(t) &= \mathbb{E} \exp \left(it \cdot \frac{1}{n} \sum_{k=1}^n \sqrt{k} X_k \right) \\
 &= \prod_{k=1}^n \mathbb{E} \exp \left(i \frac{t \sqrt{k}}{n} X_k \right) \\
 &= \prod_{k=1}^n \exp \left(-\frac{1}{2} \left(\frac{t \sqrt{k}}{n} \right)^2 \right) \\
 &= \prod_{k=1}^n \exp \left(-\frac{t^2 k}{2n^2} \right) \\
 &= \exp \left(-\frac{t^2}{2n^2} \sum_{k=1}^n k \right) \\
 &= \exp \left(-\frac{t^2}{2n^2} \cdot \frac{n(n+1)}{2} \right) \\
 &= \exp \left(-\frac{1 + \frac{1}{n}}{4} t^2 \right) \rightarrow \exp \left(-\frac{1}{4} t^2 \right)
 \end{aligned}$$

as $n \rightarrow \infty$. By Lévy's continuity theorem, this implies $Y_n \Rightarrow Y$. □

EXERCISE A.4. Let $X_0, X_1, \dots$ be integrable random variables and $\mathcal{F}_n = \sigma\{X_0, \dots, X_n\}$. Let

$$Z_0 := 0, \quad Z_n := \sum_{j=0}^{n-1} (X_{j+1} - \mathbb{E}[X_{j+1} | \mathcal{F}_j]).$$

- (i) Show that $Z_0, Z_1, \dots$ is a martingale with respect to the $\mathcal{F}_n$.
- (ii) Let $U_1, U_2, \dots$ be independent random variables, all uniformly-(0, 1) distributed. Now take $X_0 = 0$ a.s. and $X_{n+1} = X_n + U_{n+1}$. Show that a.s. we have $Z_n = X_n - \frac{1}{2}n$.
- (iii) Argue that the martingale in (ii) does not converge a.s. as $n \rightarrow \infty$? Why does this not contradict the *martingale convergence theorem*?

SOLUTION. (i) By Jensen's inequality,

$$\mathbb{E}[|\mathbb{E}[X_{j+1} | \mathcal{F}_j]|] \leq \mathbb{E}[\mathbb{E}[|X_{j+1}| | \mathcal{F}_j]] = \mathbb{E}[|X_{j+1}|].$$

Then $X_{j+1} - \mathbb{E}[X_{j+1} | \mathcal{F}_j]$ is integrable, and so is Z_n . X_{j+1} is $\mathcal{F}_{j+1} \subseteq \mathcal{F}_n$ -measurable, and $\mathbb{E}[X_{j+1} | \mathcal{F}_j]$ is $\mathcal{F}_j \subseteq \mathcal{F}_n$ -measurable. Therefore Z_n is $\mathcal{F}_n$ -measurable. Since

$$Z_{n+1} = Z_n + (X_{n+1} - \mathbb{E}[X_{n+1} | \mathcal{F}_n]).$$

For martingale property,

$$\begin{aligned}
 \mathbb{E}[Z_{n+1} | \mathcal{F}_n] &= \mathbb{E}[Z_n | \mathcal{F}_n] + \mathbb{E}[X_{n+1} | \mathcal{F}_n] - \mathbb{E}[\mathbb{E}[X_{n+1} | \mathcal{F}_n] | \mathcal{F}_n] \\
 &= Z_n + \mathbb{E}[X_{n+1}] - \mathbb{E}[X_{n+1}] = Z_n.
 \end{aligned}$$

- (ii) By $X_0 = 0$ a.s. and $X_{n+1} = X_n + U_{n+1}$, $X_n = \sum_{k=1}^n U_k$ a.s., and $\mathcal{F}_n = \sigma(X_0, \dots, X_n) = \sigma(U_1, \dots, U_n)$. Then

$$Z_n = \sum_{j=0}^{n-1} (X_{j+1} - \mathbb{E}[X_{j+1} | \mathcal{F}_j])$$

$$\begin{aligned}
&= \sum_{j=0}^{n-1} (X_{j+1} - \mathbb{E}[X_j + U_{j+1} \mid \mathcal{F}_j]) \\
&= \sum_{j=0}^{n-1} (X_{j+1} - X_j - \mathbb{E}[U_{j+1} \mid \mathcal{F}_j]) \\
&= \sum_{j=0}^{n-1} \left(U_{j+1} - \frac{1}{2} \right) \\
&= \sum_{k=1}^n U_k - \frac{n}{2} = X_n - \frac{n}{2} \quad a.s.
\end{aligned}$$

(iii) By (ii), if Z_n converged *a.s.*, we would have

$$Z_{n+1} - Z_n = U_{n+1} - \frac{1}{2} \rightarrow 0 \quad a.s.$$

Set $\varepsilon = \frac{1}{4}$. Then

$$\mathbb{P}\left(\left|U_n - \frac{1}{2}\right| > \frac{1}{4}\right) = \mathbb{P}\left(U_n < \frac{1}{4}\right) + \mathbb{P}\left(U_n > \frac{3}{4}\right) = \frac{1}{2}.$$

Let $A_n = \{|U_n - \frac{1}{2}| > \frac{1}{4}\}$. The events (A_n) are independent and $\sum_n \mathbb{P}(A_n) = \infty$. By the *second Borel-Cantelli lemma*, $\mathbb{P}(\limsup_n A_n) = 1$. So *a.s.* $|U_n - \frac{1}{2}| > \frac{1}{4}$ happens infinitely often. Therefore Z_n cannot converge *a.s.* This does not contradict the *martingale convergence theorem* because it requires uniform integrability, which is not satisfied. $\square$

EXERCISE A.5. Suppose that $\{W_t : t \geq 0\}$ is a Brownian motion. Define $X_t := tW_{\frac{1}{t}}$ if $t > 0$ and $X_0 = 0$. Show that $\{X_t : t \geq 0\}$ is again a Brownian motion.

SOLUTION. We have to check that: the defined processes are Gaussian, (almost all) sample paths are continuous, and they have the same mean and covariance functions as Brownian motion.

- Gaussian Process: Since W_t is a Brownian motion, it is a Gaussian process. For any $t_1, \dots, t_k > 0$, $(X_{t_1}, \dots, X_{t_k})$ is a linear transformation of the Gaussian vector $(W_{\frac{1}{t_1}}, \dots, W_{\frac{1}{t_k}})$. Therefore, the finite-dimensional distributions of X_t are multivariate normal.
- Moments:

$$\mathbb{E}[X_t] = \mathbb{E}\left[tW_{\frac{1}{t}}\right] = t\mathbb{E}\left[W_{\frac{1}{t}}\right] = t \cdot 0 = 0.$$

$$\text{Cov}(X_s, X_t) = \mathbb{E}[X_s X_t] = st\mathbb{E}\left[W_{\frac{1}{s}}\right]\mathbb{E}\left[W_{\frac{1}{t}}\right] = st \min\left\{\frac{1}{s}, \frac{1}{t}\right\} = \min\{s, t\}.$$

- Continuity: The function $t \mapsto \frac{1}{t}$ is continuous on $(0, \infty)$, and the sample paths of Brownian motion $u \mapsto W_u$ are continuous *a.s.* As the composition and product of continuous functions, $X_t = tW_{\frac{1}{t}}$ is continuous for all $t > 0$. Let $u = \frac{1}{t}$. Then by the *strong law of large numbers*,

$$\lim_{t \downarrow 0} X_t = \lim_{t \downarrow 0} tW_{\frac{1}{t}} = \lim_{u \rightarrow \infty} \frac{W_u}{u} = \lim_{u \rightarrow \infty} \frac{1}{u} \sum_{k=1}^n (W_k - W_{k-1}) = 0 \quad a.s.$$

Therefore, the paths of X_t are continuous at $t = 0$. $\square$

Remark. To show that $\lim_{t \downarrow 0} X_t = 0$ *a.s.* rigorously, let

$$\Omega^* = \{\omega \in \Omega \mid (W_t(\omega))_{t \geq 0} \text{ continuous, } W_0(\omega) = 0\}.$$

By assumption $\Omega \setminus \Omega^*$ is contained in a $\mathbb{P}$ -null set. Further, $(X_t)_{t>0}$ has continuous paths on Ω^* . Then $\lim_{t \downarrow 0} X_t(\omega) = 0$ if and only if for all $\epsilon > 0$ there exists $\delta_\omega > 0$ such that $|X_t(\omega)| < \epsilon$ for all $t \leq \delta_\omega$. This is true if and only if for all integers $m \geq 1$, there exists an integer n_ω , such that $|X_q(\omega)| < \frac{1}{m}$ for all $q \in \mathbb{Q}$ with $q < \frac{1}{n_\omega}$, because of continuity of $X_t(\omega)$, $t > 0$. Check that this implies

$$\left\{ \omega : \lim_{t \downarrow 0} X_t(\omega) = 0 \right\} \cap \Omega^* = \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{q \in (0, \frac{1}{n}] \cap \mathbb{Q}} \left\{ \omega : |X_q(\omega)| < \frac{1}{m} \right\} \cap \Omega^*.$$

The finite-dimensional distributions of X and W are equal. Hence

$$\mathbb{P} \left(\bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{q \in (0, \frac{1}{n}] \cap \mathbb{Q}} \left\{ \omega : |X_q(\omega)| < \frac{1}{m} \right\} \right) = \mathbb{P} \left(\bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{q \in (0, \frac{1}{n}] \cap \mathbb{Q}} \left\{ \omega : |W_q(\omega)| < \frac{1}{m} \right\} \right) = 1.$$

As a consequence $\mathbb{P}(\omega : \lim_{t \downarrow 0} X_t(\omega) = 0) = 1$.

Semester 2024–2025

EXERCISE A.6. Consider the space $\Omega = \{0, 1\}^{\mathbb{N}}$ of all infinite sequences of 0's and 1's, with the σ -algebra $\mathcal{A}$ generated by the cylinder sets. An element $\omega \in \Omega$ is typically denoted $\omega = (\omega_1, \omega_2, \dots)$. Let μ be the probability measure on $(\Omega, \mathcal{A})$ for which

$$\mu(\{\omega \in \Omega : \omega_1 = a_1, \dots, \omega_k = a_k\}) = 2^{-k},$$

for all $k = 1, 2, \dots$ and $a_i \in \{0, 1\}$.

- (i) Show that all one-point sets $\{\omega\}$ are in $\mathcal{A}$.
- (ii) Show that $\mu(\{\omega\}) = 0$ for all $\omega \in \Omega$.
- (iii) Let F be the set of all elements $\omega \in \Omega$ for which $\omega_i = 0$ for only finitely many indices i . Show that $F \in \mathcal{A}$ and that $\mu(F) = 0$.

SOLUTION. (i) Fix $y = (y_1, y_2, \dots) \in \Omega$. For $i \in \mathbb{N}_1$, write

$$c_i(y) = \{\omega \in \Omega : \omega_1 = y_1, \dots, \omega_i = y_i\}.$$

Now

$$\{y\} = \bigcap_{i=1}^{\infty} c_i(y) \in \mathcal{A}.$$

- (ii) By continuity from above for decreasing measurable sets with finite measure,

$$\mu(\{y\}) = \mu \left(\bigcap_{i=1}^{\infty} c_i(y) \right) = \lim_{i \rightarrow \infty} \mu(c_i(y)) = \lim_{i \rightarrow \infty} 2^{-i} = 0.$$

- (iii) $\omega \in F$ if and only if there exists $N \in \mathbb{N}$ such that $\omega_i = 1$ for all $i \geq N$. Then

$$F = \bigcup_{N=1}^{\infty} E_N,$$

where

$$E_N = \bigcap_{m=N}^{\infty} \{\omega \in \Omega : \omega_N = \omega_{N+1} = \dots = \omega_m = 1\}.$$

Each set is a cylinder set, hence belongs to $\mathcal{A}$. Since $\mathcal{A}$ is a σ -algebra, it is closed under countable intersections and unions, so $E_N \in \mathcal{A}$ for all N , and therefore $F \in \mathcal{A}$.

For $m \geq N$,

$$\mu(\{\omega : \omega_N = \cdots = \omega_m = 1\}) = 2^{-(m-N+1)}.$$

So by continuity from above and countable subadditivity,

$$\mu(F) = \mu\left(\bigcup_{N=1}^{\infty} E_N\right) \leq \sum_{N=1}^{\infty} \mu(E_N) = \sum_{N=1}^{\infty} \lim_{m \rightarrow \infty} 2^{-(m-N+1)} = 0.$$

□

EXERCISE A.7. The Levy distribution has density

$$f(x) = \frac{1}{\sqrt{2\pi}} \cdot \frac{e^{-\frac{1}{2x}}}{x^{\frac{3}{2}}},$$

for all $x \in (0, \infty)$. Its characteristic function is given by

$$\phi(t) = e^{\sqrt{t}(1-i)}.$$

- (i) If $X_1, \dots, X_n$ are independent and have a Levy distribution, determine the characteristic function of $\frac{X_1 + \cdots + X_n}{n}$, for all $n = 1, 2, \dots$.
- (ii) Explain why your answer in (i) does not contradict the *strong law of large numbers*.

SOLUTION. (i) Let $\bar{X}_n = \frac{1}{n} \sum_{k=1}^n X_k$. Then, since $X_1, \dots, X_n$ are *i.i.d.*, we get

$$\begin{aligned} \phi_{\bar{X}_n}(t) &= \mathbb{E}[\exp(it\bar{X}_n)] \\ &= \mathbb{E}\left[\exp\left(it \frac{X_1 + \cdots + X_n}{n}\right)\right] \\ &= \prod_{k=1}^n \mathbb{E}\left[\exp\left(\frac{it}{n} X_k\right)\right] \\ &= \mathbb{E}\left[\exp\left(\frac{it}{n} X_1\right)\right]^n \\ &= \phi_{X_1}\left(\frac{t}{n}\right)^n \\ &= \exp\left(\sqrt{\frac{t}{n}}(1-i)\right)^n \\ &= \exp\left(\sqrt{nt}(1-i)\right) = e^{\sqrt{nt}}(\cos \sqrt{nt} - i \sin \sqrt{nt}), \end{aligned}$$

which does not converge as $n \rightarrow \infty$.

- (ii) We will show that the Levy distribution has infinite mean:

$$\begin{aligned} \mathbb{E}[X_1] &= \int_{\mathbb{R}^+} x f(x) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} \frac{e^{-\frac{1}{2x}}}{x^{\frac{1}{2}}} dx \\ &= \frac{1}{\sqrt{2\pi}} \left(\int_0^1 \frac{e^{-\frac{1}{2x}}}{x^{\frac{1}{2}}} dx + \int_1^{\infty} \frac{e^{-\frac{1}{2x}}}{x^{\frac{1}{2}}} dx \right) \\ &\geq \frac{1}{\sqrt{2\pi}} \int_1^{\infty} \frac{e^{-\frac{1}{2x}}}{x^{\frac{1}{2}}} dx \\ &\geq \frac{1}{\sqrt{2\pi}} \int_1^{\infty} x^{-\frac{1}{2}} e^{-\frac{1}{2}} dx \end{aligned}$$

$$= \frac{e^{-\frac{1}{2}}}{\sqrt{2\pi}} \int_1^\infty x^{-\frac{1}{2}} dx = \infty.$$

The hypothesis of the *strong law of large numbers* is not satisfied, which justifies why $\phi_{\bar{X}_n}(t)$ does not converge.

□

EXERCISE A.8. Consider a sequence of independent random variables $X_1, X_2, \dots$ such that

$$\mathbb{P}(X_n = 1) = \frac{1}{n}, \quad \mathbb{P}(X_n = 0) = 1 - \frac{1}{n}.$$

- (i) Show that the sequence converges to 0 in probability.
- (ii) Show that the sequence does not converge to 0 *a.s.*

SOLUTION. (i) Let $\epsilon > 0$ be given. If $\epsilon > 1$, then for all n , $\mathbb{P}(|X_n - 0| > \epsilon) = 0$. As a result it trivially follows that

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - 0| > \epsilon) = 0.$$

On the other hand, for $\epsilon \leq 1$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - 0| > \epsilon) = \lim_{n \rightarrow \infty} \mathbb{P}(X_n = 1) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

- (ii) Define $E_n = \{X_n = 1\}$. Then, $\mathbb{P}(E_n) = \frac{1}{n}$, and

$$\sum_{n \geq 1} \mathbb{P}(E_n) = \sum_{n \geq 1} \frac{1}{n} = \infty.$$

Therefore, by the *second Borel-Cantelli lemma*, E_n happens infinitely often, i.e., there exists a subsequence $n_1 < n_2 < \dots$ such that $X_{n_j} = 1$. Clearly, X_n does not converge to 0 *a.s.*

□

EXERCISE A.9. Consider a sequence of independent, uniformly bounded random variables $\{X_n\}$ with $\mathbb{E}[X_1] = 0$, and $s_n \geq 0$ such that

$$\sum_{k=1}^n \mathbb{E}[X_k^2] = s_n^2 \rightarrow \infty$$

as $n \rightarrow \infty$. Show that $S_n = \sum_{k=1}^n X_k$ satisfies a *central limit theorem*; that is, we have $\frac{S_n}{s_n} \Rightarrow N$, where N is a standard normal random variable.

Hint. For an appropriate $\delta > 0$, consider Lyapunov's condition

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{s_n^{2+\delta}} \mathbb{E}[|X_k|^{2+\delta}] = 0.$$

SOLUTION. Since $\{X_k\}$ are uniformly bounded, there exists a constant $M > 0$ such that $|X_k| \leq M$ for all k . Therefore, for any $\delta > 0$,

$$\mathbb{E}[|X_k|^{2+\delta}] \leq M^\delta \mathbb{E}[X_k^2].$$

Substituting this bound into Lyapunov's condition:

$$\frac{1}{s_n^{2+\delta}} \sum_{k=1}^n \mathbb{E}[|X_k|^{2+\delta}] \leq \frac{1}{s_n^{2+\delta}} \sum_{k=1}^n M^\delta \mathbb{E}[X_k^2]$$

$$\begin{aligned}
&= \frac{M^\delta}{s_n^{2+\delta}} \sum_{k=1}^n \mathbb{E}[X_k^2] \\
&= \frac{M^\delta}{s_n^{2+\delta}} \cdot s_n^2 = \frac{M^\delta}{s_n^\delta} \rightarrow 0
\end{aligned}$$

as $n \rightarrow \infty$. *Lyapunov's central limit theorem* applies, and we conclude that $\frac{S_n}{s_n} \xrightarrow{d} \mathcal{N}(0, 1)$. $\square$

EXERCISE A.10. We have three urns (numbered urn 1, 2 and 3) and two balls. Initially, we put both balls in urn 1, and we perform the following discrete time process. At each time unit we choose one of the two balls uniformly at random (i.e., each ball is chosen with probability $\frac{1}{2}$) and place the chosen ball into one of the three urns, also uniformly chosen at random (i.e., each urn is chosen with probability $\frac{1}{3}$). Denote by X_n the number of balls in urn 1 at time n and let $\mathcal{F}_n := \sigma\{X_0, \dots, X_n\}$.

(i) Show that *a.s.* we have

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \frac{1}{2}X_n + \frac{1}{3},$$

and deduce that

$$\mathbb{E}[X_{n+1}] = \frac{1}{2}\mathbb{E}[X_n] + \frac{1}{3}.$$

(ii) Show that

$$M_n := 2^n X_n - \frac{1}{3} \sum_{k=1}^n 2^k$$

is a martingale with respect to $\mathcal{F}_n$.

(iii) Show, using the martingale (M_n) , that

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \frac{2}{3}.$$

(iv) Does $\lim_{n \rightarrow \infty} M_n$ exist *a.s.*? Relate your answer to the *martingale convergence theorem*.

SOLUTION. (i) Observe that $\mathcal{F}_n$ is generated by the π -system $\{X_1 = x_1, \dots, X_n = x_n\}$. For calculations only the value of X_n matters, so it suffices to check the events $G = \{X_n = x_n\}$, $x_n \in \{0, 1, 2\}$.

– If $X_n = 0$, we have

$$\mathbb{P}(X_{n+1} = 0) = \frac{2}{3}, \quad \mathbb{P}(X_{n+1} = 1) = \frac{1}{3},$$

so that indeed

$$\mathbb{E}[X_{n+1} \mid X_n = 0] = \frac{1}{3} = \frac{1}{2}X_n + \frac{1}{3}.$$

– If $X_n = 1$, we have

$$\mathbb{P}(X_{n+1} = 0) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3},$$

$$\mathbb{P}(X_{n+1} = 1) = \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{2},$$

$$\mathbb{P}(X_{n+1} = 2) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6},$$

so that

$$\mathbb{E}[X_{n+1} \mid X_n = 1] = \frac{1}{2} + \frac{2}{6} = \frac{1}{2}X_n + \frac{1}{3}.$$

– If $X_n = 2$, we have

$$\mathbb{P}(X_{n+1} = 1) = \frac{2}{3}, \quad \mathbb{P}(X_{n+1} = 2) = \frac{1}{3},$$

so that also

$$\mathbb{E}[X_{n+1} \mid X_n = 2] = \frac{4}{3} = \frac{1}{2}X_n + \frac{1}{3}.$$

Therefore we have shown $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \frac{1}{2}X_n + \frac{1}{3}$, taking expectations on both sides yields $\mathbb{E}[X_{n+1}] = \frac{1}{2}\mathbb{E}[X_n] + \frac{1}{3}$.

(ii) M_n is $\mathcal{F}_n$ -measurable and integrable. Using (i), we find

$$\begin{aligned} \mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}\left[2^{n+1}X_{n+1} - \frac{1}{3}\sum_{k=1}^{n+1}2^k \mid \mathcal{F}_n\right] \\ &= 2^{n+1}\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] - \frac{1}{3}\sum_{k=1}^{n+1}2^k \\ &= 2^{n+1}\left(\frac{1}{2}X_n + \frac{1}{3}\right) - \frac{1}{3}\sum_{k=1}^{n+1}2^k \\ &= 2^n X_n - \frac{1}{3}\sum_{k=1}^n 2^k = M_n. \end{aligned}$$

(iii)

$$\mathbb{E}[M_n] = 2^n \mathbb{E}[X_n] - \frac{1}{3}\sum_{k=1}^n 2^k \implies \mathbb{E}[X_n] = \frac{\mathbb{E}[M_n] + \frac{2}{3}(2^n - 1)}{2^n}.$$

M_n is a martingale, so

$$\mathbb{E}[M_n] = \mathbb{E}[M_0] = 2^0 X_0 - \frac{1}{3}\sum_{k=1}^0 2^k = 2.$$

Thus

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \lim_{n \rightarrow \infty} \frac{2 + \frac{2}{3}(2^n - 1)}{2^n} = \frac{2}{3} + \lim_{n \rightarrow \infty} \frac{4}{3 \cdot 2^n} = \frac{2}{3}.$$

(iv) By (ii),

$$M_n = 2^n X_n - \frac{1}{3}\sum_{k=1}^n 2^k = 2^n \left(X_n - \frac{2}{3}\right) + \frac{2}{3},$$

where X_n can only take values in the set $\{0, 1, 2\}$. Thus, we always have

$$\left|M_n - \frac{2}{3}\right| = 2^n \left|X_n - \frac{2}{3}\right| \geq \frac{2^n}{3} \rightarrow \infty$$

as $n \rightarrow \infty$. This implies that $|M_n| \rightarrow \infty$ and $\lim_{n \rightarrow \infty} M_n$ does not exist *a.s.*

Since we have shown that M_n does not converge to a finite limit, (M_n) is not bounded in $\mathcal{L}^1$.

Thus, the conditions of the *martingale convergence theorem* are not satisfied.

□

EXERCISE A.11. Let $\Omega = \{0, 1\}^\infty$, the space consisting of all sequences $\omega = (\omega_1, \omega_2, \dots)$, with all $\omega_i \in \{0, 1\}$. A *cylinder* is a set of the form $\{\omega \in \Omega : \omega_1 = a_1, \dots, \omega_n = a_n\}$, for some integer n , and $a_1, \dots, a_n \in \{0, 1\}$. We take the sigma-algebra $\mathcal{C}$ on Ω generated by the collection of cylinders. The probability measure μ on $(\Omega, \mathcal{C})$ is the unique measure which satisfies

$$\mu(\{\omega \in \Omega : \omega_1 = a_1, \dots, \omega_n = a_n\}) = \left(\frac{1}{2}\right)^n,$$

for all n and $a_1, \dots, a_n \in \{0, 1\}$.

- (i) Let $A \subset \Omega$ be the set of elements $\omega \in \Omega$ for which $\omega_2 \neq \omega_4$. Show that $A \in \mathcal{C}$.
- (ii) Compute $\mu(A)$.

SOLUTION. Let $C(a_1, \dots, a_n) = \{\omega \in \Omega : \omega_1 = a_1, \dots, \omega_n = a_n\}$.

- (i) We have countable unions

$$\begin{aligned} A &= \{\omega : \omega_2 = 0, \omega_4 = 1\} \cup \{\omega : \omega_2 = 1, \omega_4 = 0\} \\ &= \bigcup_{a_1, a_3 \in \{0, 1\}} C(a_1, 0, a_3, 1) \cup \bigcup_{a_1, a_3 \in \{0, 1\}} C(a_1, 1, a_3, 0) \in \mathcal{C}. \end{aligned}$$

- (ii) We can decompose A into $2 \times 2 \times 2 = 8$ disjoint cylinders, with each cylinder having measure 2^{-4} . Then $\mu(A) = \frac{1}{2}$.

□

EXERCISE A.12. (i) Show that for any real random variable $Y \geq 0$, we have

$$Y \leq \sum_{n \geq 0} \mathbb{1}_{\{Y \geq n\}},$$

and conclude that

$$\mathbb{E}[Y] \leq \sum_{n \geq 0} \mathbb{P}(Y \geq n).$$

- (ii) Show the following: for *i.i.d.* random variables $X_1, X_2, \dots$ with $\mathbb{E}[X_1] = 0$ and $\text{Var}(X_1) = \infty$, one has

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} \{|X_n| \geq \sqrt{n}\}\right) = 1.$$

SOLUTION. (i)

$$Y = \int_0^\infty \mathbb{1}_{\{Y \geq t\}} dt = \sum_{n=0}^\infty \int_n^{n+1} \mathbb{1}_{\{Y \geq t\}} dt \leq \sum_{n=0}^\infty \int_n^{n+1} \mathbb{1}_{\{Y \geq n\}} dt = \sum_{n=0}^\infty \mathbb{1}_{\{Y \geq n\}}.$$

By Tonelli's theorem,

$$\mathbb{E}[Y] \leq \mathbb{E}\left[\sum_{n=0}^\infty \mathbb{1}_{\{Y \geq n\}}\right] = \sum_{n=0}^\infty \mathbb{E}[\mathbb{1}_{\{Y \geq n\}}] = \sum_{n=0}^\infty \mathbb{P}(Y \geq n).$$

- (ii) Since $\mathbb{E}[X_1] = 0$, $\text{Var}(X_1) = \mathbb{E}[X_1^2] = \infty$. Then

$$\mathbb{E}[X_1^2] \leq \sum_{n \geq 0} \mathbb{P}(X_1^2 \geq n) = \sum_{n \geq 0} \mathbb{P}(|X_1| \geq \sqrt{n}) = \infty.$$

Since the X_n are independent, by the *second Borel-Cantelli lemma*,

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} \{|X_n| \geq \sqrt{n}\}\right) = 1.$$

□

EXERCISE A.13. Consider a sequence of random variables $X_1, X_2, \dots$, where X_0 is distributed uniformly on $[0, 1]$ and

$$X_{2n} = X_0, \quad X_{2n+1} = 1 - X_0.$$

- (i) Show that the sequence converges in distribution.
- (ii) Show that the sequence does not converge in probability.

SOLUTION. (i) For even indices, $X_{2n} = X_0$, so $X_{2n} \sim \mathcal{U}[0, 1]$. For odd indices, $1 - X_0 \sim \mathcal{U}[0, 1]$ and hence $X_{2n+1} \sim \mathcal{U}[0, 1]$ as well.

- (ii) Assume for contradiction that $X_n \xrightarrow{\mathbb{P}} X$ for some random variable X . Then every subsequence also converges in probability to the same X , we have $X_{2n} \xrightarrow{\mathbb{P}} X$, i.e.,

$$\mathbb{P}(|X_0 - X| > \epsilon) \rightarrow 0.$$

Hence $X = X_0$ a.s. Similarly, $X_{2n+1} \xrightarrow{\mathbb{P}} X$ in probability implies $X = 1 - X_0$ a.s. Combining these, $X_0 = \frac{1}{2}$ a.s., which contradicts $X_0 \sim \mathcal{U}[0, 1]$. □

EXERCISE A.14. Recall that a Poisson random variable X with parameter $\lambda > 0$ satisfies

$$\mathbb{P}(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad \forall k \in \mathbb{N}_0,$$

such that λ is the expected value and the variance of X .

- (i) Show that the characteristic function of a Poisson random variable is

$$\varphi(t) = e^{\lambda(e^{it} - 1)}.$$

- (ii) Consider $X_1, \dots, X_n$ independent Poisson random variables with parameters $\lambda_1, \dots, \lambda_n$. Show that $S_n = X_1 + \dots + X_n$ is a Poisson random variable with parameter $(\lambda_1 + \dots + \lambda_n)$.

- (iii) Show that $e^{-n} \sum_{k=0}^n \frac{n^k}{k!} \rightarrow \frac{1}{2}$ as $n \rightarrow \infty$.

SOLUTION. (i) Since $\sum_{k=0}^{\infty} \mathbb{P}(X = k) = 1$, we have $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$. Then

$$\begin{aligned} \varphi_X(t) &= \sum_{k=0}^{\infty} e^{itk} \mathbb{P}(X = k) \\ &= \sum_{k=0}^{\infty} e^{itk} \frac{e^{-\lambda} \lambda^k}{k!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^{it})^k}{k!} \\ &= e^{-\lambda} e^{\lambda e^{it}} = e^{\lambda(e^{it} - 1)}. \end{aligned}$$

- (ii) By independence,

$$\varphi_{S_n}(t) = \prod_{j=1}^n \varphi_{X_j}(t) = \prod_{j=1}^n \exp(\lambda_j(e^{it} - 1)) = \exp((\lambda_1 + \dots + \lambda_n)(e^{it} - 1)).$$

- (iii) Let $Y_n \sim \text{Poisson}(n)$. Then

$$\mathbb{P}(Y_n \leq n) = \sum_{k=0}^n \mathbb{P}(Y_n = k) = \sum_{k=0}^n \frac{e^{-n} n^k}{k!} = e^{-n} \sum_{k=0}^n \frac{n^k}{k!}.$$

By the *central limit theorem*,

$$\frac{Y_n - n}{\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Therefore,

$$\mathbb{P}(Y_n \leq n) = \mathbb{P}\left(\frac{Y_n - n}{\sqrt{n}} \leq 0\right) \rightarrow \Phi(0) = \frac{1}{2}.$$

□

EXERCISE A.15. Let $X_1, X_2, \dots$ be *i.i.d.* random variables with common distribution given by

$$\mathbb{P}(X_i = 1) = p, \quad \mathbb{P}(X_i = -1) = q = 1 - p.$$

We write $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Let $S_n = \sum_{j=1}^n X_j$.

(i) Show that $M_n := \left(\frac{q}{p}\right)^{S_n}$ is a martingale with respect to the $\mathcal{F}_n$.

(ii) Let $\lambda > 0$. Determine the constant $C = C(\lambda)$ so that $Z_n^\lambda := C^n \lambda^{S_n}$ is an $\mathcal{F}_n$ -martingale.

SOLUTION. (i) M_n is $\mathcal{F}_n$ -measurable and integrable. Then

$$\begin{aligned} \mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}\left[\left(\frac{q}{p}\right)^{S_{n+1}} \mid \mathcal{F}_n\right] \\ &= \mathbb{E}\left[\left(\frac{q}{p}\right)^{S_n} \left(\frac{q}{p}\right)^{X_{n+1}} \mid \mathcal{F}_n\right] \\ &= \left(\frac{q}{p}\right)^{S_n} \mathbb{E}\left[\left(\frac{q}{p}\right)^{X_{n+1}}\right] \\ &= M_n \mathbb{E}\left[\left(\frac{q}{p}\right)^{X_{n+1}}\right] \\ &= M_n \left(p \left(\frac{q}{p}\right)^1 + q \left(\frac{q}{p}\right)^{-1}\right) \\ &= M_n (q + p) = M_n. \end{aligned}$$

(ii)

$$Z_{n+1}^\lambda = C^{n+1} \lambda^{S_{n+1}} = C^{n+1} \lambda^{S_n + X_{n+1}} = Z_n^\lambda \cdot C \lambda^{X_{n+1}}.$$

Then by the martingale property,

$$\mathbb{E}[Z_{n+1}^\lambda \mid \mathcal{F}_n] = Z_n^\lambda C \mathbb{E}[\lambda^{X_{n+1}}] = Z_n^\lambda C (p\lambda + q\lambda^{-1}).$$

Therefore,

$$C(\lambda) = \frac{1}{p\lambda + q\lambda^{-1}}.$$

□

Semester 2023–2024

EXERCISE A.16. Let (S, Σ, μ) be a σ -finite measure space and let ν be another measure on (S, Σ) . Show that if $\nu(A) \leq \mu(A)$, for all $A \in \Sigma$, then $\nu = f\mu$ for some (a.e. uniquely determined) density function f such that $0 \leq f \leq 1$.

SOLUTION. Suppose $\mu(A) = 0$, then $\nu(A) \leq \mu(A) = 0$. Since $\nu(A) \geq 0$, we must have $\nu(A) = 0$. Hence, ν is absolutely continuous with respect to μ . By the *Radon–Nikodym theorem*, there exists a measurable function $f : S \rightarrow [0, \infty)$ such that $\nu(A) = \int_A f d\mu$ for all $A \in \Sigma$. We already know $f \geq 0$ by construction. Assume for contradiction that there exists a set $E \in \Sigma$ with $\mu(E) > 0$ such that $f(x) > 1$ for all $x \in E$. Then

$$\nu(E) = \int_E f d\mu > \int_E 1 d\mu = \mu(E),$$

which violates $\nu(A) \leq \mu(A)$. Therefore, we conclude $0 \leq f \leq 1$. $\square$

EXERCISE A.17. Let m be Lebesgue measure on $([0, 1], \mathcal{B}[0, 1])$. Consider the sequence of functions

$$f_n(x) = n^2 \sum_{j=0}^{n-1} \mathbb{1}_{A_j}, \quad x \in [0, 1], \quad n \in \mathbb{N}, \quad \text{and } A_j = \left[\frac{j}{n}, \frac{j}{n} + \frac{1}{n^3} \right),$$

and the sequence of measures

$$d\mu_n = f_n dm, \quad n \in \mathbb{N}.$$

(i) Use the *Borel–Cantelli lemma* to show that $f_n(x) \rightarrow 0$, m -a.e.

(ii) Show that

$$\lim_{n \rightarrow \infty} \left| \int_{[0,1]} g(x) d\mu_n(x) - \frac{1}{n} \sum_{j=0}^{n-1} g\left(\frac{j}{n}\right) \right| = 0.$$

(iii) Show, using a comparison between Lebesgue and Riemann integrals, that for all continuous functions $g : [0, 1] \rightarrow \mathbb{R}$ we have

$$\lim_{n \rightarrow \infty} \int_{[0,1]} g(x) d\mu_n(x) = \int_{[0,1]} g(x) dm(x).$$

SOLUTION. (i) Each A_j has length $\frac{1}{n^3}$. Since

$$\frac{j}{n} + \frac{1}{n^3} < \frac{j}{n} + \frac{1}{n} = \frac{j+1}{n},$$

the intervals $A_0, A_1, \dots, A_{n-1}$ are disjoint, so the union $B_n := \bigcup_{j=0}^{n-1} A_j$ has measure

$$m(B_n) = \sum_{j=0}^{n-1} m(A_j) = n \cdot \frac{1}{n^3} = \frac{1}{n^2}.$$

Note that

$$f_n(x) = \begin{cases} n^2, & x \in B_n \\ 0, & \text{otherwise} \end{cases}.$$

Hence we want to show that for a.e. x , $\{x \in B_n\}$ only finitely many times. The series $\sum_n m(B_n)$ converges, by the *first Borel–Cantelli lemma*,

$$m\left(\limsup_n B_n\right) = 0.$$

This implies for a.e. $x \in [0, 1]$, there exists $N \in \mathbb{N}$ such that for all $n > N$, we have $f_n(x) = 0$. Therefore, $f_n(x) \rightarrow 0$, m -a.e.

(ii) Since $d\mu_n = f dm$,

$$\begin{aligned}
 \int_{[0,1]} g(x) d\mu_n(x) &= \int_{[0,1]} g(x) f_n(x) dm(x) \\
 &= n^2 \int_{[0,1]} g(x) \sum_{j=0}^{n-1} \mathbb{1}_{A_j} dm(x) \\
 &= n^2 \sum_{j=0}^{n-1} \int_{A_j} g(x) dm(x) \\
 &= n^2 \sum_{j=0}^{n-1} \int_{A_j} \left(g(x) - g\left(\frac{j}{n}\right) \right) dm(x) + n^2 \sum_{j=0}^{n-1} g\left(\frac{j}{n}\right) \int_{A_j} dm(x) \\
 &= n^2 \sum_{j=0}^{n-1} \int_{A_j} \left(g(x) - g\left(\frac{j}{n}\right) \right) dm(x) + n^2 \sum_{j=0}^{n-1} g\left(\frac{j}{n}\right) m(A_j) \\
 &= n^2 \sum_{j=0}^{n-1} \int_{A_j} \left(g(x) - g\left(\frac{j}{n}\right) \right) dm(x) + \frac{1}{n} \sum_{j=0}^{n-1} g\left(\frac{j}{n}\right).
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \left| \int_{[0,1]} g(x) d\mu_n(x) - \frac{1}{n} \sum_{j=0}^{n-1} g\left(\frac{j}{n}\right) \right| &\leq \left| n^2 \sum_{j=0}^{n-1} \int_{A_j} \left(g(x) - g\left(\frac{j}{n}\right) \right) dm(x) \right| \\
 &= n^2 \sum_{j=0}^{n-1} \int_{A_j} \left| g(x) - g\left(\frac{j}{n}\right) \right| dm(x) \\
 &\leq n^2 \sup_{x \in A_j} \left| g(x) - g\left(\frac{j}{n}\right) \right| m(A_j) \\
 &= \frac{1}{n} \sup_{x \in A_j} \left| g(x) - g\left(\frac{j}{n}\right) \right| \rightarrow 0
 \end{aligned}$$

as $n \rightarrow \infty$.

(iii) From (ii) we have

$$\int_{[0,1]} g(x) d\mu_n(x) - \frac{1}{n} \sum_{j=0}^{n-1} g\left(\frac{j}{n}\right) \rightarrow 0$$

as $n \rightarrow \infty$. Since g is continuous on $[0, 1]$, the Riemann sums converge to the integral of g :

$$\frac{1}{n} \sum_{j=0}^{n-1} g\left(\frac{j}{n}\right) \rightarrow \int_0^1 g(x) dx = \int_{[0,1]} g(x) dm(x),$$

which completes the proof. $\square$

EXERCISE A.18. Let $X_1, \dots, X_n$ be random variables with distribution functions $F_1, \dots, F_n$ respectively. Suppose that the characteristic function of X_i is given by ϕ_i , $i = 1, \dots, n$.

- (i) Which random variable has characteristic function $\sum_{i=1}^n \lambda_i \phi_i$, where all $\lambda_i \geq 0$?
- (ii) The characteristic function of the standard exponential distribution is given by $\phi(t) = \frac{1}{1-it}$. Compute the characteristic function of a random variable with density

$$f(x) = \frac{1}{2} e^{-|x|}, \quad \forall x \in \mathbb{R}.$$

(iii) Describe which random variable has characteristic function $\frac{\lambda e^{it}}{\lambda - it}$.

SOLUTION. (i) We construct a random variable Z by selecting the distribution F_i with probability λ_i . The resulting distribution function of Z is

$$F_Z(x) = \sum_{i=1}^n \lambda_i F_i(x).$$

And the characteristic function is

$$\phi_Z(t) = \mathbb{E}[e^{itZ}] = \sum_{i=1}^n \lambda_i \mathbb{E}[e^{itX_i}] = \sum_{i=1}^n \lambda_i \phi_i(t),$$

where the λ_i should add up to 1, otherwise it will not work.

(ii) Let X_1 be exponentially distributed. Then

$$\phi_X(t) = \frac{1}{2} \phi_{X_1}(t) + \frac{1}{2} \phi_{-X_1}(t) = \frac{1}{2} \left(\frac{1}{1 - it} \right) + \frac{1}{2} \left(\frac{1}{1 + it} \right) = \frac{1}{1 + t^2}.$$

(iii) Note that an exponential random variable $Y \sim \exp(\lambda)$ has characteristic function

$$\phi_Y(t) = \frac{\lambda}{\lambda - it}.$$

Therefore, if $Z = Y + 1$, we have

$$\phi_Z(t) = \mathbb{E}[e^{it(Y+1)}] = e^{it} \mathbb{E}[e^{itY}] = e^{it} \phi_Y(t) = \frac{\lambda e^{it}}{\lambda - it}.$$

□

EXERCISE A.19. Let X be a random variable with a uniform distribution on $[-1, 1]$, defined on some probability space $(\Omega, \mathcal{A}, \mathbb{P})$.

(i) Show that $\sigma(|X|)$, the sigma-algebra generated by $|X|$, is generated by sets of the form $\{\omega \in \Omega : -a \leq X(\omega) \leq a\}$.

(ii) Let A be the event that $X < 0$. Show that

$$\mathbb{P}(A \mid \sigma(|X|)) = \frac{1}{2} \quad a.e.$$

SOLUTION. (i) The σ -algebra is generated by the pre-images of the Borel set $\mathcal{B}(|X|) = (-\infty, a]$. Hence $\sigma(|X|) = \sigma(|X| \leq a) = \sigma(\{-a \leq X \leq a\})$.

(ii) We check the integral identity

$$\int_{X \in [-a, a]} \mathbb{P}(A \mid \sigma(|X|)) d\mathbb{P} = \mathbb{P}(A \cap \{-a \leq X \leq a\}) = \mathbb{P}(\{-a \leq X \leq 0\}).$$

When the integrand is $\frac{1}{2}$, this identity is correct.

□

EXERCISE A.20. Let $X_0 = 0$ be a degenerate random variable. When $X_{k-1} = 0$,

$$X_k = \begin{cases} 1, & \frac{1}{2k} \\ -1, & \frac{1}{2k} \\ 0, & 1 - \frac{1}{k} \end{cases}.$$

When $X_{k-1} \neq 0$,

$$X_k = \begin{cases} kX_{k-1}, & \frac{1}{k} \\ 0, & 1 - \frac{1}{k} \end{cases}.$$

- (i) Show that $X_0, X_1, \dots$ is a martingale with respect to the natural sequence of sigma-algebras.
- (ii) Show that $X_k \rightarrow 0$ in probability as $k \rightarrow \infty$.
- (iii) Show that $X_0, X_1, \dots$ does not converge *a.s.* Why does this not contradict the martingale convergence theorem?

SOLUTION. (i) When $k = 0$, by definition,

$$\mathbb{E}[X_{k+1} \mid X_k = 0] = \frac{1}{2(k+1)} - \frac{1}{2(k+1)} + 0 = 0 = X_k.$$

When $k \neq 0$, similarly,

$$\mathbb{E}[X_{k+1} \mid X_k] = (k+1)X_k \frac{1}{k+1} + 0 = X_k.$$

Hence, X_k is a martingale with respect to the natural filtration $\mathcal{F}_k$.

- (ii) When $X_{k-1} = 0$, $\mathbb{P}(X_k \neq 0) = \frac{1}{2k} + \frac{1}{2k} = \frac{1}{k}$. When $X_{k-1} \neq 0$, $\mathbb{P}(X_k \neq 0) = \frac{1}{k}$. Hence, regardless of the value of X_{k-1} , we have

$$\mathbb{P}(X_k \neq 0 \mid X_{k-1} = y) \leq \frac{1}{k},$$

then

$$\begin{aligned} \mathbb{P}(X_k \neq 0) &= \sum_y \mathbb{P}(X_k \neq 0 \mid X_{k-1} = y) \mathbb{P}(X_{k-1} = y) \\ &\leq \frac{1}{k} \sum_y \mathbb{P}(X_{k-1} = y) = \frac{1}{k} \rightarrow 0 \end{aligned}$$

as $k \rightarrow \infty$, where X_k are integer-valued.

- (iii) According to (ii), if it converges, then it must converge to 0. Let (Y_k) be *i.i.d.* random variables with

$$\mathbb{P}(Y_k = 1) = \frac{1}{k}, \quad \mathbb{P}(Y_k \neq 1) = 1 - \frac{1}{k}.$$

It follows from the *second Borel-Cantelli lemma* that $\mathbb{P}(Y_k = 1 \text{ i.o.}) = 1$, since $\sum_{k=1}^{\infty} \frac{1}{k} = \infty$. By construction, the same must be true for the X_k .

Since $\sup_k \mathbb{E}[|X_k|] = \infty$, there is no violation of the *martingale convergence theorem*. □

EXERCISE A.21. Let X and Y be random variables on $(\Omega, \mathcal{F}, P)$, and suppose that $X = Y$ on some set $B \in \mathcal{F}$. Show that $\mathbb{E}[X \mid \mathcal{F}] = \mathbb{E}[Y \mid \mathcal{F}]$ *a.s.* on B .

SOLUTION. Given that X and Y are $\mathcal{F}$ -measurable, then it is immediate. In that case $\mathbb{E}[X \mid \mathcal{F}] = X$ and $\mathbb{E}[Y \mid \mathcal{F}] = Y$ *a.s.* and the claim follows. □

Remark. If you don't use that, then we have

$$\int_F \mathbb{E}[X \mid \mathcal{F}] d\mathbb{P} = \int_F X d\mathbb{P}, \quad \int_F \mathbb{E}[Y \mid \mathcal{F}] d\mathbb{P} = \int_F Y d\mathbb{P}, \quad \forall F \in \mathcal{F}.$$

This gives that

$$\int_{B'} \mathbb{E}[X | \mathcal{F}] d\mathbb{P} = \int_{B'} \mathbb{E}[Y | \mathcal{F}] d\mathbb{P}, \quad \forall B' \subseteq B.$$

This can only happen if the integrands are the same over B .

EXERCISE A.22. Suppose that μ^* is an outer measure on S and $\{A_j\}_{j=1}^\infty$ is a sequence of disjoint μ^* -measurable sets. Let $B_n := \bigcup_{j=1}^n A_j$ (and $B_0 := \emptyset$).

(i) Show by induction that

$$\mu^*(E \cap B_n) = \sum_{j=1}^n \mu^*(E \cap A_j).$$

(ii) Show that

$$\mu^*\left(E \cap \bigcup_{j=1}^\infty A_j\right) = \sum_{j=1}^\infty \mu^*(E \cap A_j),$$

for any $E \subseteq S$.

SOLUTION. (i) When $n = 1$, we have $B_1 = A_1$, then $E \cap B_1 = E \cap A_1$. Clearly

$$\mu^*(E \cap B_1) = \sum_{j=1}^1 \mu^*(E \cap A_j).$$

Assume the statement is true for $n - 1$, namely

$$\mu^*(E \cap B_{n-1}) = \sum_{j=1}^{n-1} \mu^*(E \cap A_j).$$

Observe that

$$B_n = \bigcup_{j=1}^n A_j = B_{n-1} \cup A_n.$$

Since A_n is μ^* -measurable, by the *Carathéodory's criterion*, for the set $E \cap (B_{n-1} \cup A_n) \subseteq S$, we have

$$\begin{aligned} \mu^*(E \cap (B_{n-1} \cup A_n)) &= \mu^*(E \cap (B_{n-1} \cup A_n) \cap A_n) + \mu^*(E \cap (B_{n-1} \cup A_n) \cap A_n^c) \\ &= \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1}). \end{aligned} \quad (B_{n-1} \subseteq A_n^c)$$

Then

$$\begin{aligned} \mu^*(E \cap B_n) &= \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1}) \\ &= \mu^*(E \cap A_n) + \sum_{j=1}^{n-1} \mu^*(E \cap A_j) = \sum_{j=1}^n \mu^*(E \cap A_j). \end{aligned}$$

(ii) Since $B_n = \bigcup_{j=1}^n A_j$, by countably subadditive,

$$\mu^*(E \cap B) = \mu^*\left(E \cap \bigcup_{j=1}^\infty A_j\right) \leq \sum_{j=1}^\infty \mu^*(E \cap A_j).$$

By (i), we have

$$\sum_{j=1}^n \mu^*(E \cap A_j) = \mu^*(E \cap B_n) \leq \mu^*(E \cap B).$$

Taking the limit as $n \rightarrow \infty$, we get

$$\sum_{j=1}^{\infty} \mu^*(E \cap A_j) \leq \mu^*(E \cap B).$$

Therefore,

$$\mu^*\left(E \cap \bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} \mu^*(E \cap A_j).$$

□

EXERCISE A.23. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

(i) Suppose that there exists a positive sequence $a_n \rightarrow 0$ as $n \rightarrow \infty$ so that

$$\sum_{n \in \mathbb{N}} \mathbb{P}(|X_n| > a_n) < \infty.$$

Show that under this assumption, $X_n \rightarrow 0$ a.s.

(ii) Show by providing a counterexample that the converse of (i) is false.

SOLUTION. (i) Let $A_n := \{|X_n| > a_n\}$. Then $\sum_{n \in \mathbb{N}} \mathbb{P}(A_n) < \infty$. By the first Borel-Cantelli lemma, $\mathbb{P}(\limsup_n A_n) = 0$, which means only finitely many of the events A_n occur. Then for a.e. $\omega \in \Omega$, there exists N such that for all $n > N$: $|X_n(\omega)| \leq a_n \rightarrow 0$ as $n \rightarrow \infty$. Therefore $X_n \rightarrow 0$ a.s.

(ii) Let $(\Omega, \mathcal{F}, \mathbb{P}) = ((0, 1], \mathcal{B}(0, 1], \lambda)$, where λ is Lebesgue measure. Define

$$A_n := \left(0, \frac{1}{n}\right], \quad X_n := \mathbb{1}_{A_n}.$$

Fix $\omega \in (0, 1]$. Choose $N > \frac{1}{\omega}$. Then for all $n \geq N$ we have $\frac{1}{n} < \omega$, hence $\omega \notin (0, \frac{1}{n}] = A_n$, so $X_n(\omega) = 0$ for all $n \geq N$. Therefore $X_n \rightarrow 0$ a.s. However,

$$\sum_{n \in \mathbb{N}} \mathbb{P}(|X_n| > a_n) = \sum_{n \in \mathbb{N}} \mathbb{P}(A_n) = \sum_{n \in \mathbb{N}} \lambda\left(0, \frac{1}{n}\right] = \sum_{n \in \mathbb{N}} \frac{1}{n} = \infty.$$

□

EXERCISE A.24. Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, let $A, A_1, A_2, \dots, B \in \mathcal{F}$ and let $\mathcal{G} \subseteq \mathcal{F}$.

(i) Show that if $A \subseteq B$, then

$$\mathbb{P}(B \cap A^c \mid \mathcal{G}) = \mathbb{P}(B \mid \mathcal{G}) - \mathbb{P}(A \mid \mathcal{G}), \quad \mathbb{P}\text{-a.s.}$$

(ii) Show that if $A_n \uparrow A$, then

$$\mathbb{P}(A_n \mid \mathcal{G}) \uparrow \mathbb{P}(A \mid \mathcal{G}), \quad \mathbb{P}\text{-a.s.}$$

SOLUTION. (i) Since $A \subseteq B$,

$$B = (B \cap A) \cup (B \cap A^c) = A \cup (B \cap A^c).$$

Then $\mathbb{1}_B = \mathbb{1}_A + \mathbb{1}_{B \cap A^c}$. Taking conditional expectations and using linearity,

$$\mathbb{E}[\mathbb{1}_{B \cap A^c} \mid \mathcal{G}] = \mathbb{E}[\mathbb{1}_B - \mathbb{1}_A \mid \mathcal{G}] = \mathbb{E}[\mathbb{1}_B \mid \mathcal{G}] - \mathbb{E}[\mathbb{1}_A \mid \mathcal{G}].$$

By the definition of conditional probability, this is exactly

$$\mathbb{P}(B \cap A^c \mid \mathcal{G}) = \mathbb{P}(B \mid \mathcal{G}) - \mathbb{P}(A \mid \mathcal{G}), \quad \mathbb{P}\text{-a.s.}$$

- (ii) Since $A_n \uparrow A$, we have $\mathbb{1}_{A_n} \uparrow \mathbb{1}_A$ pointwise. Let $Y_n := \mathbb{E}[\mathbb{1}_{A_n} \mid \mathcal{G}]$, by monotonicity (Y_n) is a non-decreasing sequence, so $Y := \lim_n Y_n$ exists. For $G \in \mathcal{G}$, by the *monotone convergence theorem*,

$$\begin{aligned} \int_G Y d\mathbb{P} &= \int_G \lim_{n \rightarrow \infty} Y_n d\mathbb{P} \\ &= \lim_{n \rightarrow \infty} \int_G Y_n d\mathbb{P} \\ &= \lim_{n \rightarrow \infty} \int_G \mathbb{1}_{A_n} d\mathbb{P} \\ &= \int_G \mathbb{1}_A d\mathbb{P} = \int_G \mathbb{E}[\mathbb{1}_A \mid \mathcal{G}] d\mathbb{P}. \end{aligned}$$

It follows that $Y = \mathbb{E}[\mathbb{1}_A \mid \mathcal{G}]$, $\mathbb{P}$ -a.s. Therefore, $\mathbb{E}[\mathbb{1}_{A_n} \mid \mathcal{G}] \uparrow \mathbb{E}[\mathbb{1}_A \mid \mathcal{G}]$, and hence $\mathbb{P}(A_n \mid \mathcal{G}) \uparrow \mathbb{P}(A \mid \mathcal{G})$, $\mathbb{P}$ -a.s.

□

EXERCISE A.25. The Cauchy distribution has density

$$f(x) = \frac{1}{\pi(1+x^2)},$$

for all $x \in \mathbb{R}$. Its characteristic function is given by $\phi(t) = e^{-|t|}$.

- (i) If $X_1, \dots, X_n$ are independent and have a Cauchy distribution, determine the distribution of $\frac{X_1 + \dots + X_n}{n}$, for all $n = 1, 2, \dots$
- (ii) Explain why your answer in (i) does not violate the *weak law of large numbers*.

SOLUTION. (i) Let $\bar{X}_n = \frac{1}{n}(X_1 + \dots + X_n)$. Then

$$\begin{aligned} \phi_{\bar{X}_n}(t) &= \mathbb{E}\left[e^{it\bar{X}_n}\right] \\ &= \prod_{k=1}^n \mathbb{E}\left[e^{it\frac{X_k}{n}}\right] \\ &= \left(\phi\left(\frac{t}{n}\right)\right)^n \\ &= \left(e^{-|\frac{t}{n}|}\right)^n = e^{-|t|}. \end{aligned}$$

So the distribution does not change.

- (ii) Since

$$\begin{aligned} \mathbb{E}[|X|] &= \int_{-\infty}^{\infty} |x| \frac{1}{\pi(1+x^2)} dx \\ &= \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx \\ &= \frac{1}{\pi} \ln(1+x^2) \Big|_0^{\infty} = \infty, \end{aligned}$$

the assumption of the *weak law of large numbers* is not satisfied.

□

EXERCISE A.26. Consider the triangular array X_{nk} of random variables, with $n = 1, 2, \dots$ and $k = 1, \dots, r_n$. Let

$$S_n = \sum_{k=1}^{r_n} X_{nk}, \quad s_n^2 = \sum_{k=1}^{r_n} \mathbb{E}[X_{nk}^2].$$

Suppose that $|X_{nk}| = m_n$ a.s. and that $\frac{m_n}{s_n} \rightarrow 0$ as $n \rightarrow \infty$. Prove a *central limit theorem* for S_n .

SOLUTION. Since $|X_{nk}| = m_n$ a.s.,

$$\begin{aligned} \frac{1}{s_n^{2+\delta}} \sum_{k=1}^{r_n} \mathbb{E}[|X_{nk}|^{2+\delta}] &= \frac{1}{s_n^{2+\delta}} \sum_{k=1}^{r_n} \mathbb{E}[m_n^\delta X_{nk}^2] \\ &= \frac{m_n^\delta}{s_n^{2+\delta}} \sum_{k=1}^{r_n} \mathbb{E}[X_{nk}^2] \\ &= \frac{m_n^\delta s_n^2}{s_n^{2+\delta}} = \left(\frac{m_n}{s_n}\right)^\delta \rightarrow 0, \end{aligned}$$

where $\frac{m_n}{s_n} \rightarrow 0$ as $n \rightarrow \infty$. Hence the Lyapunov condition holds and it proves the central limit theorem for S_n . $\square$

EXERCISE A.27. Suppose you play the following game. You start with £1. You flip a fair coin repeatedly; if heads comes up you lose £1, and if tails comes up you win £1. The rule is that you stop playing as soon as you have either £0 or £10.

- (i) Formulate this set-up with a martingale, together with corresponding stopping times.
- (ii) Use martingale theory to prove that the probability that after playing this game you have £10 is equal to $\frac{1}{10}$.

SOLUTION. (i) We define a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $X_1, X_2, \dots$ be a sequence of *i.i.d.* random variables representing the outcome of each coin flip, such that:

$$\mathbb{P}(X_n = 1) = \frac{1}{2}, \quad \mathbb{P}(X_n = -1) = \frac{1}{2}.$$

Let S_n denote the wealth after n flips, with the initial wealth $S_0 = 1$. Then

$$S_n = S_0 + \sum_{i=1}^n X_i.$$

Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ be the natural filtration generated by the random variables, with $\mathcal{F}_0 = \{\emptyset, \Omega\}$. S_n is adapted to $\mathcal{F}_n$ and is integrable, with the martingale property:

$$\begin{aligned} \mathbb{E}[S_{n+1} | \mathcal{F}_n] &= \mathbb{E}[S_n + X_{n+1} | \mathcal{F}_n] \\ &= S_n + \mathbb{E}[X_{n+1} | \mathcal{F}_n] \\ &= S_n + \mathbb{E}[X_{n+1}] = S_n. \end{aligned}$$

Thus, S_n is a martingale. We define the stopping time T as:

$$T = \inf\{n \geq 0 : S_n = 0, \text{ or } S_n = 10\}.$$

- (ii) Since the random walk on a finite interval reaches the boundary with probability 1, $S_T \in \{0, 10\}$. Let

$$p = \mathbb{P}(S_T = 10), \quad \mathbb{P}(S_T = 0) = 1 - p.$$

Then

$$\mathbb{E}[S_T] = 10 \cdot \mathbb{P}(S_T = 10) + 0 \cdot \mathbb{P}(S_T = 0) = 10p = 1,$$

which implies $p = \frac{1}{10}$.

□

SECTION A.2

| Class Tests

Semester 2025–2026

EXERCISE A.28. Consider the space $\Omega = \{1, 2, \dots\}$ with σ -field the collection of all subsets of Ω . Let $\mathbb{P}$ be a probability measure on this σ -field. Assume that there is an infinite sequence $A_1, A_2, \dots$ of independent events, all with probability $\frac{1}{2}$.

- (i) For all $\omega \in \Omega$, ω is contained in either $A_1 \cap A_2$, $A_1^c \cap A_2$, $A_1 \cap A_2^c$ or $A_1^c \cap A_2^c$. Use this to show that $\mathbb{P}(\{\omega\}) \leq \frac{1}{4}$.
- (ii) Show that $\mathbb{P}(\{\omega\}) = 0$ for all $\omega \in \Omega$ and explain why this is a contradiction.

SOLUTION. (i) Note that

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1^c \cap A_2) = \mathbb{P}(A_1 \cap A_2^c) = \mathbb{P}(A_1^c \cap A_2^c) = \frac{1}{4},$$

by independence. The singleton $\{\omega\}$ is contained in exactly one of them, so its probability cannot be larger than $\frac{1}{4}$.

- (ii) By the same token, for every k , $\{\omega\}$ is contained in precisely one set of the form $B_1 \cap B_2 \cap \dots \cap B_k$, where each B_i is either A_i or A_i^c . Since each such intersection has probability 2^{-k} we conclude that $\mathbb{P}(\{\omega\}) \leq 2^{-k}$ for all k , hence $\mathbb{P}(\{\omega\}) = 0$. Since

$$\sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) = 1,$$

this is a contradiction. Hence a sequence as assumed cannot exist. □

EXERCISE A.29. On the field $\mathcal{B}_0$ in $(0, 1]$, let $Q(A)$ be 1 if A contains an interval of the form $(\frac{1}{2}, \frac{1}{2} + \epsilon]$ (where ϵ can depend on A), and $Q(A) = 0$ otherwise. Show that Q is finitely additive but not countably additive.

SOLUTION. First we show that Q is finitely additive. Let A and B be in $\mathcal{B}_0$ with $A \cap B = \emptyset$. Then

- $Q(A) = Q(B) = 1$ is impossible, since in that case the intersection would contain an interval $(\frac{1}{2}, \frac{1}{2} + \min(\epsilon_A, \epsilon_B)]$.
- If $Q(A) = Q(B) = 0$, then $A \cup B$ does not contain an interval as described, hence $Q(A \cup B) = 0$.
- If $Q(A) = 1$ and $Q(B) = 0$, then $A \cup B$ contains an interval of the required form and $Q(A \cup B) = 1$.

To show that Q is not countably additive, take

$$A_n := \left(\frac{1}{2} + \frac{1}{n+1}, \frac{1}{2} + \frac{1}{n} \right], \quad n = 2, 3, \dots$$

Then $Q(A_n) = 0$ for all n but

$$\bigcup_{n=2}^{\infty} A_n = \left(\frac{1}{2}, 1 \right] \implies Q\left(\bigcup_{n=2}^{\infty} A_n\right) = 1.$$

□

EXERCISE A.30. Let $A_1, A_2, \dots$ be an infinite sequence of events in a probability space. Let

$$A^* := \limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k.$$

Show that $\mathbb{P}(A_n - A^*) \rightarrow 0$ as $n \rightarrow \infty$.

SOLUTION. Let $B_n := \bigcup_{k=n}^{\infty} A_k$. Then $B_n \downarrow \bigcap_{n=1}^{\infty} B_n = \limsup_n A_n = A^*$. By continuity from above of probability measures,

$$\mathbb{P}(B_n) \downarrow \mathbb{P}(A^*) \implies \mathbb{P}(B_n - A^*) = \mathbb{P}(B_n) - \mathbb{P}(A^*) \rightarrow 0.$$

Since $A_n \subset B_n$, we have $A_n \setminus A^* \subset B_n \setminus A^*$, hence

$$\mathbb{P}(A_n - A^*) \leq \mathbb{P}(B_n - A^*) \rightarrow 0,$$

as $n \rightarrow \infty$. □

EXERCISE A.31. Let $X_1, \dots, X_n$ be identically distributed integrable random variables and write $S_n = X_1 + \dots + X_n$. Show that *a.s.* we have

$$\mathbb{E}[X_1 | S_n] = \frac{S_n}{n}.$$

SOLUTION. Naturally, $\mathbb{E}[S_n | S_n] = S_n$. Since the X_i are identically distributed, $\mathbb{E}[X_i | S_n] = \mathbb{E}[X_1 | S_n]$. Using [Proposition 6.11\(ii\)](#), we derive

$$\begin{aligned} S_n &= \mathbb{E}[S_n | S_n] \\ &= \mathbb{E}\left[\sum_i X_i | S_n\right] \\ &= \sum_i \mathbb{E}[X_i | S_n] = n\mathbb{E}[X_1 | S_n], \end{aligned}$$

yielding the result. □

EXERCISE A.32. Let X and Y be independent random variables, both uniformly distributed on $(0, 1)$. Show that *a.s.* $\mathbb{E}[XY | X] = \frac{X}{2}$.

SOLUTION. $\frac{X}{2}$ is clearly measurable $\sigma(X)$ and integrable.

$$\mathbb{E}[XY | X] = X\mathbb{E}[Y | X] = X\mathbb{E}[Y] = \frac{X}{2}.$$

The first identity is possible by [Theorem 6.12](#), the second by independence, and the third follows from the distribution of Y . □

Remark. Alternatively, we can let $G \in \sigma(X)$ and write

$$\begin{aligned} \int_G XY d\mathbb{P} &= \int_G \mathbf{1}_G XY d\mathbb{P} \\ &= \int_G \mathbf{1}_G X d\mathbb{P} \int_{\Omega} Y d\mathbb{P} \\ &= \int_G X d\mathbb{P} \cdot \frac{1}{2} = \int_G \frac{X}{2} d\mathbb{P}, \end{aligned}$$

using the independence of $\sigma(X)$ and $\sigma(Y)$ in the second line.

EXERCISE A.33. Let X and Y be independent random variables, and suppose that X has distribution μ on $(\mathbb{R}, \mathcal{B})$. *Fubini's theorem* implies that for Borel sets $H \subset \mathbb{R}$ and $J \subset \mathbb{R}^2$ we have

$$\mathbb{P}(X \in H, (X, Y) \in J) = \int_H \mathbb{P}((x, Y) \in J) d\mu(x).$$

Now let

$$f(x) = \mathbb{P}((x, Y) \in J).$$

Show that $f(X) = \mathbb{P}((X, Y) \in J \mid X)$ a.s.

SOLUTION. First, note that f is $\sigma(X)$ -measurable by construction. We write

$$\begin{aligned} \int_{X \in H} f(X) d\mathbb{P} &= \int_{X \in H} f(X(\omega)) \mathbb{P}(d\omega) \\ &= \mathbb{E}[\mathbf{1}_{\{X \in H\}} f(X)] \\ &= \int_H f(x) d\mu(x) \\ &= \int_H \mathbb{P}((x, Y) \in J) d\mu(x) = \mathbb{P}(X \in H, (X, Y) \in J). \end{aligned}$$

This proves the claim. $\square$

Semester 2024–2025

EXERCISE A.34. Let $\mathbb{P}$ be a probability measure on a sigma-field $\mathcal{F}$, on a space Ω . Let $A, B \in \mathcal{F}$ and define

$$C := \{\omega \in \Omega : \omega \text{ is in exactly one of } A \text{ and } B\}.$$

Show that

$$\mathbb{P}(C) = \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B).$$

SOLUTION. We write $A \cup B$ as a disjoint union:

$$A \cup B = (A \cap B^c) \cup (B \cap A^c) \cup (A \cap B).$$

Since

$$C = (A \cap B^c) \cup (B \cap A^c),$$

we see that $C \in \mathcal{F}$ and that

$$\mathbb{P}(C) = \mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B).$$

$\square$

EXERCISE A.35. Consider $(0, 1]$ and the Borel sigma-field $\mathcal{B}$ in $(0, 1]$. A measure μ on $\mathcal{B}$ is called *atom free* if $\mu(B) > 0$ implies that there exists $A \subset B$ such that $0 < \mu(A) < \mu(B)$.

- (i) Give an example of a measure μ on $\mathcal{B}$ which is not atom free.
- (ii) Let λ be Lebesgue measure on $(0, 1]$ and let $B \subset (0, 1]$ be a Borel set such that $\lambda(B) > 0$. Define $f : (0, 1] \rightarrow [0, 1]$ as

$$f(x) := \lambda(B \cap (0, x]).$$

Show that f is continuous.

- (iii) Show that λ is atom free.

SOLUTION. (i) For instance the probability measure ν given by $\nu(A) = \mathbb{1}_{\{1 \in A\}}$, then $\nu(\{1\}) = 1$. The only measurable subsets of $\{1\}$ are $\emptyset$ and $\{1\}$, with ν -measures 0 and 1 respectively. Therefore, there is no $A \subset \{1\}$ such that $0 < \nu(A) < 1$.

(ii) Fix $x, y \in (0, 1]$. Without loss of generality assume $x < y$. Then

$$f(y) - f(x) = \lambda(B \cap (0, y]) - \lambda(B \cap (0, x]) = \lambda(B \cap (x, y]).$$

Since $B \cap (x, y] \subset (x, y]$, we have

$$0 \leq f(y) - f(x) = \lambda(B \cap (x, y]) \leq \lambda((x, y]) = y - x.$$

Thus $|f(y) - f(x)| \leq |y - x|$ for all x and y , so f is Lipschitz and therefore continuous on $(0, 1]$.

(iii) $f(0) = 0$, $f(1) = \lambda(B) > 0$ and f is continuous. By the *intermediate value theorem*, for every $0 < \alpha < f(1)$ there is $y \in (0, 1]$ so that $f(y) = \alpha$. Let $A := B \cap (0, y]$, then $A \subset B$, $A \in \mathcal{B}$ and $\lambda(A) = \alpha$.

□

EXERCISE A.36. Consider a sequence of random variables $(X_n)_{n=1}^\infty$ on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$.¹ Show that

$$\mathbb{E} \left[\sum_{n=1}^{\infty} X_n \right] = \sum_{n=1}^{\infty} \mathbb{E}[X_n],$$

in each of the following cases:

(i) All X_n are non-negative.

(ii) $\mathbb{E}[\sum_{n=1}^{\infty} |X_n|] < \infty$.

SOLUTION. Denote $S_n = \sum_{k=1}^n X_k$.

(i) If $X_n \geq 0$ for all $n \in \mathbb{N}$, the sequence of random variables $(S_n)_{n \geq 1}$ is non-negative and non-decreasing. Hence $S_n \uparrow \sum_{k=1}^{\infty} X_k \in \mathbb{R} \cup \{\infty\}$ a.e. Thus by the *monotone convergence theorem*,

$$\begin{aligned} \mathbb{E} \left[\sum_{k=1}^{\infty} X_k \right] &= \mathbb{E} \left[\lim_{n \rightarrow \infty} S_n \right] \\ &= \int \lim_{n \rightarrow \infty} S_n(\omega) \mathbb{P}(d\omega) \\ &= \lim_{n \rightarrow \infty} \int S_n(\omega) \mathbb{P}(d\omega) \\ &= \lim_{n \rightarrow \infty} \mathbb{E} \left[\sum_{k=1}^n X_k \right] \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \mathbb{E}[X_k] = \sum_{k=1}^{\infty} \mathbb{E}[X_k] \in \mathbb{R} \cup \{\infty\}. \end{aligned}$$

(ii)

$$\mathbb{E} \left[\sum_{n=1}^{\infty} |X_n| \right] < \infty \implies \sum_{n=1}^{\infty} |X_n| < \infty \quad a.e.$$

Since absolute convergence in $\mathbb{R}$ implies $\sum_{n=1}^{\infty} X_n$ convergence in $\mathbb{R}$, we have $S_n \xrightarrow{n \rightarrow \infty} \sum_{k=1}^{\infty} X_k \in \mathbb{R}$ a.e. Then

$$|S_n| = \left| \sum_{k=1}^n X_k \right| \leq \sum_{k=1}^n |X_k| \leq \sum_{k=1}^{\infty} |X_k|, \quad \forall n \in \mathbb{N}.$$

¹ X_n are measurable functions from Ω to $\mathbb{R} \cup \{-\infty, \infty\}$.

Hence, by the *dominated convergence theorem*,

$$\mathbb{E}\left[\sum_{k=1}^{\infty} X_k\right] = \mathbb{E}\left[\lim_{n \rightarrow \infty} S_n\right] = \lim_{n \rightarrow \infty} \mathbb{E}[S_n] = \sum_{k=1}^{\infty} \mathbb{E}[X_k].$$

□

EXERCISE A.37. Consider a sequence of random variables $(X_n)_{n=1}^{\infty}$ on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

(i) Suppose that there exists a positive sequence $a_n \rightarrow 0$ as $n \rightarrow \infty$ such that

$$\sum_{n \in \mathbb{N}} \mathbb{P}(|X_n| > a_n) < \infty.$$

Conclude that, under this assumption, $X_n \rightarrow 0$ a.s.

(ii) Show by a suitable counterexample that the converse of (i) is false.

SOLUTION. (i) Define $A_n = \{\omega \in \Omega : |X_n(\omega)| > a_n\}$. By assumption, $\sum_{n \in \mathbb{N}} \mathbb{P}(A_n) < \infty$. Hence, by the *first Borel-Cantelli Lemma*,

$$\mathbb{P}(|X_n| > a_n \text{ i.o.}) = \mathbb{P}(A_n \text{ i.o.}) = 0.$$

In other words, for almost all $\omega \in \Omega$, there is an $N \in \mathbb{N}$ such that $|X_n| \leq a_n$ for all $n \geq N$. Since $a_n \rightarrow 0$, $X_n \rightarrow 0$ a.s.

(ii) Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}(\Omega)$, $\mathbb{P} = \lambda$, i.e., the Lebesgue measure. Set $X_n := \mathbb{1}_{[0, \frac{1}{n}]}$. For $\omega > 0$ there is an $N \in \mathbb{N}$ such that $X_n(\omega) = \mathbb{1}_{[0, \frac{1}{n}]}(\omega) = 0$ for all $n \geq N$, i.e., $X_n \rightarrow 0$ a.s. Take any positive sequence (a_n) with $a_n \rightarrow 0$. Then

$$\begin{aligned} \sum_{n \in \mathbb{N}} \mathbb{P}(|X_n| > a_n) &= \sum_{n \in \mathbb{N}} \mathbb{P}(\omega : \mathbb{1}_{[0, \frac{1}{n}]}(\omega) > a_n > 0) \\ &= \sum_{n \in \mathbb{N}} \lambda\left(\left[0, \frac{1}{n}\right]\right) \geq \sum_{n \geq K} \frac{1}{n} = \infty, \end{aligned}$$

where $a_n < 1$ for all $n \geq K$.

□

EXERCISE A.38. Consider the unit interval with the Borel sigma-field and Lebesgue measure $\mathbb{P}$. Let $\mathcal{G}$ be the sigma-field of all sets that are either countable or co-countable (that is, their complement is countable). Show that $\mathbb{P}(A)$ is a version of $\mathbb{P}(A \mid \mathcal{G})$.

SOLUTION. $\mathbb{P}(A)$ is clearly measurable since it is a constant function. We check whether

$$\int_G \mathbb{P}(A) d\mathbb{P} = \mathbb{P}(A \cap G), \quad G \in \mathcal{G}.$$

Note that if G is countable, $\mathbb{P}(G) = 0$ and both sides are 0, while if G is co-countable, both sides are equal to $\mathbb{P}(A)$. This yields the desired result. □

EXERCISE A.39. Let the random variables U and V be independent and uniformly distributed on $(0, 1)$. Let $X = \min\{U, V\}$ and $Y = \max\{U, V\}$. Show that $\frac{Y}{2}$ is a version of $\mathbb{E}[X \mid Y]$.

SOLUTION. Clearly $\frac{1}{2}Y$ is integrable. We check whether

$$\int_G \frac{1}{2}Y d\mathbb{P} = \int_G X d\mathbb{P}, \quad G \in \sigma(Y).$$

By [Theorem 6.10](#), it suffices to check this for the sets $\{Y \leq a\}$, $a \in (0, 1)$.² A unified approach is to express the integral in terms of u and v . That yields

$$\int_{\{Y \leq a\}} \frac{1}{2}Y d\mathbb{P} = \int_0^a \int_0^u \frac{u}{2} dv du + \int_0^a \int_u^a \frac{v}{2} dv du = \frac{1}{3}a^3,$$

and similarly

$$\int_{\{Y \leq a\}} X d\mathbb{P} = \int_0^a \int_0^u v dv du + \int_0^a \int_u^a u dv du = \frac{1}{3}a^3.$$

□

² There are several ways to check this, e.g. by showing that the density of Y is equal to $2y$. This approach is however more difficult for the right hand side of the equation.

SECTION A.3

| Assignments

Semester 2023–2024

EXERCISE A.40. (i) Let $(X, \mathcal{A})$ be a measurable space and let $x, y \in X$. Consider the measures $\mu = \delta_x$ and $\nu = \delta_y$ on $(X, \mathcal{A})$. Is $\mu = \nu$ equivalent to $x = y$?

(ii) Is there a σ -finite measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\mu((a, b]) = \infty$ whenever $a < b$?

(iii) For a sequence $(x_n)_{n \geq 1}$ in $\mathbb{R}$ define a measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ by $\mu = \sum_{n \geq 1} \delta_{x_n}$. For which sequences $(x_n)_{n \geq 1}$ is this measure σ -finite?

SOLUTION. (i) By the definition of the Dirac delta measure, for any $A \subseteq \mathcal{A}$ we have

$$\delta_x(A) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}, \quad \delta_y(A) = \begin{cases} 1 & y \in A \\ 0 & y \notin A \end{cases}.$$

If $x = y$, for any countable set $A \subseteq \mathcal{A}$, suppose $x \in A$ (which implies $y \in A$ since $x = y$), we have $\delta_x(A) = \delta_y(A) = 1$. Otherwise, $\delta_x(A) = \delta_y(A) = 0$ when $x, y \notin A$. In both cases $\delta_x(A) = \delta_y(A)$.

If $x \neq y$, then there exists a set $A = \{x\}$ such that $x \in A$ and $y \notin A$ (or vice versa), and we have $1 = \delta_x(A) \neq \delta_y(A) = 0$, which implies $\delta_x(A) \neq \delta_y(A)$.

Therefore, we conclude $\delta_x(A) = \delta_y(A)$; i.e., $\mu = \nu$ is equivalent to $x = y$.

(ii) No. We can consider a measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$:

$$\mu(E) = \begin{cases} 0 & \text{if } E = \emptyset \\ \infty & \text{otherwise} \end{cases},$$

which is well-defined and satisfies the definition of $\mu((a, b]) = \infty$.

If we take E_n to be any non-empty subset of $\mathbb{R}$, then $\mu(E_n) = \infty$. If we take E_n to be empty sets, we have $\mu(E_n) = 0$, but the union of countably many empty sets is still the empty set and cannot cover $\mathbb{R}$.

For a measure to be σ -finite on $\mathbb{R}$, there must exist a countable collection of sets E_n such that $\mu(E_n) < \infty$ for all n and $\bigcup_{n=1}^{\infty} E_n = \mathbb{R}$. Thus, any countable union of $(a, b]$, i.e., any open interval, should be assigned a finite value, which contradicts the definition. Therefore, the measure μ is not σ -finite on $\mathbb{R}$.

(iii) By the definition of the Dirac delta measure, for any Borel set $A \subseteq \mathbb{R}$ we have

$$\delta_{x_n}(A) = \begin{cases} 1 & x_n \in A \\ 0 & x_n \notin A \end{cases}.$$

If the sequence $(x_n)_{n \geq 1}$ has only finitely many distinct terms, then μ is σ -finite. This is because we can cover each distinct term with a Borel set of measure 1 using the Dirac delta measure. The sum of the measures of these sets is finite.

If the sequence $(x_n)_{n \geq 1}$ has infinitely many distinct terms but they are countable, then μ is also σ -finite. We can cover each term x_n with a Borel set A_n such that $A_n = \{x_n\}$ and $\mu(A_n) = 1$. The union of these sets will cover all the terms in the sequence which is countable.

If the sequence $(x_n)_{n \geq 1}$ has uncountably many distinct terms, then μ is not σ -finite. This is because no countable collection of sets with finite measure can cover an uncountable set.

□

EXERCISE A.41. (i) Describe the relation between *Borel-Cantelli lemmas* and *Kolmogorov's 0-1 law*. Construct an example where $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$ and $\mathbb{P}(A_n \text{ i.o.}) = p$, where $p \in (0, 1)$. Does this contradict the *second Borel-Cantelli lemma*?

(ii) Let $A_1, A_2, \dots$ be any independent sequence of events and let

$$S_x = \left\{ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \mathbb{1}_{A_j} \leq x \right\}.$$

Prove that for each $x \in \mathbb{R}$ we have that $\mathbb{P}(S_x) \in \{0, 1\}$.

SOLUTION. (i) The Borel-Cantelli lemmas provide conditions under which a sequence of events will occur infinitely often and consist of two parts. The first part states that if the sum of the probabilities of a sequence of events is finite, then almost surely only a finite number of those events will occur. The second part, which applies only to independent events, states that if the sum of the probabilities is infinite, then almost surely infinitely many of those events will occur. Kolmogorov's 0-1 law, on the other hand, deals with tail events of a sequence of independent random variables. These are events that depend only on the outcomes of the random variables from some point onward, regardless of how far into the sequence that point is. The law states that the probability of such a tail event occurring is either 0 or 1.

Consider a sequence of independent events A_n such that

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots = \infty.$$

By the second part of the Borel-Cantelli lemma, we can conclude that A_n occurs infinitely often with probability 1. This is consistent with Kolmogorov's 0-1 law, which tells us that the probability of a tail event (like A_n occurring infinitely often) is either 0 or 1. In this case, the Borel-Cantelli lemma specifies that the probability is 1.

(ii) We define

$$\mathcal{T} := \bigcap_{n=1}^{\infty} \sigma(A_j)_{j \geq n}$$

as the tail σ -algebra generated by the events $A_n, A_{n+1}, \dots$. For any fixed $m \geq 1$, we can express S_x as

$$S_x = \left\{ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^{m-1} \mathbb{1}_{A_j} + \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=m}^n \mathbb{1}_{A_j} \leq x \right\},$$

which is equivalent to considering the set

$$S_x = \left\{ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=m}^n \mathbb{1}_{A_j} \leq x \right\}.$$

Since $\mathbb{1}_{A_m}$ is $\sigma(A_j)_{j \geq m}$ -measurable, it follows that $\frac{1}{n} \sum_{j=m}^n \mathbb{1}_{A_j}$ is also $\sigma(A_j)_{j \geq m}$ -measurable; hence both

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{j=m}^n \mathbb{1}_{A_j}, \quad \text{and} \quad \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{j=m}^n \mathbb{1}_{A_j}$$

are $\mathcal{T}$ -measurable. Given that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{j=m}^n \mathbb{1}_{A_j} \leq \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \mathbb{1}_{A_j} \leq \limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{j=m}^n \mathbb{1}_{A_j},$$

if the limit exists, the middle term must be $\mathcal{T}$ -measurable. By Kolmogorov's 0-1 law, any event in the tail σ -algebra $\mathcal{T}$ has probability either 0 or 1. Since S_x is $\mathcal{T}$ -measurable, it follows that $\mathbb{P}(S_x) \in \{0, 1\}$.

□

EXERCISE A.42. Let (S, Σ, μ) be a measure space. Suppose that $f \in \Sigma^+(S, \Sigma)$ and $f_n \in \Sigma^+(S, \Sigma)$ are such that $f_n \rightarrow f$ pointwise as $n \rightarrow \infty$, and

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu < \infty.$$

(i) Use *Fatou's lemma* to prove that for any $E \in \Sigma$,

$$\int_E f d\mu = \lim_{n \rightarrow \infty} \int_E f_n d\mu.$$

(ii) Show that this conclusion is false if we drop the assumption that

$$\lim_{n \rightarrow \infty} \int f_n d\mu < \infty.$$

SOLUTION. (i) Consider the non-negative function $f_n - f$. By Fatou's lemma, we have

$$\int \liminf_{n \rightarrow \infty} (f_n - f) d\mu \leq \liminf_{n \rightarrow \infty} \int (f_n - f) d\mu.$$

Given that $f_n \rightarrow f$ pointwise, we have $\liminf_{n \rightarrow \infty} (f_n - f) = 0$. Therefore,

$$0 \leq \liminf_{n \rightarrow \infty} \int (f_n - f) d\mu = \liminf_{n \rightarrow \infty} \int f_n d\mu - \int f d\mu = 0.$$

Using the given condition $\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$, we deduce that for any $E \in \Sigma$,

$$\int_E f d\mu = \lim_{n \rightarrow \infty} \int_E f_n d\mu.$$

(ii) Consider the sequence of functions

$$f_n(x) = \begin{cases} n, & 0 < x < \frac{1}{n}, \\ 0, & \text{otherwise.} \end{cases}$$

on the measure space $[0, 1]$. We have $\lim_{n \rightarrow \infty} f_n = 0$ pointwise, but $\int f_n d\mu = 1$ for all n , whereas $\int f d\mu = 0$.

□

EXERCISE A.43. Let (S, Σ, μ) be a measure space and let $p \in [1, \infty)$. Let $\mathcal{T}$ be the set of simple functions defined on S .

(i) Show that $f \in \mathcal{T} \cap \mathcal{L}^p(\mu)$ if and only if $f \in \mathcal{T}$ and $\mu\{f \neq 0\} < \infty$.

(ii) Does it hold that $\mathcal{T} \cap \mathcal{L}^p(\mu) = \mathcal{T} \cap \mathcal{L}^1(\mu)$ for all $p \in [1, \infty)$?

SOLUTION. (i) Write the simple function f in its canonical form

$$f = \sum_{i=1}^n a_i \mathbb{1}_{A_i},$$

where the sets A_i are pairwise disjoint and the coefficients a_i are distinct and nonzero. Then

$$\|f\|_p^p = \int_S |f|^p d\mu = \sum_{i=1}^n |a_i|^p \mu(A_i).$$

Thus $\|f\|_p < \infty$ if and only if $\mu(A_i) < \infty$ for every i . Since $\{f \neq 0\} = \bigcup_{i=1}^n A_i$, this is equivalent to $\mu\{f \neq 0\} < \infty$.

(ii) Yes. By part (i), for every $p \in [1, \infty)$,

$$f \in \mathcal{T} \cap \mathcal{L}^p(\mu) \iff f \in \mathcal{T} \text{ and } \mu\{f \neq 0\} < \infty.$$

The right-hand condition does not depend on p , so

$$\mathcal{T} \cap \mathcal{L}^p(\mu) = \mathcal{T} \cap \mathcal{L}^1(\mu).$$

□

Remark. The equivalence in part (ii) is special to simple functions with support of finite measure; it does not assert that $\mathcal{L}^1(\mu) = \mathcal{L}^p(\mu)$ in general.

EXERCISE A.44. On the measure space $((0, \infty), \mathcal{B}(0, \infty), \lambda)$, define the sequence of functions

$$f_n(x) := e^{-nx} - 2e^{-2nx}, \quad x > 0, \quad n \geq 1. \quad (\text{A.1})$$

(i) Can we use Tonelli-Fubini's theorem to infer that

$$\sum_{n=1}^{\infty} \int_0^{\infty} f_n(x) dx = \int_0^{\infty} \left(\sum_{n=1}^{\infty} f_n(x) \right) dx?$$

(ii) Does the answer change if we replace the sequence $(f_n)_{n \geq 1}$ defined in (A.1) with a sequence of positive functions with $\lambda(f_n) < \infty$?

SOLUTION. (i) No. Since $e^{-nx} - 2e^{-2nx}$ may take negative values when $x > 0$ and $n \geq 1$, we cannot apply Tonelli's theorem. Fubini's theorem requires us to check whether the integral of the absolute value converges. We therefore calculate

$$\int_0^{\infty} |e^{-nx} - 2e^{-2nx}| dx.$$

Solving $e^{-nx} - 2e^{-2nx} = 0$ gives $x = \frac{\ln 2}{n}$. We split the integral:

$$\int_0^{\infty} |e^{-nx} - 2e^{-2nx}| dx = \int_0^{\frac{\ln 2}{n}} (2e^{-2nx} - e^{-nx}) dx + \int_{\frac{\ln 2}{n}}^{\infty} (e^{-nx} - 2e^{-2nx}) dx. \quad (\text{A.2})$$

We have

$$\begin{aligned} \int_0^{\frac{\ln 2}{n}} (2e^{-2nx} - e^{-nx}) dx &= \int_0^{-2\ln 2} -\frac{2e^u - e^{u/2}}{2n} du \\ &= \frac{1}{2n} \int_{-2\ln 2}^0 (2e^u - e^{u/2}) du \\ &= \frac{1}{2n} \left(\frac{3}{2} - 1 \right) = \frac{1}{4n}. \end{aligned} \quad (\text{A.3})$$

Also, we have

$$\int_{\frac{\ln 2}{n}}^{\infty} (e^{-nx} - 2e^{-2nx}) dx = \left[-\frac{1}{n} (e^{-nx} - e^{-2nx}) \right]_{\frac{\ln 2}{n}}^{\infty}$$

$$= 0 - \left(-\frac{1}{4n}\right) = \frac{1}{4n}. \quad (\text{A.4})$$

Substituting (A.3) and (A.4) into (A.2), we have

$$\int_0^\infty |e^{-nx} - 2e^{-2nx}| dx = \frac{1}{2n}.$$

Hence, $\sum_{n=1}^\infty \frac{1}{2n}$ diverges. Therefore, the series of integrals of $|f_n(x)|$ does not converge and we cannot use Fubini's theorem.

- (ii) Yes. If we replace $(f_n)_{n \geq 1}$ with a sequence of positive functions $(\tilde{f}_n)_{n \geq 1}$, then Tonelli's theorem applies directly. However, although $\lambda(\tilde{f}_n) < \infty$, that is,

$$\int |\tilde{f}_n(x)| d\lambda(x) = \int_0^\infty |\tilde{f}_n(x)| dx < \infty,$$

we may still not be able to apply Fubini's theorem because the sum of the integrals may diverge. For example, let

$$\tilde{f}_n(x) = e^{-nx} - 2e^{-2nx}, \quad x \in \left[\frac{\ln 2}{n}, \infty\right).$$

Then

$$\sum_{n=1}^\infty \int_{\ln 2/n}^\infty |e^{-nx} - 2e^{-2nx}| dx = \sum_{n=1}^\infty \frac{1}{4n},$$

which diverges. Thus Fubini's theorem does not apply even though the individual functions are non-negative and integrable. $\square$

EXERCISE A.45. Let $S_1 = (0, 1)$, $\Sigma_1 = \mathcal{B}(S_1)$ and μ_1 is the Lebesgue measure. Let $S_2 = (0, 1)$, $\Sigma_2 = \mathcal{P}(S_2)$ and let μ_2 be the counting measure on (S_2, Σ_2) . Define

$$f(s_1, s_2) = \begin{cases} 1, & s_1 = s_2 \\ 0, & \text{else} \end{cases}.$$

- (i) Justify that

$$\int_{S_1} \int_{S_2} f(s_1, s_2) d\mu_2(s_2) d\mu_1(s_1) = 1, \quad \int_{S_2} \int_{S_1} f(s_1, s_2) d\mu_1(s_1) d\mu_2(s_2) = 0.$$

- (ii) Does this contradict Fubini's theorem?

SOLUTION. (i) For

$$\int_{S_1} \int_{S_2} f(s_1, s_2) d\mu_2(s_2) d\mu_1(s_1),$$

the inner integral counts the sum of $f(s_1, s_2)$ over all $s_2 \in S_2$. For fixed s_1 , exactly one point, $s_2 = s_1$, contributes a value of 1. Therefore,

$$\int_{S_1} \int_{S_2} f(s_1, s_2) d\mu_2(s_2) d\mu_1(s_1) = \int_{S_1} 1 d\mu_1(s_1) = 1.$$

In the second iterated integral, the differentials should be ordered as

$$\int_{S_2} \int_{S_1} f(s_1, s_2) d\mu_1(s_1) d\mu_2(s_2).$$

For fixed s_2 , the inner integral is the Lebesgue measure of the singleton $\{s_2\}$ and is therefore zero. Hence the second iterated integral is zero.

- (ii) No. Fubini's theorem requires both measures to be σ -finite. The counting measure μ_2 is not σ -finite because S_2 is uncountable, so Fubini's theorem does not apply.

□

EXERCISE A.46. Let (S, Σ, μ) be a measure space. Assume that $S = \bigcup_{n=1}^{\infty} E_n$, where $(E_n)_{n \geq 1}$ are pairwise disjoint measurable sets with $\mu(E_n) < \infty$, for all $n \geq 1$. Define another measure ν on Σ by

$$\nu(A) = \sum_{n=1}^{\infty} 2^{-n} \frac{\mu(A \cap E_n)}{\mu(E_n) + 1}.$$

- (i) Prove that ν is a finite measure on (S, Σ) .
(ii) Prove that $\nu \ll \mu$ and that $\mu \ll \nu$.
(iii) Find explicitly two positive integrable functions f and g so that for all $A \in \Sigma$,

$$\nu(A) = \int_A f d\mu, \quad \mu(A) = \int_A g d\nu.$$

SOLUTION. (i) Since $S = \bigcup_{n=1}^{\infty} E_n$, we have

$$\nu(S) = \sum_{n=1}^{\infty} 2^{-n} \frac{\mu(E_n)}{\mu(E_n) + 1} < \sum_{n=1}^{\infty} 2^{-n} = 1.$$

Hence, ν is a finite measure on (S, Σ) .

- (ii) " $\nu \ll \mu$ " If $\mu(E) = 0$, then $\mu(E \cap E_n) = 0$ for every n , and thus

$$\nu(E) = \sum_{n=1}^{\infty} 2^{-n} \frac{\mu(E \cap E_n)}{\mu(E_n) + 1} = 0.$$

Therefore, $\nu \ll \mu$.

" $\mu \ll \nu$ " If $\nu(E) = 0$, then every non-negative term in the defining series for $\nu(E)$ is zero. Hence, $\mu(E \cap E_n) = 0$ for every n , and

$$\mu(E) = \mu\left(\bigcup_{n=1}^{\infty} (E \cap E_n)\right) = \sum_{n=1}^{\infty} \mu(E \cap E_n) = 0.$$

Therefore, $\mu \ll \nu$.

- (iii) For

$$\nu(A) = \int_A f d\mu, \tag{A.5}$$

the Radon–Nikodym derivative of ν with respect to μ is

$$f = \sum_{n=1}^{\infty} 2^{-n} \frac{\mathbb{1}_{E_n}}{\mu(E_n) + 1}.$$

Indeed,

$$\int_A f d\mu = \sum_{n=1}^{\infty} 2^{-n} \frac{1}{\mu(E_n) + 1} \int_A \mathbb{1}_{E_n} d\mu = \nu(A).$$

For the reverse identity,

$$\mu(A) = \int_A g d\nu, \tag{A.6}$$

the Radon–Nikodym derivative of μ with respect to ν is

$$g = f^{-1} = \sum_{n=1}^{\infty} 2^n (\mu(E_n) + 1) \mathbb{1}_{E_n}.$$

Since the sets E_n are pairwise disjoint,

$$\begin{aligned} \int_A g d\nu &= \sum_{n=1}^{\infty} 2^n (\mu(E_n) + 1) \nu(A \cap E_n) \\ &= \sum_{n=1}^{\infty} \mu(A \cap E_n) = \mu(A). \end{aligned}$$

Therefore, (A.6) is verified. The two functions f and g are integrable with respect to μ and ν , respectively. □

EXERCISE A.47. Suppose that $X_n \Rightarrow X$ and that $Y_n \rightarrow c$ in probability, where c is a constant. Suppose that all random variables are defined on the same probability space. Show that $X_n Y_n \Rightarrow cX$.

SOLUTION. To show $X_n Y_n \Rightarrow cX$, decompose $X_n Y_n = X_n(Y_n - c) + cX_n$. It remains to show that $X_n(Y_n - c) \rightarrow 0$ in probability. For fixed $\varepsilon > 0$ and any $R > 0$, we have

$$\mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \leq \mathbb{P}(|X_n| > R) + \mathbb{P}\left(|Y_n - c| > \frac{\varepsilon}{R}\right).$$

Since $Y_n \rightarrow c$ in probability, the second term tends to zero. Since $X_n \Rightarrow X$, the sequence (X_n) is tight, so the first term can be made uniformly small by choosing R sufficiently large. It follows that

$$\mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \rightarrow 0,$$

and hence $X_n Y_n \Rightarrow cX$. □

EXERCISE A.48. Suppose that $X_n \Rightarrow X$. Show that for all open sets G we have

$$\liminf_{n \rightarrow \infty} \mathbb{P}(X_n \in G) \geq \mathbb{P}(X \in G).$$

SOLUTION. We begin with stating and proving the following lemma.

LEMMA A.1.

$X_n \Rightarrow X$ implies $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every bounded continuous function f .

Proof. Let f be a bounded continuous function. Without loss of generality, assume that $|f(x)| \leq 1$ for all x . Then f is uniformly continuous on a sufficiently large compact set. Let I be such a compact set with $\mathbb{P}(X \notin I) < \varepsilon$. Partition $I = \bigcup_j I_j$ into finitely many subintervals such that the variation of f on each I_j is less than ε . For each I_j , choose a point x_j and define

$$f_\varepsilon = \sum_j f(x_j) \mathbb{1}_{I_j}.$$

The triangle inequality then gives

$$\begin{aligned} |\mathbb{E}f(X_n) - \mathbb{E}f(X)| &\leq |\mathbb{E}f(X_n) - \mathbb{E}f_\varepsilon(X_n)| + |\mathbb{E}f_\varepsilon(X_n) - \mathbb{E}f_\varepsilon(X)| \\ &\quad + |\mathbb{E}f_\varepsilon(X) - \mathbb{E}f(X)|. \end{aligned}$$

Each term can be made arbitrarily small as $n \rightarrow \infty$. Therefore, $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$. □

For any open set G , define the bounded continuous functions

$$f_m(x) = \min \{1, md(x, G^c)\}.$$

Then $0 \leq f_m(x) \uparrow \mathbb{1}_G(x)$. By the monotone convergence theorem,

$$\mathbb{E}f_m(X) \uparrow \mathbb{E}\mathbb{1}_G(X) = \mathbb{P}(X \in G).$$

Since $X_n \Rightarrow X$, the preceding lemma gives $\mathbb{E}f_m(X_n) \rightarrow \mathbb{E}f_m(X)$. Moreover, $f_m \leq \mathbb{1}_G$, so for every m ,

$$\liminf_{n \rightarrow \infty} \mathbb{P}(X_n \in G) \geq \liminf_{n \rightarrow \infty} \mathbb{E}f_m(X_n) = \mathbb{E}f_m(X).$$

Taking $m \rightarrow \infty$ proves the result. $\square$

EXERCISE A.49. Consider the triangular distribution with density $1 - |x|$, for $x \in (-1, 1)$. Show that the corresponding characteristic function is $\frac{2(1 - \cos t)}{t^2}$.

Hint: if X and Y are uniformly distributed on $(-\frac{1}{2}, \frac{1}{2})$ then $X + Y$ has a triangular distribution.

SOLUTION. The characteristic function of a uniform distribution on $(-\frac{1}{2}, \frac{1}{2})$ is

$$\phi_X(t) = \int_{-1/2}^{1/2} e^{itx} dx = \frac{e^{it/2} - e^{-it/2}}{it} = \frac{\sin(t/2)}{t/2}.$$

Therefore,

$$\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t) = \left(\frac{\sin(t/2)}{t/2}\right)^2 = \frac{2(1 - \cos t)}{t^2}.$$

Since $X + Y$ has the triangular density $1 - |x|$ on $(-1, 1)$, this is the required characteristic function. $\square$

EXERCISE A.50. Suppose that the random variables X and Y are independent and that $X + Y$ and X have the same distribution. What can you say, using characteristic functions, about the distribution of Y ?

SOLUTION. The characteristic functions of X and Y are

$$\phi_X(t) = \mathbb{E}[e^{itX}], \quad \phi_Y(t) = \mathbb{E}[e^{itY}].$$

Since $X + Y$ and X have the same distribution and X and Y are independent,

$$\phi_X(t) = \phi_{X+Y}(t) = \phi_X(t)\phi_Y(t).$$

Because $\phi_X(0) = 1$ and ϕ_X is continuous, it is nonzero in a neighbourhood of zero. Hence, $\phi_Y(t) = 1$ throughout that neighbourhood. This implies that $Y = 0$ almost surely, so Y has the degenerate distribution at zero. $\square$

EXERCISE A.51 (Bayes' Rule). On a measure space $(\Omega, \mathcal{F}, \mathbb{P})$, $\mathcal{G} \subset \mathcal{F}$ and $G \in \mathcal{G}$, show that we have

$$\mathbb{P}(G | A) = \frac{\int_G \mathbb{P}(A | \mathcal{G}) d\mathbb{P}}{\int_\Omega \mathbb{P}(A | \mathcal{G}) d\mathbb{P}}.$$

SOLUTION. By definition,

$$\mathbb{P}(G | A) = \frac{\mathbb{P}(A \cap G)}{\mathbb{P}(A)}.$$

Since $G \in \mathcal{G}$,

$$\mathbb{P}(A \cap G) = \int_G \mathbb{P}(A | \mathcal{G}) d\mathbb{P}.$$

Also,

$$\mathbb{P}(A) = \mathbb{E}[\mathbb{1}_A] = \mathbb{E}[\mathbb{E}[\mathbb{1}_A \mid \mathcal{G}]] = \int_{\Omega} \mathbb{P}(A \mid \mathcal{G}) d\mathbb{P}.$$

Substituting these identities gives the result. $\square$

EXERCISE A.52. Let $X_1, X_2, \dots, X_n$ be integrable *i.i.d.* random variables, and let $S_n = X_1 + \dots + X_n$. Show that $\mathbb{E}[X_1 \mid \sigma(S_n)] = \frac{S_n}{n}$ with probability 1.

SOLUTION. Since $X_1, \dots, X_n$ are identically distributed and exchangeable, we have

$$\mathbb{E}[X_k \mid \sigma(S_n)] = \mathbb{E}[X_1 \mid \sigma(S_n)], \quad k = 1, \dots, n.$$

By linearity of conditional expectation,

$$\begin{aligned} S_n &= \mathbb{E}[S_n \mid \sigma(S_n)] \\ &= \sum_{k=1}^n \mathbb{E}[X_k \mid \sigma(S_n)] = n\mathbb{E}[X_1 \mid \sigma(S_n)]. \end{aligned}$$

Therefore, $\mathbb{E}[X_1 \mid \sigma(S_n)] = \frac{S_n}{n}$ almost surely. $\square$

EXERCISE A.53. Give an example on $\Omega = \{a, b, c\}$ in which

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_1] \mid \mathcal{F}_2] \neq \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_2] \mid \mathcal{F}_1].$$

SOLUTION. Let $\Omega = \{a, b, c\}$ with the uniform probability measure, and let

$$\mathcal{F}_1 = \{\emptyset, \Omega, \{a\}, \{b, c\}\}, \quad \mathcal{F}_2 = \{\emptyset, \Omega, \{b\}, \{a, c\}\}.$$

Define $X(a) = 1$, $X(b) = 2$, and $X(c) = 3$. Then

$$\mathbb{E}[X \mid \mathcal{F}_1](a) = 1, \quad \mathbb{E}[X \mid \mathcal{F}_1](b) = \mathbb{E}[X \mid \mathcal{F}_1](c) = \frac{5}{2},$$

whereas $\mathbb{E}[X \mid \mathcal{F}_2] = 2$. Consequently,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_1] \mid \mathcal{F}_2] \neq \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_2] \mid \mathcal{F}_1].$$

$\square$

References

- [1] Patrick Billingsley. *Probability and Measure*. John Wiley & Sons, New York, 3rd edition, 1995.
- [2] Peter J. C. Spreij. Measure theoretic probability. Lecture notes, 2021. Minor revisions by S. G. Cox.